ITCSI030 Cyber Security Practical Project
Project Title: University Network Design

New Era Institute of Vocational
And Continuing Education

Subject: ITCSI 030 Cyber Security Practical Project

Project Title: University Network Design

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Project Supervisor: Mr. Gan

Preface

Designing a university network is a multifaceted endeavor that requires meticulous attention to every aspect of ensure optimal performance, scalability, and security. Create a network that meets the diverse needs of students, faculty, staff, and administrations.

Principles and goals required for university network design. The delicate balance between security design. The delicate balance between security and accessibility emphasizes the fluid nature of technology, requiring flexibility to adapt to changing needs. Collaboration among stakeholders was considered critical to aligning the network design with the overall mission and goals of the institution.

In this project, I will design a university network to provide a relatively safe and stable network for the campus students and teachers. The university network in the project is divided into main campus and branch campus. The main campus has 7 floors, and the branch campus has 5 floors. There are approximately 800 faculty and staff and approximately 4,200 students on the main and branch campus.

Finally, if I were to use hardware for my final year project, it would be very expensive. So I will use Cisco Packet Tracer to simulate the university's network environment to prove that these configurations are also feasible in a real network.

Acknowledge

Frist, I would like to thank my project supervisor Mr. Gan. In the process of completing this project, I encountered many difficulties, and Supervisor Gan not only helped me solve these problems, but also gave me valuable guidance and support. His expertise and experience provide me with strong support and enable me to better complete my project. Thanks to Mr. Gan for his patient guidance and selfless help.

I am grateful to all the teachers who have taught me. Thank you for your selfless teaching and patient guidance, which have enabled me to continue to improve academically. Thank you again for your support and careful teaching. I will cherish this help and continue to study hard.

Finally, I want to thank my friends. When I encounter difficulties in completing my assignment and tutorial, you always go out of your way to help me and give me valuable suggestions and suggestions for improvement. Your friendship and support are an integral part of my learning journey. Thank you for helping me through the academic challenges together. I will remember your friendship and transform this gratitude into better efforts in return for your help and support. Thank you again for your companionship and encouragement along the way.

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Charter 1: Introduction

Nowadays, the internet has penetrated into every aspect of people's lives and has become an important part of the operation of society, and university life is no exception. The internet plays an indispensable role in universities, providing students with rich academic resources, social platforms, and convenient ways to obtain information.

The internet provides campus students with a wide range of academic resources, allowing them to more easily access materials, papers, and reference books in various disciplines. Online libraries, academic databases, and open course platforms make learning more flexible and diverse.

The internet also provides campus students with a rich and colorful social platform and promotes communication and cooperation. Through social media, students can easily connect with classmates, teachers, and other scholars around the world to share academic insights, project experiences, and interesting life stories.

Finally, the internet has become an indispensable tool for campus students in today's era to learn, communicate, and obtain information.

1.1 Motivation

I will choose to design a university network as my final year project because I know that a university network is a complex, sophisticated, and huge network system that is responsible for connecting various resources and information inside and outside the school. I want to have a deeper understanding of how this network system ensures efficiency, security, and reliability. Understanding the architecture, protocols, and security measures of university networks was a goal of explorations.

By challenging myself to design a viable university network, I believe this will develop my ability to solve real-world problems. This is not only a technical test, but also requires me to comprehensively consider multiple factors such as system integration, user experience, and information security. I firmly believe that through this project, I can not only satisfy my curiosity about university networks, but also lay a solid technical foundation for my future career.

1.2 Project Scope

In my project, the university is mainly divided into main campus and branch campuses. The main campus has 7 floors, and the branch campus has 5 floors. These are approximately 800 faculty and staff and approximately 4,200 students on the main and branch campuses.

In this project, the technical configuration involved in the two campuses will include DHCP, ACL, NAT, VPN , Routing, Dial peer, Telephony-service and SSH.

Equipment for both campuses includes routers, multilayer switch, switch, firewall, WLC controllers, access point, and IP phones.

1.3 Project Objectives

The project will meet the following objectives:

- DHCP: auto assign IP to all devices.
- ACL: Block unknow network traffic and authorize specific users.
- VLAN: Isolate faculty, staff, students, and guest networks.
- NAT: Convert the real IP of the internal network to the virtual IP of the external network.
- AAA Server: AAA is used to verify the Wi-Fi account password and FTP account password.
- CCTV System: Improve security levels in server rooms and campuses.
- Door Access system: Ensure only authorized personnel have access to restricted areas.
- WLC controller: Used to control access points.
- VPN: Encrypted network connections for main campus and branch campuses.
- Firewall: Provide a more secure network for the intranet.

Chapter 2: Literature Review

2.1Comparison with Existing Projects

First, I found the reference material on YouTube. This is my main reference video[1]. He connected two campus networks. The title is "Packet Tracer Campus Network Design and Implementation Project." In this video, I observed that he successfully established the connection between the main campus and the branch campus and conducted a successful ping test.

Second, I needed a VPN in my design, so I found a second reference video [2]. He uses VPN to encrypt data, The title is “Getting Started with VPN in Cisco Packet Tracer 7.3”. In this video, I observed that he successfully encrypted the IP data communicated between the routers at both ends by setting up a VPN.

Third, I needed to use NAT in my design, so I found a third reference video [3]. This video shows how to configure NAT using a firewall to block direct access to the internal network from the external network while allowing access to the Internet through NAT. The video is titled "Configuring the ASA Firewall Using Packet Tracer." In this video, I observed that he configured a firewall, effectively blocking ping requests from the external network to the internal network. At the same time, he set up a network address translation NAT to allow the external network to access the internet by connecting to the NAT configured in the firewall.

Fourth, for my design needs, I found a fourth reference video [4] that explains how to configure a Syslog server in Cisco Packet Tracer. The video title is "How to Configure Syslog Server in Cisco Packet Tracer | Technology Hakim #Syslog Configuration CCNA". In this video, I observed him setting up a Syslog server to save log information from the router.

Finally, I needed two different emails to communicate with each other, so I found a fifth reference video [5]. The title of the video is "Creating SMTP in Cisco Packet Tracer (two emails Gmail and yahoo)". In this video, I observe him configuring two email servers so that two different domain names can communicate with each other via email.

2.2 Project improvements

Although the YouTube [1] project has connected and pinged the networks of the main campus and branch campus, I think the security level of vide YouTube [1]is low. In my project, I will add an ASA firewall to protect the server room, set up ACLs to block unknown networks, and add a VPN on the router connecting the main and branch campus for encryption.

In addition to the main references above, I also consulted other resources, mainly borrowing their techniques and concepts without making many modifications.

Charter 3: Methodology

3.1 Work Plan

1. Search online for final year project topics. (2 Weeks)
- Confirmed to do university network design.
- Search the internet for information about university network design.

2. Design university network (2 Weeks)
- Designing Main Campus network
- Designing Branch Campus network

3. Configure VLAN and configure DHCP on the university network. (1 Weeks)
- Configure VLAN for each department and student.
- Configure a DHCP server to automatically assign IP to devices of various departments and students.

4. Configure WLC controller and configure AAA Server on the university network. (2 Weeks)
- Configure WI-FI for students and manage it using WLC.
- Configure AAA Server for student WI-FI authentication.

5. Configure Firewall, ACL, VPN on University.(2 Weeks)
- At Server Room Configure firewall.
- Configure ACL on the router to deny student network ftp and ssh
- Configure ACL on the router to allow only IT Department can ssh.
- Configure VPN between Main Router and Branch Router to encrypt data.

6. Configure NAT on the external network. (1 week)
- Configure Firewall on the external network.
- Configure NAT at the ISP router convert the web server IP into a virtual network.
- Configure NAT on the firewall and convert the Web Server IP converted by the ISP router into a virtual network so that the external network can access the internet.

7. Testing university network. (1 Weeks)

- Test that the main campus network can ping the branch campus network.
- Test whether the ACL is valid.
- Tests from the main campus can be emailed to the branch campus.
- Tests from the main campus can be connected to branch campus via the internet.
- Test whether the VPN encrypts data from two campuses.
- Test whether the external network can access the internet and whether the internal network cannot be ping.
- Test that all firewalls, routers, and multilayer switch are functioning properly.

8. Writing Final year project documentation. (2 weeks)

- Draw the floor plan and draw the network diagram.
- Writing project Report and prepare PPT.

3.2 Gantt Chart

University Network Design	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10	Week11	Week12	Week13
1. Search for final year project topics													
2. Design university network													
3. Configure VLAN and DHCP													
4. Configure WLC Controller and AAA Server													
5. Configure firewall, ACL, VPN.													
6. Configure Nat on external network.													
7. Testing university network													
8. Writing Final year project documentation.													

3.3 Project Resources

1. Hardware Resources:
The hard ware I will be using for this project consists of a laptop.

2. Software:
In this project , I will be using software including Cisco packet tracer, Draw.io, Microsoft Word, and Microsoft PowerPoint, Microsoft Excel , and google chrome software to complete my final project.

3. Access to information:
Internet – Google.com/ Youtube.com/Git Hub/ Bard and ChatGPT.

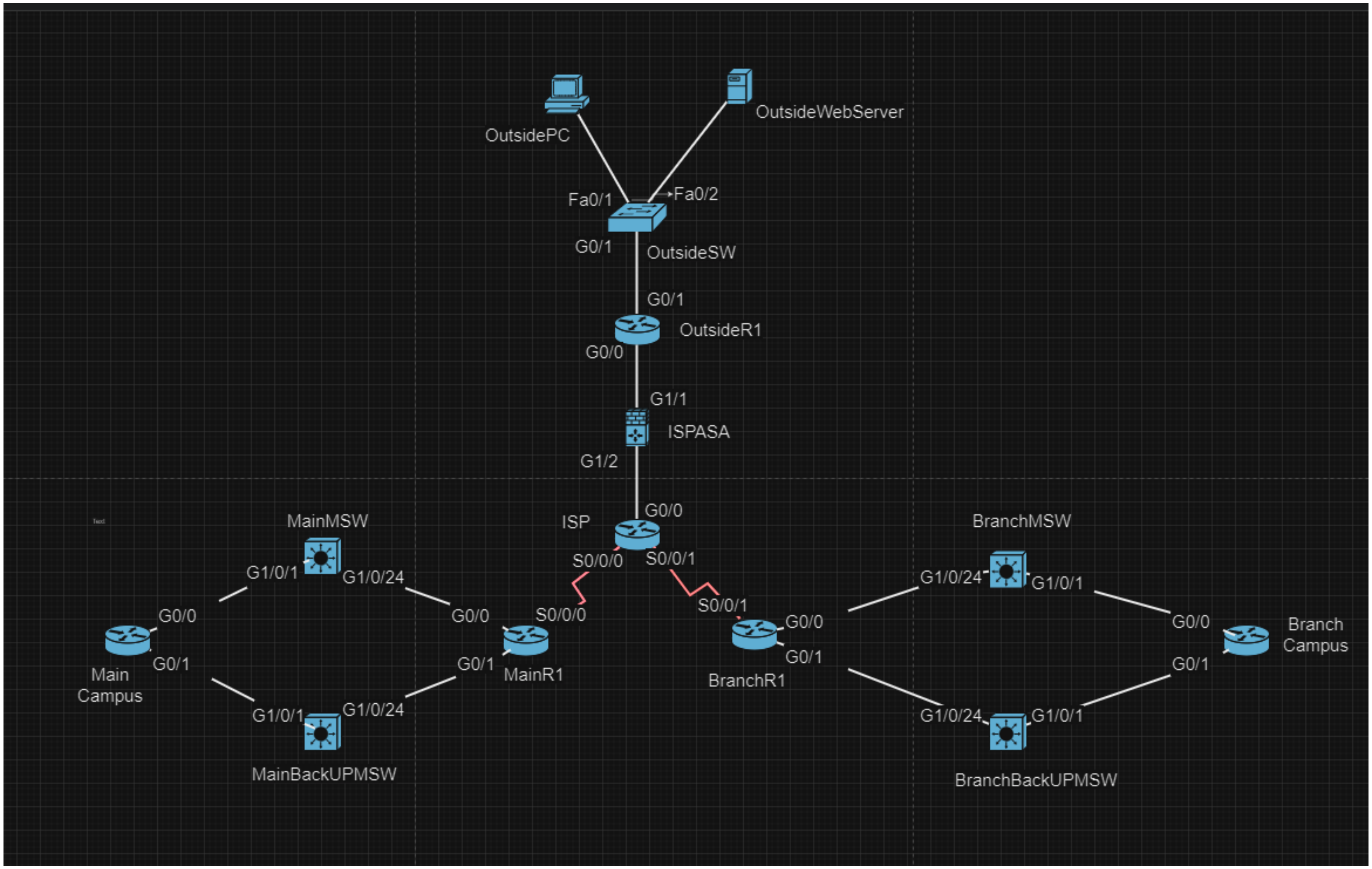
4.Human Resources:
Project Supervisor :

1. Mr. Gan
2. Mr. Daniel Yap

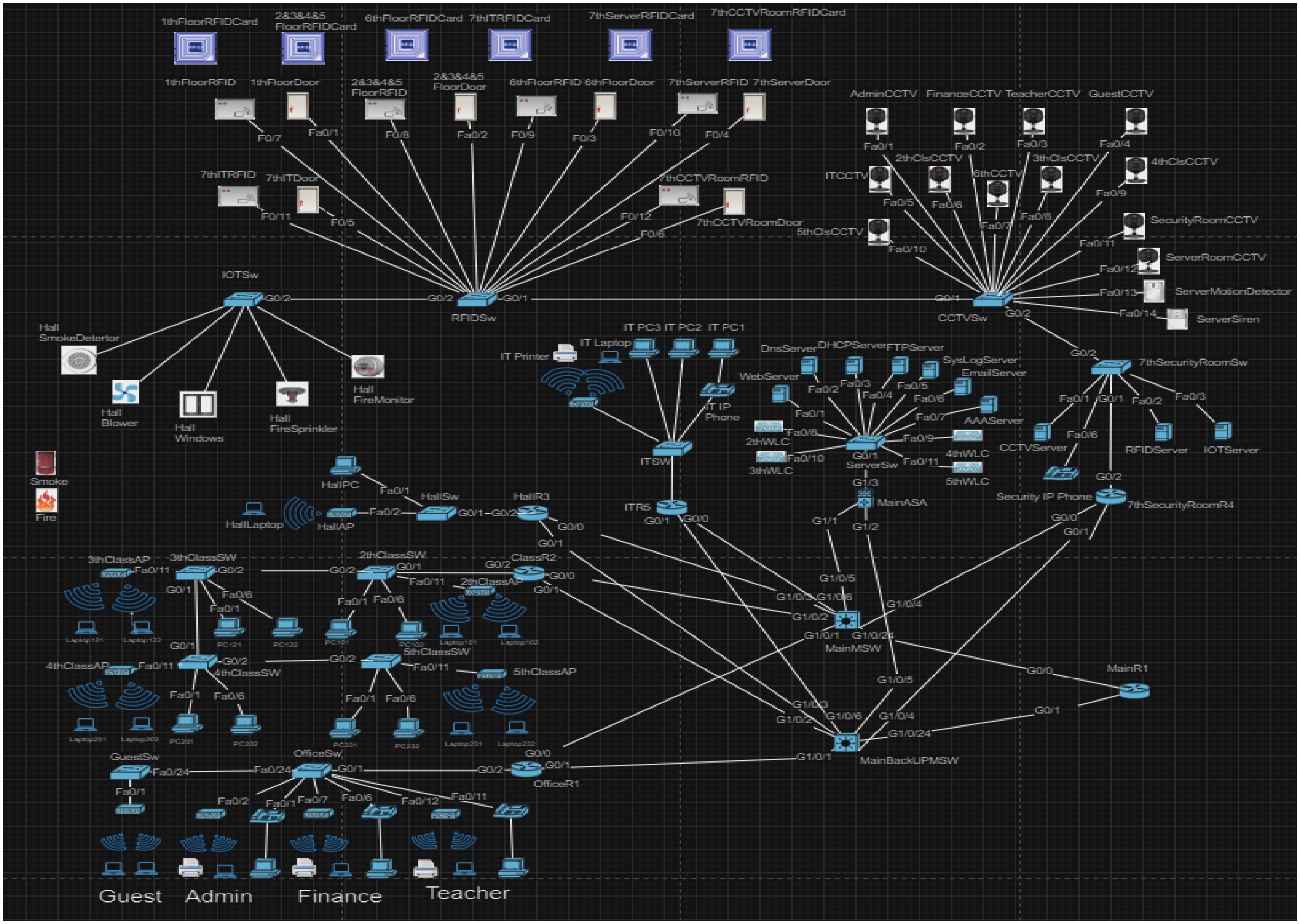
User involve: Lee Han Liang

Chapter 4: Design and Implementation

4.1Topology



Main Campus


1thFloorRFIDCard
2&3&4&5 FloorRFIDCard
6thFloorRFIDCard
7thITRFIDCard
7thServerRFIDCard
7thCCTVRoomRFIDCard
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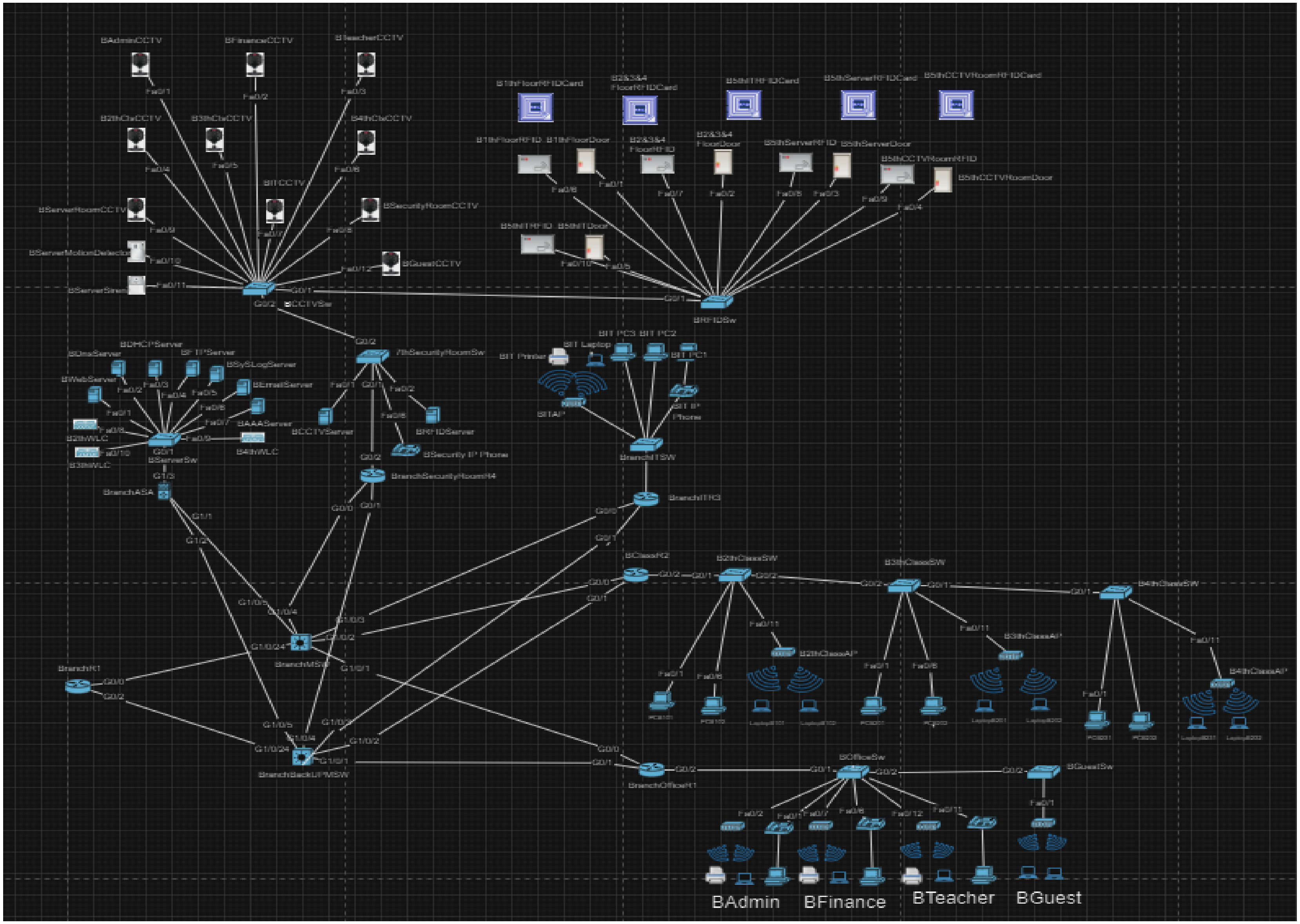
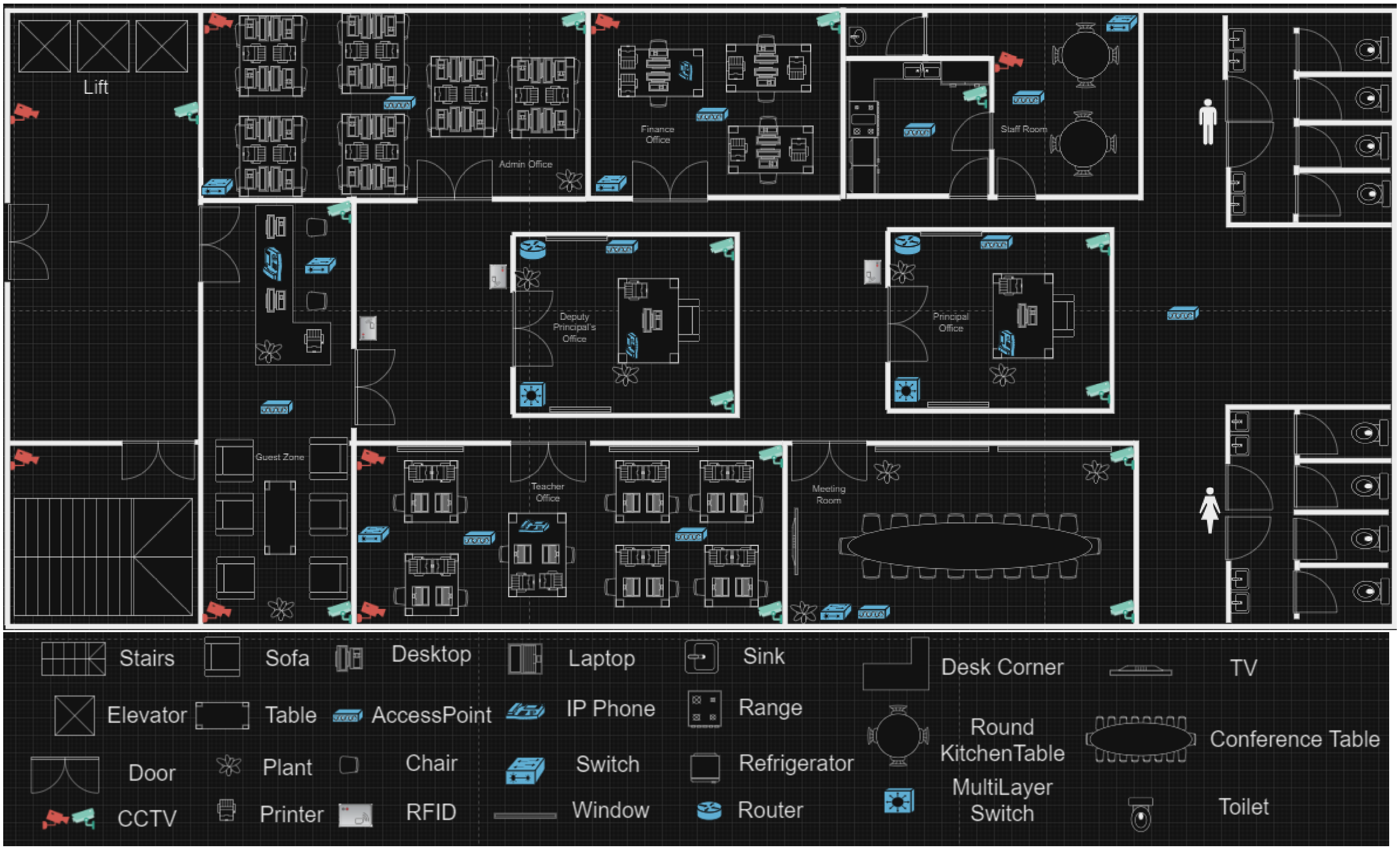
1thFloorDoor
2&3&4&5 FloorRFID
2&3&4&5 FloorDoor
6thFloorRFID
6thFloorDoor
7thServerRFID
7thServerDoor
7thITRFID
7thITDoor
7thCCTVRoomRFID
7thCCTVRoomDoor
AdminCCTV
FinanceCCTV
TeacherCCTV
GuestCCTV
2thClsCCTV
3thClsCCTV
4thClsCCTV
ITCCTV
6thCCTV
5thClsCCTV
SecurityRoomCCTV
ServerRoomCCTV
ServerMotionDetector
ServerSiren
IOTSw
RFIDSw
CCTVSw
Hall SmokeDetertor
Hall Blower
Hall Windows
Hall FireSprinkler
Hall FireMonitor
Smoke
Fire
IT Printer
IT Laptop
IT PC3
IT PC2
IT PC1
IT IP Phone
ITSW
ITR5
DnsServer
DHCPServer
FTPServer
SysLogServer
EmailServer
WebServer
AAAServer
2thWLC
3thWLC
4thWLC
5thWLC
ServerSw
MainASA
7thSecurityRoomSw
CCTVServer
RFIDServer
IOTServer
Security IP Phone
7thSecurityRoomR4
HallPC
HallSw
HallR3
HallLaptop
HallAP
3thClassAP
3thClassSW
2thClassSW
ClassR2
2thClassAP
4thClassAP
4thClassSW
5thClassSW
5thClassAP
MainMSW
MainR1
MainBackUPMSW
GuestSw
OfficeSw
OfficeR1
Guest
Admin
Finance
Teacher

Branch Campus



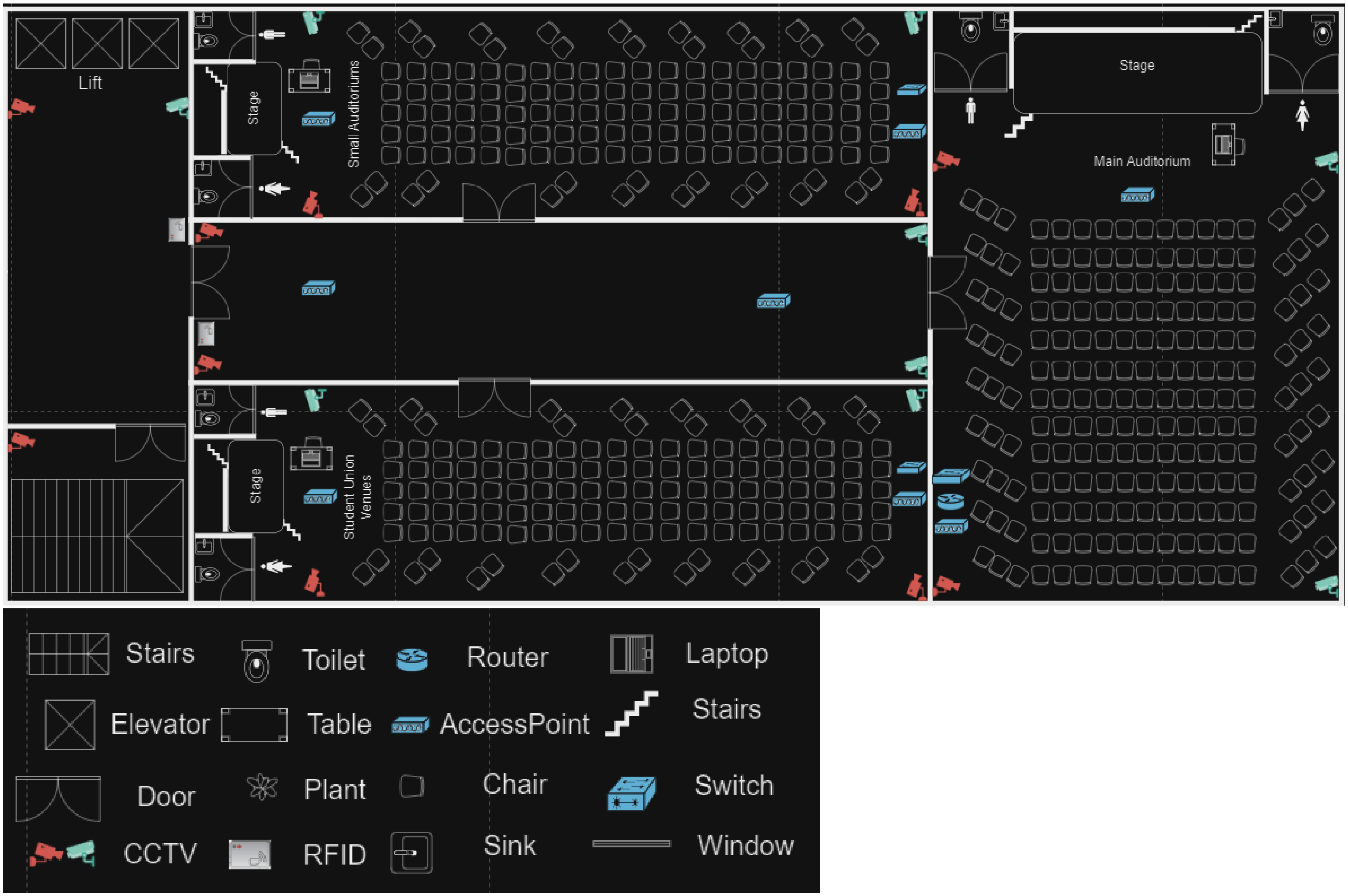
4.2 Floor Plan

Main Campus 1 Floor



Main Campus 2&3&4&5 Floor Plan(The second, third, fourth and fifth floors all have the same design.)

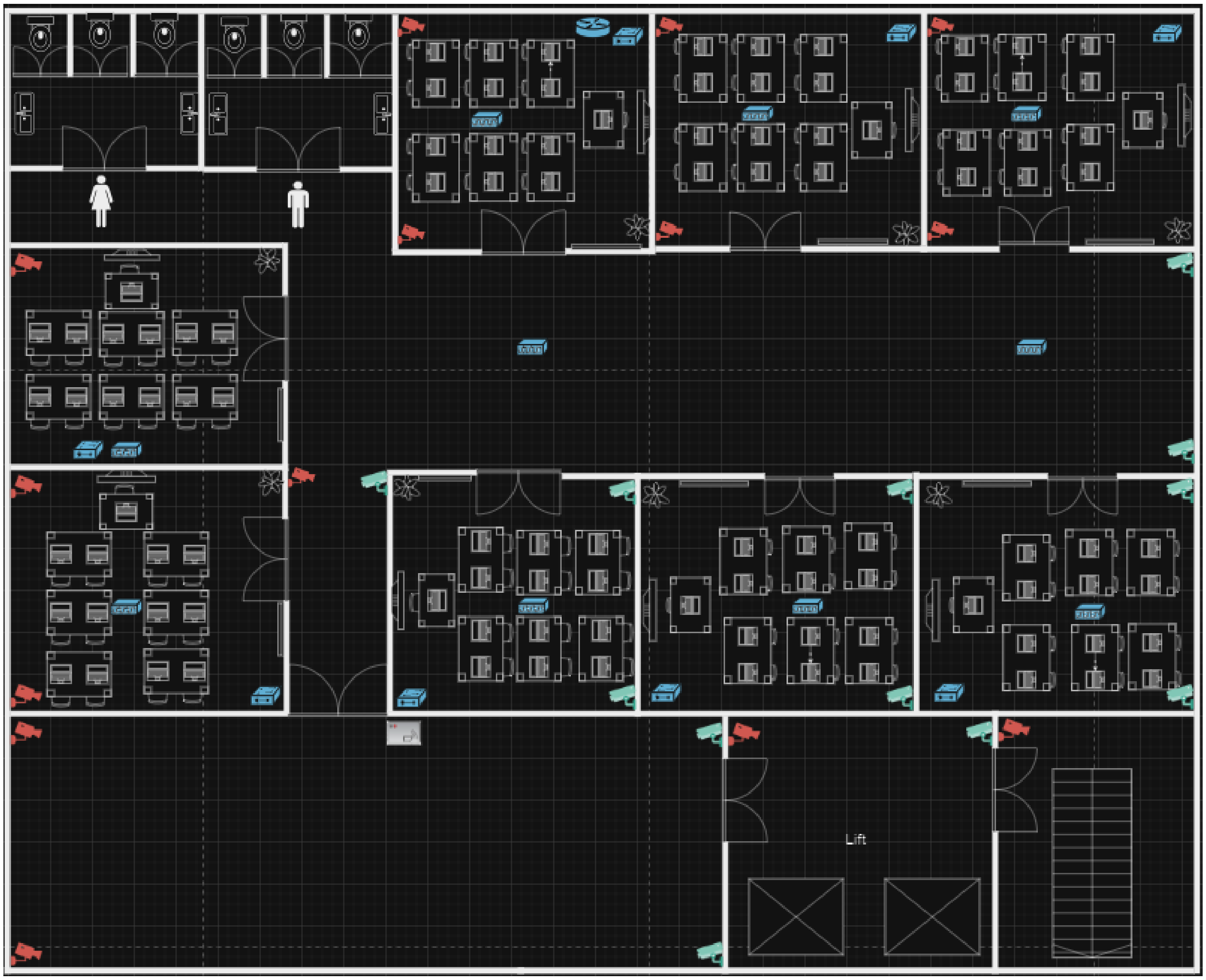
Main Campus 6 Floor Plan

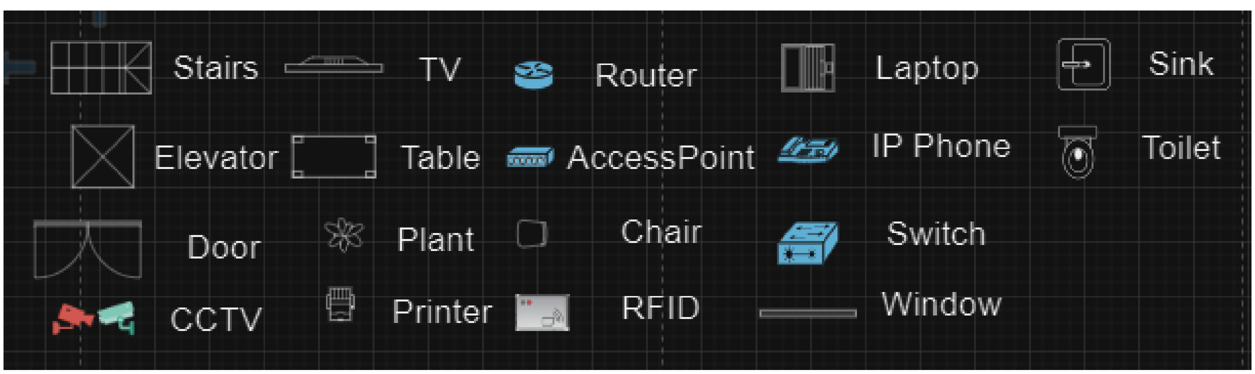


Main Campus 7 Floor Plan

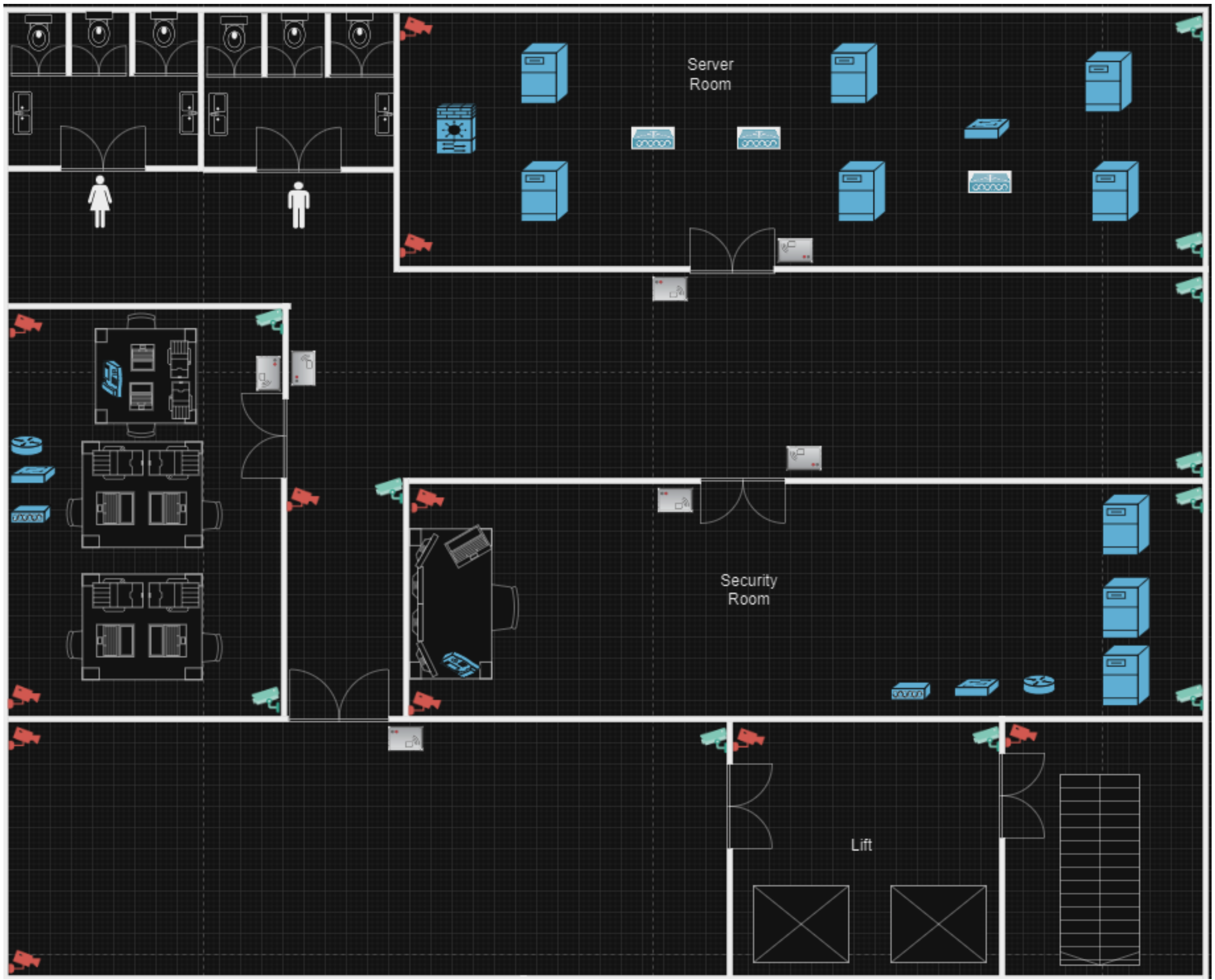
Branch Campus 1 Floor Plan

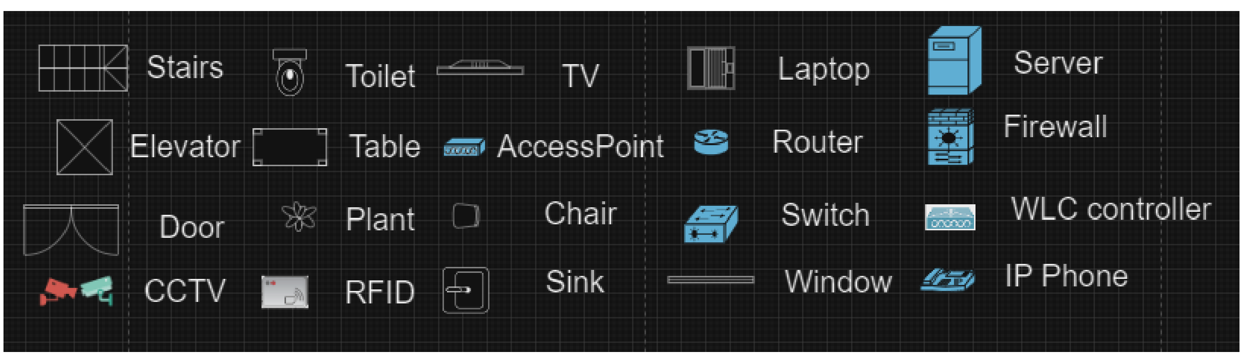
Branch Campus 2&3&4 Floor Plan(The second, third, and fourth floors all have the same design.)





Branch Campus 5 Floor Plan





4.3 Budget

Name	Quantity	Price	Total Price
Router 2911	13	RM7102.53	RM 92,332.89
Multilayer Switch	4	RM22,312.15	RM 89,248.60
Switch 2960-24tt	24	RM7047.79	RM 169,146.96
Server	19	RM 29,141.96	RM 553,697.24
Firewall	3	RM 17,761.17	RM 53,283.51
IP Phone	10	RM2,118.05	RM 21,180.50
WLC Controller	7	RM 18,413.61	RM 128,895.27
Access Point	18	RM2,486.13	RM 44,750.34
Printer	8	RM2,259.00	RM 18,072.00
RFID Reader and door	11	RM1,604.05	RM 17,644.55
CCTV	22	RM1,320.94	RM 29,060.68
Motion Detector	2	RM1,143.33	RM 2,286.66
Siren	2	RM2,024.24	RM 4,048.48
Smoke Detector	1	RM269.00	RM269.00
Blower	1	RM340.11	RM340.11
Window	1	RM124.00	RM124.00
Fire Sprinkler	1	RM219.00	Rm219.00
Fire Monitor	1	RM 869.86	RM 869.86
Desktop	26	RM3,899.00	RM101,374.00
Laptop	27	RM3,564.96	RM96,253.92
		Total Amount:	RM1,423,095.57

4.4 IP Table

Device	Interface	IP Address	Subnet Mask	Default-Gateway	VLAN
OfficeR1	G0/0	192.168.50.2	/24	-	-
	G0/1	192.168.55.2	/24	-	-
	G0/2.4	192.168.4.1	/24	-	VLAN4
	G0/2.5	192.168.5.1	/24	-	VLAN5
	G0/2.6	192.168.6.1	/24	-	VLAN6
	G0/2.8	192.168.8.1	/24	-	VLAN8
	G0/2.61	192.168.61	/24	-	VLAN61
ClassR2	G0/0	192.168.51.2	/24	-	-
	G0/1	192.168.56.2	/24	-	-
	G0/2.11	192.168.11.1	/24	-	VLAN11
	G0/2.12	192.168.12.1	/24	-	VLAN12
	G0/2.13	192.168.13.1	/24	-	VLAN13
	G0/2.14	192.168.14.1	/24	-	VLAN14
	G0/2.95	192.168.95.1	/24	-	VLAN95
	G0/2.96	192.168.96.1	/24	-	VLAN96
	G0/2.97	192.168.97.1	/24	-	VLAN97
	G0/2.98	192.168.98.1	/24	-	VLAN98
	G0/2.101	192.168.101.1	/24	-	VLAN101
	G0/2.102	192.168.102.1	/24	-	VLAN102
	G0/2.121	192.168.121.1	/24	-	VLAN121
	G0/2.122	192.168.122.1	/24	-	VLAN122
	G0/2.201	192.168.201.1	/24	-	VLAN201
	G0/2.202	192.168.202.1	/24	-	VLAN202
	G0/2.231	192.168.231.1	/24	-	VLAN231
	G0/2.232	192.168.232.1	/24	-	VLAN232
HallR3	G0/0	192.168.52.2	/24	-	-
	G0/1	192.168.57.2	/24	-	-
	G0/2.9	192.168.9.1	/24	-	VLAN9
7thSecurityRoomR4	G0/0	192.168.53.2	/24	-	-
	G0/1	192.168.58.2	/24	-	-
	G0/2.10	192.168.10.1	/24	-	VLAN10
	G0/2.15	192.168.15.1	/24	-	VLAN15

	G0/2.16	192.168.16.1	/24	-	VLAN16
	G0/2.17	192.168.17.1	/24	-	VLAN17
	G0/2.63	192.168.63.1	/24	-	VLAN63
MainASA	G1/1	10.1.1.1	/30	-	-
	G1/2	10.2.2.1	/30	-	-
	G1/3	192.168.2.1	/24	-	-
ITR5	G0/0	192.168.54.2	/24	-	-
	G0/1	192.168.59.2	/24	-	-
	G0/2.7	192.168.7.1	/24	-	VLAN7
	G0/2.62	192.168.62.1	/24	-	VLAN62
MainMSW	G1/0/1	192.168.50.1	/24	-	-
	G1/0/2	192.168.51.1	/24	-	-
	G1/0/3	192.168.52.1	/24	-	-
	G1/0/4	192.168.53.1	/24	-	-
	G1/0/5	10.1.1.2	/30	-	-
	G1/0/6	192.168.54.1	/24	-	-
	G1/0/24	192.168.1.2	/24	-	-
MainBackUpMSW	G1/0/1	192.168.55.1	/24	-	-
	G1/0/2	192.168.56.1	/24	-	-
	G1/0/3	192.168.57.1	/24	-	-
	G1/0/4	192.168.58.1	/24	-	-
	G1/0/5	10.2.2.2	/30	-	-
	G1/0/6	192.168.59.1	/24	-	-
	G1/0/24	192.168.3.2	/24	-	-
MainR1	G0/0	192.168.1.1	/24	-	-
	G0/1	192.168.3.1	/24	-	-
	S0/0/0	10.3.3.1	/30	-	-
ISP	G0/0	10.11.11.2	/30	-	-
	S0/0/0	10.3.3.2	/30	-	-
	S0/0/1	10.5.5.2	/30	-	-
ISPASA	G1/1	209.165.100.1	/24	-	-
	G1/2	10.11.11.1	/30	-	-
OutsideR1	G0/0	209.165.100.2	/24	-	-
	G0/1	123.123.123.1	/24	-	-
BranchR1	G0/0	172.16.1.1	/24	-	-
	G0/1	172.16.3.1	/24	-	-

	S0/0/1	10.5.5.1	/30	-	-
BranchMSW	G1/0/1	172.16.50.1	/24	-	-
	G1/0/2	172.16.51.1	/24	-	-
	G1/0/3	172.16.52.1	/24	-	-
	G1/0/4	172.16.53.1	/24	-	-
	G1/0/5	10.9.9.2	/30	-	-
	G1/0/24	172.16.1.2	/24	-	-
BranchBackUpMSW	G1/0/1	172.16.54.1	/24	-	-
	G1/0/2	172.16.55.1	/24	-	-
	G1/0/3	172.16.56.1	/24	-	-
	G1/0/4	172.16.57.1	/24	-	-
	G1/0/5	10.10.10.2	/30	-	-
	G1/0/24	172.16.3.2	/24	-	-
BranchOfficeR1	G0/0	172.16.50.2	/24	-	-
	G0/1	172.16.54.2	/24	-	-
	G0/2.4	172.16.4.1	/24	-	VLAN4
	G0/2.5	172.16.5.1	/24	-	VLAN5
	G0/2.6	172.16.6.1	/24	-	VLAN6
	G0/2.8	172.16.8.1	/24	-	VLAN8
	G0/2.61	172.16.61.1	/24	-	VLAN61
BranchClassR2	G0/0	172.16.51.2	/24	-	-
	G0/1	172.16.55.2	/24	-	-
	G0/2.11	172.16.11.1	/24	-	VLAN11
	G0/2.12	172.16.12.1	/24	-	VLAN12
	G0/2.13	172.16.13.1	/24	-	VLAN13
	G0/2.96	172.16.96.1	/24	-	VLAN96
	G0/2.97	172.16.97.1	/24	-	VLAN97
	G0/2.98	172.16.98.1	/24	-	VLAN98
	G0/2.101	172.16.101.1	/24	-	VLAN101
	G0/2.102	172.16.102.1	/24	-	VLAN102
	G0/2.201	172.16.201.1	/24	-	VLAN201
	G0/2.202	172.16.202.1	/24	-	VLAN202
	G0/2.231	172.16.231.1	/24	-	VLAN231
	G0/2.232	172.16.232.1	/24	-	VLAN232
BranchITR3	G0/0	172.16.52.2	/24	-	-
	G0/1	172.16.56.2	/24	-	-

	G0/2.7	172.16.7.1	/24	-	VLAN7
	G0/2.63	172.16.63.1	/24	-	VLAN63
BranchSecurityRoomR4	G0/0	172.16.53.2	/24	-	-
	G0/1	172.16.57.2	/24	-	-
	G0/2.10	172.16.10.1	/24	-	VLAN10
	G0/2.14	172.16.14.1	/24	-	VLAN14
	G0/2.15	172.16.15.1	/24	-	VLAN15
	G0/2.62	172.16.62.1	/24	-	VLAN62
Admin PC	F0/1	DHCP Assign	/24	192.168.4.1	VLAN4
Admin IP Phone	F0/1	DHCP Assign	/24	192.168.61.1	VLAN61
Admin Laptop	F0/2	DHCP Assign	/24	192.168.4.1	VLAN4
Finance PC	F0/6	DHCP Assign	/24	192.168.5.1	VLAN5
Finance Laptop	F0/7	DHCP Assign	/24	192.168.5.1	VLAN5
Finance IP Phone	F0/6	DHCP Assign	/24	192.168.61.1	VLAN61
Teacher PC	F0/11	DHCP Assign	/24	192.168.6.1	VLAN6
Teacher Laptop	F0/12	DHCP Assign	/24	192.168.6.1	VLAN6
Teacher IP Phone	F0/11	DHCP Assign	/24	192.168.61.1	VLAN61
Guest Laptop	F0/1	DHCP Assign	/24	192.168.8.1	VLAN8
PC101	F0/1	DHCP Assign	/24	192.168.101.1	VLAN101
PC102	F0/6	DHCP Assign	/24	192.168.102.1	VLAN102
2thClassAP	F0/11	DHCP Assign	/24	192.168.98.1	VLAN98
Laptop101-102	Wireless	DHCP Assign	/24	192.168.11.1	VLAN11
PC121	F0/1	DHCP Assign	/24	192.168.121.1	VLAN121
PC122	F0/6	DHCP Assign	/24	192.168.122.1	VLAN122
3thClassAP	F0/11	DHCP Assign	/24	192.168.97.1	VLAN97
Laptop121-122	Wireless	DHCP Assign	/24	192.168.12.1	VLAN12
PC201	F0/1	DHCP Assign	/24	192.168.201.1	VLAN201
PC202	F0/6	DHCP Assign	/24	192.168.202.1	VLAN202
4thClassAP	F0/11	DHCP Assign	/24	192.168.96.1	VLAN96
Laptop201-202	Wireless	DHCP Assign	/24	192.168.13.1	VLAN13
PC231	F0/1	DHCP Assign	/24	192.168.231.1	VLAN231
PC232	F0/6	DHCP Assign	/24	192.168.232.1	VLAN232
5thClassAP	F0/11	DHCP Assign	/24	192.168.96.1	VLAN96
Laptop231-232	Wireless	DHCP Assign	/24	192.168.14.1	VLAN14
HallPC	F0/1	DHCP Assign	/24	192.168.9.1	VLAN9
HallLaptop	F0/2	DHCP Assign	/24	192.168.9.1	VLAN9

IT PC1-2-3	F0/1-3	DHCP Assign	/24	192.168.7.1	VLAN7
IT IP Phone	F0/1	DHCP Assign	/24	192.168.62.1	VLAN62
IT Laptop	Wireless	DHCP Assign	/24	192.168.7.1	VLAN7
Web Server	F0/1	192.168.2.2	/24	192.168.2.1	-
DNS Server	F0/2	192.168.2.3	/24	192.168.2.1	-
DHCP Server	F0/3	192.168.2.4	/24	192.168.2.1	-
FTP Server	F0/4	192.168.2.5	/24	192.168.2.1	-
Syslog Server	F0/5	192.168.2.6	/24	192.168.2.1	-
Email Server	F0/6	192.168.2.7	/24	192.168.2.1	-
AAA Server	F0/7	192.168.2.8	/24	192.168.2.1	-
2thWLC	F0/8	192.168.2.9	/24	192.168.2.1	-
3thWLC	F0/9	192.168.2.10	/24	192.168.2.1	-
4thWLC	F0/10	192.168.2.11	/24	192.168.2.1	-
5thWLC	F0/11	192.168.2.12	/24	192.168.2.1	-
CCTV Server	F0/1	192.168.10.2	/24	192.168.10.1	VLAN10
RFID Server	F0/2	192.168.10.3	/24	192.168.10.1	VLAN10
IoT Server	F0/3	192.168.10.4	/24	192.168.10.1	VLAN10
Security IP Phone	F0/6	DHCP Assign	/24	192.168.63.1	VLAN63
Admin CCTV	F0/1	DHCP Assign	/24	192.168.15.1	VLAN15
Finance CCTV	F0/2	DHCP Assign	/24	192.168.15.1	VLAN15
Teacher CCTV	F0/3	DHCP Assign	/24	192.168.15.1	VLAN15
Guest CCTV	F0/4	DHCP Assign	/24	1921.68.15.1	VLAN15
ITCCTV	F0/5	DHCP Assign	/24	192.168.15.1	VLAN15
2thClassCCTV	F0/6	DHCP Assign	/24	192.168.15.1	VLAN15
3thClassCCTV	F0/7	DHCP Assign	/24	192.168.15.1	VLAN15
4thClassCCTV	F0/8	DHCP Assign	/24	192.168.15.1	VLAN15
5thClassCCTV	F0/9	DHCP Assign	/24	192.168.15.1	VLAN15
6thHallCCTV	F0/10	DHCP Assign	/24	192.168.15.1	VLAN15
SecurityRoomCCTV	F0/11	DHCP Assign	/24	192.168.15.1	VLAN15
ServerRoomCCTV	F0/12	DHCP Assign	/24	192.168.15.1	VLAN15
ServerMotionDetector	F0/13	DHCP Assign	/24	192.168.15.1	VLAN15
ServerSiren	F0/14	DHCP Assign	/24	192.168.15.1	VLAN15
1thFloorDoor	F0/1	DHCP Assign	/24	192.168.16.1	VLAN16
2&3&4&5thFloorDoor	F0/2	DHCP Assign	/24	192.168.16.1	VLAN16
6thFloorDoor	F0/3	DHCP Assign	/24	192.168.16.1	VLAN16
7thITDoor	F0/4	DHCP Assign	/24	192.168.16.1	VLAN16

7thServerDoor	F0/5	DHCP Assign	/24	192.168.16.1	VLAN16
7thCCTVDoor	F0/6	DHCP Assign	/24	192.168.16.1	VLAN16
1thFloorRFID	F0/7	DHCP Assign	/24	192.168.16.1	VLAN16
2&3&4&5thFloorRFID	F0/8	DHCP Assign	/24	192.168.16.1	VLAN16
6thFloorRFID	F0/9	DHCP Assign	/24	192.168.16.1	VLAN16
7thITRFID	F0/10	DHCP Assign	/24	192.168.16.1	VLAN16
7thServerRFID	F0/11	DHCP Assign	/24	192.168.16.1	VLAN16
7thCCTVRFID	F0/12	DHCP Assign	/24	192.168.16.1	VLAN16
Hall Smoke Detector	F0/1	DHCP Assign	/24	192.168.17.1	VLAN17
Hall Blower	F0/2	DHCP Assign	/24	192.168.17.1	VLAN17
Hall Window	F0/3	DHCP Assign	/24	192.168.17.1	VLAN17
Hall Fire Sprinkler	F0/4	DHCP Assign	/24	192.168.17.1	VLAN17
Hall Fire Monitor	F0/5	DHCP Assign	/24	192.168.17.1	VLAN17
PC BAdmin	F0/1	DHCP Assign	/24	172.16.4.1	VLAN4
Laptop BAdmin	F0/2	DHCP Assign	/24	172.16.4.1	VLAN4
BAdmin IP Phone	F0/1	DHCP Assign	/24	172.16.61.1	VLAN61
BFinance PC	F0/6	DHCP Assign	/24	172.16.5.1	VLAN5
BFinance Laptop	F0/7	DHCP Assign	/24	172.16.5.1	VLAN5
BFinance IP Phone	F0/6	DHCP Assign	/24	172.16.61.1	VLAN61
BTeacher PC	F0/11	DHCP Assign	/24	172.16.6.1	VLAN6
BTeacher Laptop	F0/12	DHCP Assign	/24	172.16.6.1	VLAN6
BTeacher IP Phone	F0/11	DHCP Assign	/24	172.16.61.1	VLAN61
PC B101	F0/1	DHCP Assign	/24	172.16.101.1	VLAN101
PC B102	F0/6	DHCP Assign	/24	172.16.102.1	VLAN102
B2thClassAp	F0/11	DHCP Assign	/24	172.16.98.1	VLAN98
Laptop B101-B102	Wireless	DHCP Assign	/24	172.16.11.1	VLAN11
PC B201	F0/1	DHCP Assign	/24	172.16.201.1	VLAN201
PC B202	F0/6	DHCP Assign	/24	172.16.202.1	VLAN202
B3thClassAP	F0/11	DHCP Assign	/24	172.16.97.1	VLAN97
Laptop B201-B202	Wireless	DHCP Assign	/24	172.16.12.1	VLAN12
PC B231	F0/1	DHCP Assign	/24	172.16.231.1	VLAN231
PC B232	F0/2	DHCP Assign	/24	172.16.232.1	VLAN232
B4thClassAP	F0/11	DHCP Assign	/24	172.16.96.1	VLAN96
Laptop B231-232	Wireless	DHCP Assign	/24	172.16.13.1	VLAN13
B ITPC1-2-3	F0/1-5	DHCP Assign	/24	172.16.7.1	VLAN7
B IT IP Phone	F0/1	DHCP Assign	/24	172.16.63.1	VLAN63

B IT Laptop	Wireless	DHCP Assign	/24	172.16.7.1	VLAN7
B Webserver	F0/1	172.16.2.2	/24	172.16.2.1	-
B DNS Server	F0/2	172.16.2.3	/24	172.16.2.1	-
B DHCP Server	F0/3	172.16.2.4	/24	172.16.2.1	-
B FTP Server	F0/4	172.16.2.5	/24	172.16.2.1	-
B Syslog Server	F0/5	172.16.2.6	/24	172.16.2.1	-
B Email Server	F0/6	172.16.2.7	/24	172.16.2.1	-
B AAA Server	F0/7	172.16.2.8	/24	172.16.2.1	-
B2thWLC	F0/8	172.16.2.9	/24	172.16.2.1	-
B3thWLC	F0/9	172.16.2.10	/24	172.16.2.1	-
B4thWLC	F0/10	172.16.2.11	/24	172.16.2.1	-
B CCTV Server	F0/1	172.16.10.2	/24	172.16.10.1	VLAN10
B RFID Server	F0/2	172.16.10.3	/24	172.16.10.1	VLAN10
B Security IP Phone	F0/3	DHCP Assign	/24	172.16.62.1	VLAN62
BAdmin CCTV	F0/1	DHCP Assign	/24	172.16.14.1	VLAN14
BFinance CCTV	F0/2	DHCP Assign	/24	172.16.14.1	VLAN14
BTeacher CCTV	F0/3	DHCP Assign	/24	172.16.14.1	VLAN14
B2thClassCCTV	F0/4	DHCP Assign	/24	172.16.14.1	VLAN14
B3thClassCCTV	F0/5	DHCP Assign	/24	172.16.14.1	VLAN14
B4thClassCCTV	F0/6	DHCP Assign	/24	172.16.14.1	VLAN14
BSecurity CCTV	F0/7	DHCP Assign	/24	172.16.14.1	VLAN14
BIT CCTV	F0/8	DHCP Assign	/24	172.16.14.1	VLAN14
BServer CCTV	F0/9	DHCP Assign	/24	172.16.14.1	VLAN14
BServer Motion	F0/10	DHCP Assign	/24	172.16.14.1	VLAN14
BServer Siren	F0/11	DHCP Assign	/24	172.16.14.1	VLAN14
BGuest CCTV	F0/12	DHCP Assign	/24	172.16.14.1	VLAN14
B1thFloorDoor	F0/1	DHCP Assign	/24	172.16.151	VLAN15
B2&3&4thClassDoor	F0/2	DHCP Assign	/24	172.16.151	VLAN15
B5thSecurityDoor	F0/3	DHCP Assign	/24	172.16.151	VLAN15
B5thServerDoor	F0/4	DHCP Assign	/24	172.16.151	VLAN15
B5thITDoor	F0/5	DHCP Assign	/24	172.16.151	VLAN15
B1thFloorRFID	F0/6	DHCP Assign	/24	172.16.151	VLAN15
B2&3&4thClassRFID	F0/7	DHCP Assign	/24	172.16.151	VLAN15
B5thSecurityRFID	F0/8	DHCP Assign	/24	172.16.151	VLAN15
5thServerRFID	F0/9	DHCP Assign	/24	172.16.151	VLAN15
5thITRFID	F0/10	DHCP Assign	/24	172.16.151	VLAN15

4.5 Configuration

<table>
<tr><th>Device</th><th>Configure</th><th>Command</th></tr>
<tr><td rowspan="2">OfficeR1</td><td>IPv4 , sub interface ,and helper address</td><td>interface GigabitEthernet0/0
 ip address 192.168.50.2 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.55.2 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/2.4
 encapsulation dot1Q 4
 ip address 192.168.4.1 255.255.255.0
 ip helper-address 192.168.2.4
 ip access-group icmp_office out
!
interface GigabitEthernet0/2.5
 encapsulation dot1Q 5
 ip address 192.168.5.1 255.255.255.0
 ip helper-address 192.168.2.4
 ip access-group icmp_office out
!
interface GigabitEthernet0/2.6
 encapsulation dot1Q 6
 ip address 192.168.6.1 255.255.255.0
 ip helper-address 192.168.2.4
 ip access-group icmp_office out
!
interface GigabitEthernet0/2.8
 encapsulation dot1Q 8
 ip address 192.168.8.1 255.255.255.0
 ip helper-address 192.168.2.4
 ip access-group guest_acl in
 ip access-group icmp_office out
!
interface GigabitEthernet0/2.61
 encapsulation dot1Q 61
 ip address 192.168.61.1 255.255.255.0
!</td></tr>
<tr><td>Dynamic Route</td><td>router eigrp 1
 network 192.168.4.0
 network 192.168.5.0
 network 192.168.6.0
 network 192.168.61.0
 network 192.168.50.0
 network 192.168.8.0
 network 192.168.55.0
!</td></tr>
</table>

	DHCP Pool	`ip dhcp excluded-address 192.168.61.1` `!` `ip dhcp pool VoIP` ` network 192.168.61.0 255.255.255.0` ` default-router 192.168.61.1` ` option 150 ip 192.168.61.1` `!`
	ACL	`ip access-list extended guest_acl` ` deny tcp any any eq ftp` ` permit ip any any` `ip access-list extended icmp_office` ` permit icmp any 192.168.4.0 0.0.0.255 echo-reply` ` permit icmp any 192.168.5.0 0.0.0.255 echo-reply` ` permit icmp any 192.168.6.0 0.0.0.255 echo-reply` ` deny icmp any 192.168.8.0 0.0.0.255 echo-reply` ` permit ip any any` `access-list 22 permit 192.168.7.0 0.0.0.255` `access-list 22 deny any` `!`
	Telephony-service	`telephony-service` ` max-ephones 3` ` max-dn 3` ` ip source-address 192.168.61.1 port 2000` ` auto assign 1 to 3` `!` `ephone-dn 1` ` number 1001` `!` `ephone-dn 2` ` number 1002` `!` `ephone-dn 3` ` number 1003` `!` `ephone 1` ` device-security-mode none` ` mac-address 000C.CFB4.1147` ` type 7960` ` button 1:3` `!` `ephone 2` ` device-security-mode none` ` mac-address 0001.96D7.744B` ` type 7960` ` button 1:1` `!` `ephone 3` ` device-security-mode none` ` mac-address 0000.0C75.5246` ` type 7960` ` button 1:2`

	Dial-peer	`dial-peer voice 1 voip` ` destination-pattern 200.` ` session target ipv4:192.168.62.1` `!` `dial-peer voice 2 voip` ` destination-pattern 300.` ` session target ipv4:192.168.63.1` `!` `dial-peer voice 4 voip` ` destination-pattern 200.` ` session target ipv4:192.168.59.2` `!` `dial-peer voice 5 voip` ` destination-pattern 300.` ` session target ipv4:192.168.58.2` `!`
	Syslog	`logging 192.168.2.6`
	SSH	`line vty 0 4` ` access-class 22 in` ` login local` ` transport input ssh` `line vty 5 15` ` access-class 22 in` ` login local` ` transport input ssh` `!`

Device	Configure	Command
ClassR2	IPv4 , sub interface , and helper address	! interface GigabitEthernet0/0 ip address 192.168.51.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/1 ip address 192.168.56.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/2 no ip address duplex auto speed auto ! interface GigabitEthernet0/2.11 encapsulation dot1Q 11 ip address 192.168.11.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.12 encapsulation dot1Q 12 ip address 192.168.12.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.13 encapsulation dot1Q 13 ip address 192.168.13.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.14 encapsulation dot1Q 14 ip address 192.168.14.1 255.255.255.0

		`` interface GigabitEthernet0/2.95 encapsulation dot1Q 95 ip address 192.168.95.1 255.255.255.0 ip helper-address 192.168.2.4 ! interface GigabitEthernet0/2.96 encapsulation dot1Q 96 ip address 192.168.96.1 255.255.255.0 ip helper-address 192.168.2.4 ! interface GigabitEthernet0/2.97 encapsulation dot1Q 97 ip address 192.168.97.1 255.255.255.0 ip helper-address 192.168.2.4 ! interface GigabitEthernet0/2.98 encapsulation dot1Q 98 ip address 192.168.98.1 255.255.255.0 ip helper-address 192.168.2.4 ! interface GigabitEthernet0/2.101 encapsulation dot1Q 101 ip address 192.168.101.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.102 encapsulation dot1Q 102 ip address 192.168.102.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.121 encapsulation dot1Q 121 ip address 192.168.121.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ```

		interface GigabitEthernet0/2.122 encapsulation dot1Q 122 ip address 192.168.122.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.201 encapsulation dot1Q 201 ip address 192.168.201.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.202 encapsulation dot1Q 202 ip address 192.168.202.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.231 encapsulation dot1Q 231 ip address 192.168.231.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in ! interface GigabitEthernet0/2.232 encapsulation dot1Q 232 ip address 192.168.232.1 255.255.255.0 ip helper-address 192.168.2.4 ip access-group Class-ACL in
	Dynamic Route	router eigrp 1 network 192.168.101.0 network 192.168.102.0 network 192.168.121.0 network 192.168.122.0 network 192.168.201.0 network 192.168.202.0 network 192.168.231.0 network 192.168.232.0 network 192.168.51.0 network 192.168.56.0 network 192.168.11.0 network 192.168.12.0 network 192.168.13.0 network 192.168.14.0 network 192.168.95.0 network 192.168.96.0 network 192.168.97.0 network 192.168.98.0
	ACL	ip access-list extended Class-ACL deny tcp any any eq ftp permit ip any any access-list 22 permit 192.168.7.0 0.0.0.255 access-list 22 deny any

	SSH	`line vty 0 4` ` access-class 22 in` ` login local` ` transport input ssh` `line vty 5 15` ` access-class 22 in` ` login local` ` transport input ssh`
	Syslog	`logging 192.168.2.6`

Device	Configure	Command
HallR3	IPv4 ,sub interface, and helper address	`interface GigabitEthernet0/0` ` ip address 192.168.52.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/1` ` ip address 192.168.57.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2` ` no ip address` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2.9` ` encapsulation dot1Q 9` ` ip address 192.168.9.1 255.255.255.0` ` ip helper-address 192.168.2.4` ` ip access-group Hall-ACL in` `!`
	Dynamic Route	`router eigrp 1` ` network 192.168.9.0` ` network 192.168.52.0` ` network 192.168.57.0`
	ACL	`access-list 22 permit 192.168.7.0 0.0.0.255` `access-list 22 deny any` `ip access-list extended Hall-ACL` ` deny tcp any any eq ftp` ` permit ip any any`
	SSH	`line vty 0 4` ` access-class 22 in` ` login local` ` transport input ssh` `line vty 5 15` ` access-class 22 in` ` login local` ` transport input ssh` `!`
	Syslog	`logging 192.168.2.6`

Device	Configure	Command
SecurityR4	IPv4 ,sub interface, and helper address	`interface GigabitEthernet0/0` ` ip address 192.168.53.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/1` ` ip address 192.168.58.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2` ` no ip address` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2.10` ` encapsulation dot1Q 10` ` ip address 192.168.10.1 255.255.255.0` ` ip access-group Security_ACL in` `!` `interface GigabitEthernet0/2.15` ` encapsulation dot1Q 15` ` ip address 192.168.15.1 255.255.255.0` ` ip helper-address 192.168.2.4` `!` `interface GigabitEthernet0/2.16` ` encapsulation dot1Q 16` ` ip address 192.168.16.1 255.255.255.0` ` ip helper-address 192.168.2.4` `!` `interface GigabitEthernet0/2.17` ` encapsulation dot1Q 17` ` ip address 192.168.17.1 255.255.255.0` ` ip helper-address 192.168.2.4` `!` `interface GigabitEthernet0/2.63` ` encapsulation dot1Q 63` ` ip address 192.168.63.1 255.255.255.0`
	Dynamic Route	`router eigrp 1` ` network 192.168.53.0` ` network 192.168.58.0` ` network 192.168.10.0` ` network 192.168.15.0` ` network 192.168.16.0` ` network 192.168.17.0` ` network 192.168.63.0`
	DHCP Pool	`ip dhcp excluded-address 192.168.63.1` `!` `ip dhcp pool SecurityVoice` ` network 192.168.63.0 255.255.255.0` ` default-router 192.168.63.1` ` option 150 ip 192.168.63.1`
	ACL	`ip access-list extended Security_ACL` ` deny tcp any any eq ftp` ` permit ip any any` `access-list 22 permit 192.168.7.0 0.0.0.255` `access-list 22 deny any`

	Telephony-service	telephony-service max-ephones 1 max-dn 1 ip source-address 192.168.63.1 port 2000 auto assign 1 to 1 ! ephone-dn 1 number 3001 ! ephone 1 device-security-mode none mac-address 0004.9A7A.1043 type 7960 button 1:1 !
	Dial-peer	dial-peer voice 2 voip destination-pattern 100. session target ipv4:192.168.61.1 ! dial-peer voice 3 voip destination-pattern 200. session target ipv4:192.168.62.1 ! dial-peer voice 5 voip destination-pattern 100. session target ipv4:192.168.55.2 ! dial-peer voice 6 voip destination-pattern 200. session target ipv4:192.168.59.2 !
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh !
	Syslog	logging 192.168.2.6

Device	Configure	Command
ITR5	IPv4 ,sub interface, and Helper address	interface GigabitEthernet0/0 ip address 192.168.54.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/1 ip address 192.168.59.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/2 no ip address duplex auto speed auto ! interface GigabitEthernet0/2.7 encapsulation dot1Q 7 ip address 192.168.7.1 255.255.255.0 ip helper-address 192.168.2.4 ! interface GigabitEthernet0/2.62 encapsulation dot1Q 62 ip address 192.168.62.1 255.255.255.0 !
	Dynamic Route	router eigrp 1 network 192.168.7.0 network 192.168.62.0 network 192.168.54.0 network 192.168.59.0 !
	DHCP Pool	ip dhcp excluded-address 192.168.62.1 ! ip dhcp pool ITVoIP network 192.168.62.0 255.255.255.0 default-router 192.168.62.1 option 150 ip 192.168.62.1 !
	Telephony-service	telephony-service max-ephones 1 max-dn 1 ip source-address 192.168.62.1 port 2000 auto assign 1 to 1 ! ephone-dn 1 number 2001 ! ephone 1 device-security-mode none mac-address 0003.E488.A17A type 7960 button 1:1 !

	Dial-peer	dial-peer voice 1 voip destination-pattern 100. session target ipv4:192.168.61.1 ! dial-peer voice 3 voip destination-pattern 300. session target ipv4:192.168.63.1 ! dial-peer voice 4 voip destination-pattern 100. session target ipv4:192.168.55.2 ! dial-peer voice 6 voip destination-pattern 300. session target ipv4:192.168.58.2
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 192.168.2.6

Device	Configure	Command
MainASA	IPv4 ,security level, and nameif	interface GigabitEthernet1/1 nameif MainCampus security-level 100 ip address 10.1.1.1 255.255.255.252 ! interface GigabitEthernet1/2 nameif MainCampus2 security-level 100 ip address 10.2.2.1 255.255.255.252 ! interface GigabitEthernet1/3 nameif Server security-level 100 ip address 192.168.2.1 255.255.255.0 !
	Dynamic Route	router eigrp 1 network 10.1.1.0 0.0.0.255 network 10.2.2.0 0.0.0.255 network 192.168.2.0 !
	SSH, Class map , Policy-map	aaa authentication ssh console LOCAL ! username asa password eU1NEhFdUPn4NplY encrypted ! class-map inspection_default match default-inspection-traffic ! policy-map type inspect dns preset_dns_map parameters message-length maximum 512 policy-map global_policy class inspection_default inspect dns preset_dns_map inspect ftp inspect tftp ! service-policy global_policy global ! telnet timeout 5 ssh 192.168.7.0 255.255.255.0 MainCampus ssh 192.168.7.0 255.255.255.0 MainCampus2 ssh timeout 5
	ACL	access-list main extended permit icmp any host 192.168.2.2 access-list main extended permit tcp any host 192.168.2.2 eq www access-list main extended permit tcp any host 192.168.2.3 eq domain access-list main extended permit ip any any ! ! access-group main in interface MainCampus access-group main in interface MainCampus2

<table>
<tr><th>Device</th><th>Configure</th><th>Command</th></tr>
<tr><td>MainMSW and MainBackUpMSW</td><td>IPv4</td><td>MainMSW
<pre>
interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.50.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/2
 no switchport
 ip address 192.168.51.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/3
 no switchport
 ip address 192.168.52.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/4
 no switchport
 ip address 192.168.53.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/5
 no switchport
 ip address 10.1.1.2 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/6
 no switchport
 ip address 192.168.54.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/24
 no switchport
 ip address 192.168.1.2 255.255.255.0
 duplex auto
 speed auto
!
</pre>MainBackUpMSW</td></tr>
</table>

		interface GigabitEthernet1/0/1 no switchport ip address 192.168.55.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/2 no switchport ip address 192.168.56.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/3 no switchport ip address 192.168.57.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/4 no switchport ip address 192.168.58.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/5 no switchport ip address 10.2.2.2 255.255.255.252 duplex auto speed auto ! interface GigabitEthernet1/0/6 no switchport ip address 192.168.59.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/24 no switchport ip address 192.168.3.2 255.255.255.0 duplex auto speed auto
	Dynamic Route	MainMSW router eigrp 1 network 192.168.50.0 network 192.168.51.0 network 192.168.52.0 network 192.168.53.0 network 192.168.54.0 network 10.1.1.0 0.0.0.255 network 192.168.1.0 auto-summary MainBackUpMSW router eigrp 1 network 192.168.55.0 network 192.168.56.0 network 192.168.57.0 network 192.168.58.0 network 192.168.59.0 network 10.2.2.0 0.0.0.255 network 192.168.3.0 auto-summary

	ACL	`access-list 22 permit 192.168.7.0 0.0.0.255` `access-list 22 deny any`
	SSH	`line vty 0 4` ` access-class 22 in` ` login local` ` transport input ssh` `line vty 5 15` ` access-class 22 in` ` login local` ` transport input ssh`
	Syslog	`logging 192.168.2.6`

Device	Configure	Command
MainR 1	IPv4 and Crypto map	interface GigabitEthernet0/0 ip address 192.168.1.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/1 ip address 192.168.3.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/2 no ip address duplex auto speed auto shutdown ! interface Serial0/0/0 ip address 10.3.3.1 255.255.255.252 crypto map VPN-MAP
	Default Route	router eigrp 1 network 192.168.1.0 network 192.168.3.0 network 10.3.3.0 0.0.0.3 network 10.4.4.0 0.0.0.3 !
	VPN	crypto isakmp policy 10 encr aes 128 authentication pre-share group 5 ! crypto isakmp key vpn address 10.5.5.1 ! ! ! crypto ipsec transform-set VPN-P2 esp-aes 128 esp-sha-hmac ! crypto map VPN-MAP 10 ipsec-isakmp description VPN connect to BranchR1 set peer 10.5.5.1 set pfs group5 set transform-set VPN-P2 match address 100
	ACL	access-list 22 permit 192.168.7.0 0.0.0.255 access-list 22 deny any access-list 100 permit ip 192.168.0.0 0.0.255.255 172.16.0.0 0.0.255.255
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 192.168.2.6

Device	Configure	Command
ISP	IPv4 and NAT inside Outside	interface GigabitEthernet0/0 ip address 10.11.11.2 255.255.255.252 ip nat outside duplex auto speed auto ! interface GigabitEthernet0/1 no ip address duplex auto speed auto shutdown ! interface GigabitEthernet0/2 no ip address duplex auto speed auto shutdown ! interface Serial0/0/0 ip address 10.3.3.2 255.255.255.252 ip nat inside clock rate 2000000 ! interface Serial0/0/1 ip address 10.5.5.2 255.255.255.252 ip nat inside
	Dynamic Route and redistribute static	router eigrp 1 redistribute static network 10.3.3.0 0.0.0.3 network 10.5.5.0 0.0.0.3
	Default Route	ip route 0.0.0.0 0.0.0.0 10.11.11.1
	NAT	ip nat inside source list 1 interface Serial0/0/0 overload ip nat inside source list 2 interface Serial0/0/1 overload ip nat inside source static 192.168.2.2 10.11.11.4 ip nat inside source static 172.16.2.2 10.11.11.5 ip nat inside source static 192.168.2.3 10.11.11.6 ip nat inside source static 172.16.2.3 10.11.11.7
	ACL	access-list 1 permit 192.168.0.0 0.0.255.255 access-list 2 permit 172.16.0.0 0.0.255.255

<table>
<tr><th>Device</th><th>Configure</th><th>Command</th></tr>
<tr><td rowspan="4">ISPASA</td><td>IPv4, Security level, and nameif</td><td>interface GigabitEthernet1/1
nameif outside
security-level 0
ip address 209.165.100.1 255.255.255.0
!
interface GigabitEthernet1/2
nameif inside
security-level 100
ip address 10.11.11.1 255.255.255.252</td></tr>
<tr><td>Static Route and Default Route</td><td>route outside 0.0.0.0 0.0.0.0 209.165.100.2 1
route inside 10.0.0.0 255.0.0.0 10.11.11.2 1</td></tr>
<tr><td>Object network , and NAT</td><td>object network Branch
host 10.5.5.2
nat (inside,outside) static 209.165.100.8
object network Branch-DNS
host 10.11.11.7
nat (inside,outside) static 209.165.100.6
object network Branch-server
host 10.11.11.5
nat (inside,outside) static 209.165.100.4
object network Main
host 10.3.3.2
nat (inside,outside) static 209.165.100.7
object network Main-DNS
host 10.11.11.6
nat (inside,outside) static 209.165.100.5
object network Main-server
host 10.11.11.4
nat (inside,outside) static 209.165.100.3</td></tr>
<tr><td>Access-list and Access-group</td><td>access-list outside extended permit icmp any host 10.11.11.4
access-list outside extended permit tcp any host 10.11.11.4 eq www
access-list outside extended permit icmp any host 10.11.11.5
access-list outside extended permit tcp any host 10.11.11.5 eq www
access-list outside extended permit tcp any host 10.11.11.6 eq domain
access-list outside extended permit tcp any host 10.11.11.7 eq domain
access-list outside extended permit ip any any
!
!
access-group outside in interface outside</td></tr>
</table>

<table>
<tr><th>Device</th><th>Configure</th><th>Command</th></tr>
<tr><td rowspan="2">Outside Router</td><td>IPv4</td><td>interface GigabitEthernet0/0
ip address 209.165.100.2 255.255.255.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 123.123.123.1 255.255.255.0
duplex auto
speed auto</td></tr>
<tr><td>Static route</td><td>ip route 209.165.100.0 255.255.255.0 209.165.100.1</td></tr>
</table>

Device	Configure	Command
BranchR1	IPv4 , and crypto map	interface GigabitEthernet0/0 ip address 172.16.1.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/1 ip address 172.16.3.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/2 no ip address duplex auto speed auto shutdown ! interface Serial0/0/0 no ip address clock rate 2000000 shutdown ! interface Serial0/0/1 ip address 10.5.5.1 255.255.255.252 clock rate 2000000 crypto map VPN-MAP
	Dynamic Route	router eigrp 1 network 172.16.1.0 0.0.0.255 network 172.16.3.0 0.0.0.255 network 10.5.5.0 0.0.0.3 network 10.6.6.0 0.0.0.3
	VPN	crypto isakmp policy 10 encr aes 128 authentication pre-share group 5 ! crypto isakmp key vpn address 10.3.3.1 ! ! ! crypto ipsec transform-set VPN-P2 esp-aes 128 esp-sha-hmac ! crypto map VPN-MAP 10 ipsec-isakmp description VPN connect to MainR1 set peer 10.3.3.1 set pfs group5 set transform-set VPN-P2 match address 100
	ACL	access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any access-list 100 permit ip 172.16.0.0 0.0.255.255 192.168.0.0 0.0.255.255
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 172.16.2.6

Device	Configure	Command
BrancMSW and BranchBackUpMSW	IPv4	BranchMSW interface GigabitEthernet1/0/1 no switchport ip address 172.16.50.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/2 no switchport ip address 172.16.51.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/3 no switchport ip address 172.16.52.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/4 no switchport ip address 172.16.53.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/5 no switchport ip address 10.9.9.2 255.255.255.252 duplex auto speed auto ! interface GigabitEthernet1/0/24 no switchport ip address 172.16.1.2 255.255.255.0 duplex auto speed auto ! BranchBackUpMSW

		interface GigabitEthernet1/0/1 no switchport ip address 172.16.54.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/2 no switchport ip address 172.16.55.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/3 no switchport ip address 172.16.56.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/4 no switchport ip address 172.16.57.1 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet1/0/5 no switchport ip address 10.10.10.2 255.255.255.252 duplex auto speed auto interface GigabitEthernet1/0/24 no switchport ip address 172.16.3.2 255.255.255.0 duplex auto speed auto !
	Dynamic Route	BranchMSW router eigrp 1 network 172.16.50.0 0.0.0.255 network 172.16.51.0 0.0.0.255 network 172.16.52.0 0.0.0.255 network 172.16.53.0 0.0.0.255 network 10.9.9.0 0.0.0.255 network 172.16.1.0 0.0.0.255 auto-summary BranchBackUpMSW router eigrp 1 network 172.16.54.0 0.0.0.255 network 172.16.55.0 0.0.0.255 network 172.16.56.0 0.0.0.255 network 172.16.57.0 0.0.0.255 network 10.10.10.0 0.0.0.255 network 172.16.3.0 0.0.0.255 auto-summary
	ACL	access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any

	SSH	`line vty 0 4` ` access-class 22 in` ` login local` ` transport input ssh` `line vty 5 15` ` access-class 22 in` ` login local` ` transport input ssh`
	Syslog	`logging 172.16.2.6`

Device	Configure	Command
BOfficeR1	IPv4, Sub interface, access-group, and helper address	`interface GigabitEthernet0/0` ` ip address 172.16.50.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/1` ` ip address 172.16.54.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2` ` no ip address` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2.4` ` encapsulation dot1Q 4` ` ip address 172.16.4.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Branch_Office_ACL out` `!` `interface GigabitEthernet0/2.5` ` encapsulation dot1Q 5` ` ip address 172.16.5.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Branch_Office_ACL out` `!` `interface GigabitEthernet0/2.6` ` encapsulation dot1Q 6` ` ip address 172.16.6.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Branch_Office_ACL out` `!` `interface GigabitEthernet0/2.8` ` encapsulation dot1Q 8` ` ip address 172.16.8.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Branch_guest_ACL in` ` ip access-group Branch_Office_ACL out` `!` `interface GigabitEthernet0/2.61` ` encapsulation dot1Q 61` ` ip address 172.16.61.1 255.255.255.0` `!`
	Dynamic Route	`router eigrp 1` ` network 172.16.4.0 0.0.0.255` ` network 172.16.5.0 0.0.0.255` ` network 172.16.6.0 0.0.0.255` ` network 172.16.8.0 0.0.0.255` ` network 172.16.50.0 0.0.0.255` ` network 172.16.54.0 0.0.0.255` ` network 172.16.61.0 0.0.0.255` `!`
	DHCP Pool and excluded address	`ip dhcp excluded-address 172.16.61.1` `!` `ip dhcp pool OfficeVoIP` ` network 172.16.61.0 255.255.255.0` ` default-router 172.16.61.1` ` option 150 ip 172.16.61.1`

	ACL	ip access-list extended Branch_Office_ACL permit icmp any 172.16.4.0 0.0.0.255 echo-reply permit icmp any 172.16.5.0 0.0.0.255 echo-reply permit icmp any 172.16.6.0 0.0.0.255 echo-reply deny icmp any 172.16.8.0 0.0.0.255 echo-reply permit ip any any ip access-list extended Branch_guest_ACL deny tcp any any eq ftp permit ip any any access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any
	Telephony-service	telephony-service max-ephones 3 max-dn 3 ip source-address 172.16.61.1 port 2000 auto assign 1 to 3 ! ephone-dn 1 number 4001 ! ephone-dn 2 number 4002 ! ephone-dn 3 number 4003 !
	Dial-peer	dial-peer voice 1 voip destination-pattern 500. session target ipv4:172.16.63.1 ! dial-peer voice 2 voip destination-pattern 600. session target ipv4:172.16.62.1 ! dial-peer voice 4 voip destination-pattern 500. session target ipv4:172.16.56.2 ! dial-peer voice 5 voip destination-pattern 600. session target ipv4:172.16.57.2 !
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 172.16.2.6

Device	Configure	Command
BClassR2	IPv4, Sub Interface, access-group, and helper address.	`interface GigabitEthernet0/0` ` ip address 172.16.51.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/1` ` ip address 172.16.55.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2` ` no ip address` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2.11` ` encapsulation dot1Q 11` ` ip address 172.16.11.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Class_ACL in` `!` `interface GigabitEthernet0/2.12` ` encapsulation dot1Q 12` ` ip address 172.16.12.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Class_ACL in` `!` `interface GigabitEthernet0/2.13` ` encapsulation dot1Q 13` ` ip address 172.16.13.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Class_ACL in` `!` `interface GigabitEthernet0/2.96` ` encapsulation dot1Q 96` ` ip address 172.16.96.1 255.255.255.0` ` ip helper-address 172.16.2.4` `!` `interface GigabitEthernet0/2.97` ` encapsulation dot1Q 97` ` ip address 172.16.97.1 255.255.255.0` ` ip helper-address 172.16.2.4` `!` `interface GigabitEthernet0/2.98` ` encapsulation dot1Q 98` ` ip address 172.16.98.1 255.255.255.0` ` ip helper-address 172.16.2.4` `!` `interface GigabitEthernet0/2.101` ` encapsulation dot1Q 101` ` ip address 172.16.101.1 255.255.255.0` ` ip helper-address 172.16.2.4` ` ip access-group Class_ACL in` `!`

		interface GigabitEthernet0/2.102 encapsulation dot1Q 102 ip address 172.16.102.1 255.255.255.0 ip helper-address 172.16.2.4 ip access-group Class_ACL in ! interface GigabitEthernet0/2.201 encapsulation dot1Q 201 ip address 172.16.201.1 255.255.255.0 ip helper-address 172.16.2.4 ip access-group Class_ACL in ! interface GigabitEthernet0/2.202 encapsulation dot1Q 202 ip address 172.16.202.1 255.255.255.0 ip helper-address 172.16.2.4 ip access-group Class_ACL in ! interface GigabitEthernet0/2.231 encapsulation dot1Q 231 ip address 172.16.231.1 255.255.255.0 ip helper-address 172.16.2.4 ip access-group Class_ACL in ! interface GigabitEthernet0/2.232 encapsulation dot1Q 232 ip address 172.16.232.1 255.255.255.0 ip helper-address 172.16.2.4 ip access-group Class_ACL in
	Dynamic Route	router eigrp 1 network 172.16.11.0 0.0.0.255 network 172.16.12.0 0.0.0.255 network 172.16.13.0 0.0.0.255 network 172.16.96.0 0.0.0.255 network 172.16.97.0 0.0.0.255 network 172.16.98.0 0.0.0.255 network 172.16.101.0 0.0.0.255 network 172.16.102.0 0.0.0.255 network 172.16.202.0 0.0.0.255 network 172.16.201.0 0.0.0.255 network 172.16.231.0 0.0.0.255 network 172.16.232.0 0.0.0.255 network 172.16.51.0 0.0.0.255 network 172.16.55.0 0.0.0.255
	ACL	ip access-list extended Class_ACL deny tcp any any eq ftp permit ip any any access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 172.16.2.6

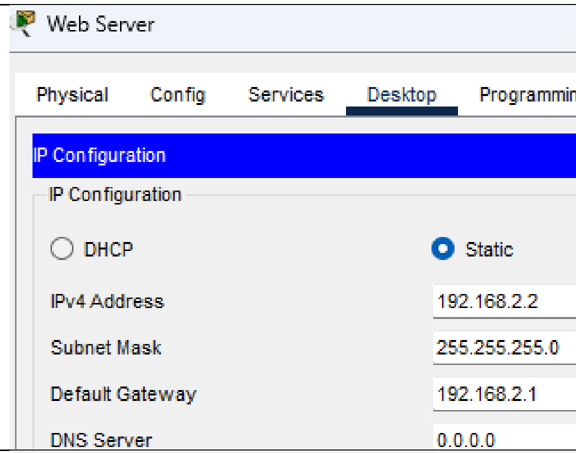
Device	Configure	Command
BranchITR3	IPv4,sub interface, and helper address.	`interface GigabitEthernet0/0` ` ip address 172.16.52.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/1` ` ip address 172.16.56.2 255.255.255.0` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2` ` no ip address` ` duplex auto` ` speed auto` `!` `interface GigabitEthernet0/2.7` ` encapsulation dot1Q 7` ` ip address 172.16.7.1 255.255.255.0` ` ip helper-address 172.16.2.4` `!` `interface GigabitEthernet0/2.63` ` encapsulation dot1Q 63` ` ip address 172.16.63.1 255.255.255.0`
	Dynamic Route	`router eigrp 1` ` network 172.16.7.0 0.0.0.255` ` network 172.16.52.0 0.0.0.255` ` network 172.16.56.0 0.0.0.255` ` network 172.16.63.0 0.0.0.255`
	DHCP Pool and excluded address	`ip dhcp excluded-address 172.16.61.1` `!` `ip dhcp pool ITVoIP` ` network 172.16.63.0 255.255.255.0` ` default-router 172.16.63.1` ` option 150 ip 172.16.63.1`
	Telephony-service	`telephony-service` ` max-ephones 1` ` max-dn 1` ` ip source-address 172.16.63.1 port 2000` ` auto assign 1 to 1` `!` `ephone-dn 1` ` number 5001`
	Dial-peer	`dial-peer voice 1 voip` ` destination-pattern 400.` ` session target ipv4:172.16.61.1` `!` `dial-peer voice 3 voip` ` destination-pattern 600.` ` session target ipv4:172.16.62.1` `!` `dial-peer voice 4 voip` ` destination-pattern 400.` ` session target ipv4:172.16.54.2` `!` `dial-peer voice 6 voip` ` destination-pattern 600.` ` session target ipv4:172.16.57.2`

	ACL	access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 172.16.2.6

Device	Configure	Command
BranchSecurityRoomR4	IPv4, Sub interface, access-group, and helper address	interface GigabitEthernet0/0 ip address 172.16.53.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/1 ip address 172.16.57.2 255.255.255.0 duplex auto speed auto ! interface GigabitEthernet0/2 no ip address duplex auto speed auto ! interface GigabitEthernet0/2.10 encapsulation dot1Q 10 ip address 172.16.10.1 255.255.255.0 ip access-group Branch_Security_ACL in ! interface GigabitEthernet0/2.14 encapsulation dot1Q 14 ip address 172.16.14.1 255.255.255.0 ip helper-address 172.16.2.4 ! interface GigabitEthernet0/2.15 encapsulation dot1Q 15 ip address 172.16.15.1 255.255.255.0 ip helper-address 172.16.2.4 ! interface GigabitEthernet0/2.62 encapsulation dot1Q 62 ip address 172.16.62.1 255.255.255.0
	Dynamic Route	router eigrp 1 network 172.16.10.0 0.0.0.255 network 172.16.14.0 0.0.0.255 network 172.16.15.0 0.0.0.255 network 172.16.62.0 0.0.0.255 network 172.16.53.0 0.0.0.255 network 172.16.57.0 0.0.0.255

	DHCP Pool and excluded address	ip dhcp excluded-address 172.16.62.1 ! ip dhcp pool CCTVRoomVoIP network 172.16.62.0 255.255.255.0 default-router 172.16.62.1 option 150 ip 172.16.62.1
	ACL	ip access-list extended Branch_Security_ACL deny tcp any any eq ftp permit ip any any access-list 22 permit 172.16.7.0 0.0.0.255 access-list 22 deny any
	Telephony-service	telephony-service max-ephones 1 max-dn 1 ip source-address 172.16.62.1 port 2000 auto assign 1 to 1 ! ephone-dn 1 number 6001
	Dial-peer	dial-peer voice 2 voip destination-pattern 400. session target ipv4:172.16.61.1 ! dial-peer voice 3 voip destination-pattern 500. session target ipv4:172.16.63.1 ! dial-peer voice 5 voip destination-pattern 400. session target ipv4:172.16.54.2 ! dial-peer voice 6 voip destination-pattern 500. session target ipv4:172.16.56.2
	SSH	line vty 0 4 access-class 22 in login local transport input ssh line vty 5 15 access-class 22 in login local transport input ssh
	Syslog	logging 172.16.2.6

Device	Configure	Command
BranchASA	IPv4 , security level, and nameif	interface GigabitEthernet1/1 nameif BranchCampus security-level 100 ip address 10.9.9.1 255.255.255.252 ! interface GigabitEthernet1/2 nameif BranchCampus2 security-level 100 ip address 10.10.10.1 255.255.255.252 ! interface GigabitEthernet1/3 nameif BranchServer security-level 100 ip address 172.16.2.1 255.255.255.0
	Dynamic Route	router eigrp 1 network 172.16.2.0 0.0.0.255 network 10.9.9.0 0.0.0.255 network 10.10.10.0 0.0.0.255
	ACL and access-group	access-list branch extended permit icmp any host 172.16.2.2 access-list branch extended permit tcp any host 172.16.2.2 eq www access-list branch extended permit tcp any host 172.16.2.3 eq domain access-list branch extended permit ip any any ! ! access-group branch in interface BranchCampus access-group branch in interface BranchCampus2
	SSH, class-map, and policy map	aaa authentication ssh console LOCAL ! username branchasa password eU1NEhFdUPn4NplY encrypted ! class-map inspection_default match default-inspection-traffic ! policy-map type inspect dns preset_dns_map parameters message-length maximum 512 policy-map global_policy class inspection_default inspect dns preset_dns_map inspect ftp inspect tftp ! service-policy global_policy global ! telnet timeout 5 ssh 172.16.7.0 255.255.255.0 BranchCampus ssh 172.16.7.0 255.255.255.0 BranchCampus2 ssh timeout 5

Device	Configure	Command
Web Server	IPv4	

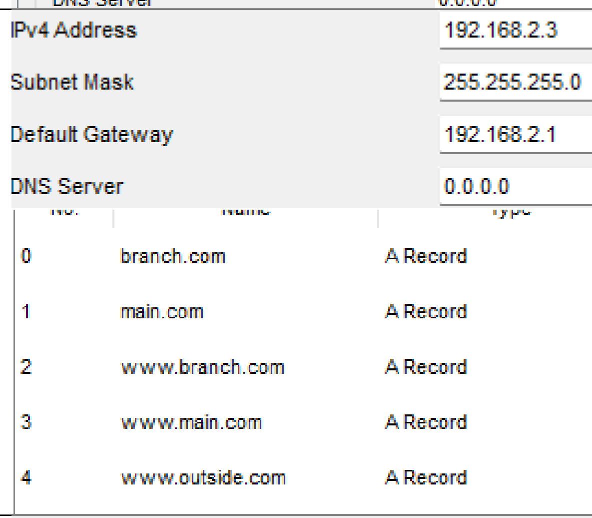
Web Server

Physical | Config | Services | Desktop | Programmin

IP Configuration

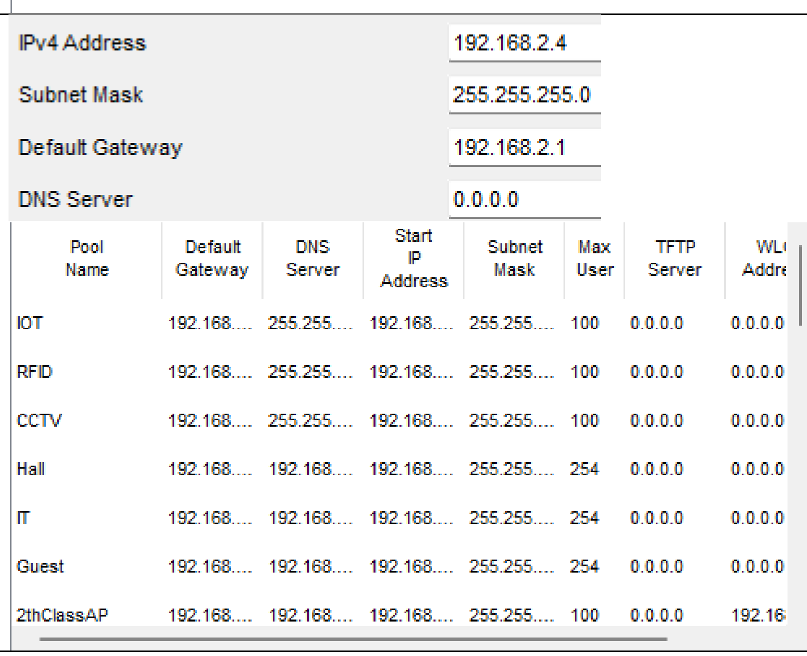
IP Configuration

DHCP	Static
IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0

Device	Configure	Command
DNS Server	IPv4 and DNS	

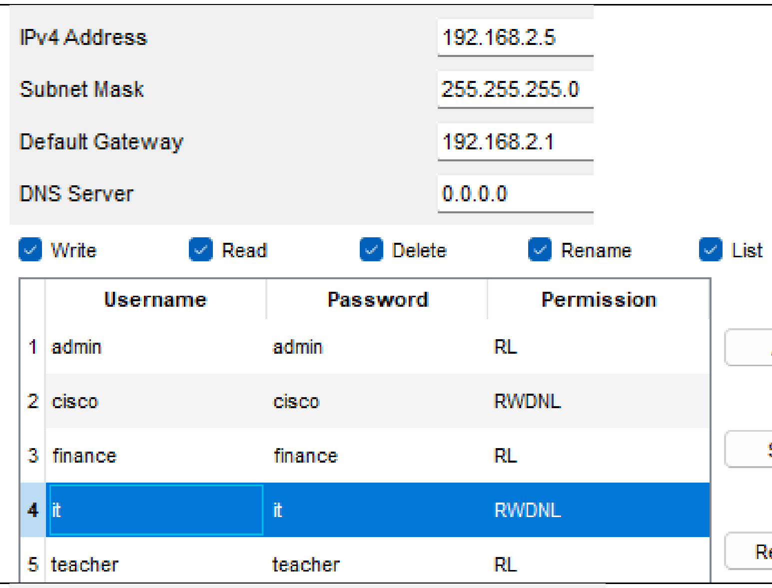
IPv4 Address	192.168.2.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0

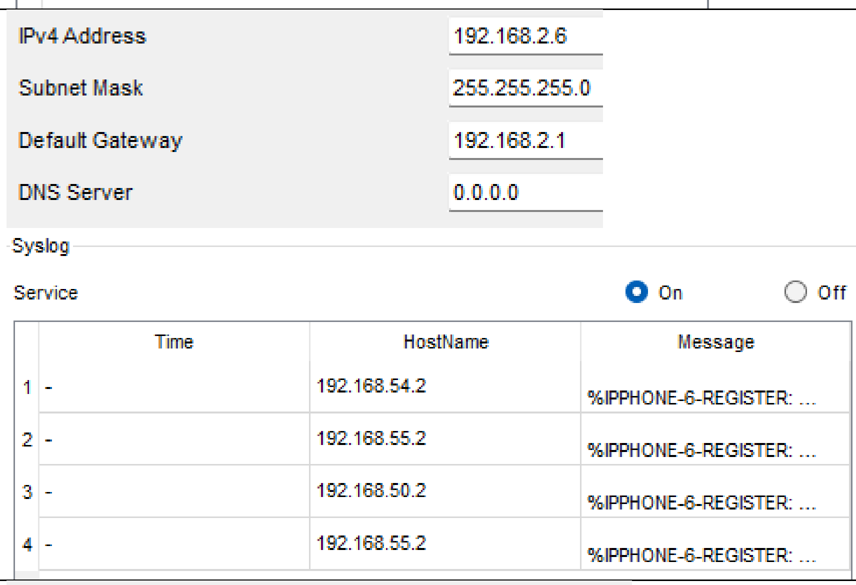
0	branch.com	A Record	172.16.2.7
1	main.com	A Record	192.168.2.6
2	www.branch.com	A Record	172.16.2.2
3	www.main.com	A Record	192.168.2.2
4	www.outside.com	A Record	123.123.123.2

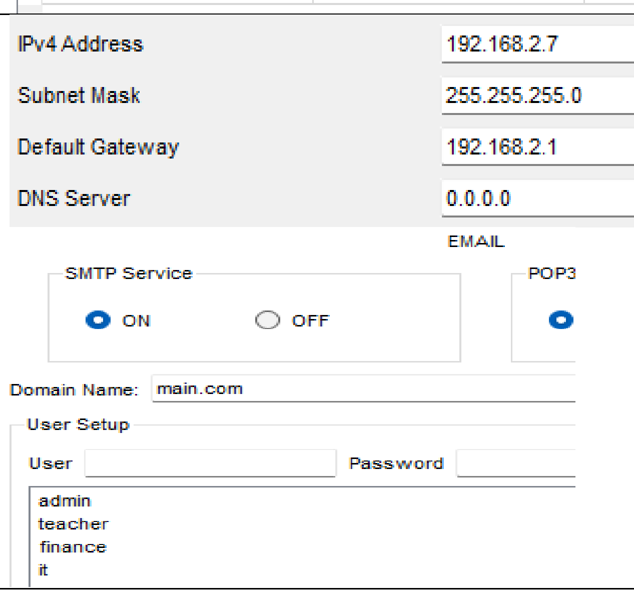
Device	Configure	Command
DHCP Server	IPv4 and DHCP	

IPv4 Address	192.168.2.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WL Addre
IOT	192.168....	255.255....	192.168....	255.255....	100	0.0.0.0	0.0.0.0
RFID	192.168....	255.255....	192.168....	255.255....	100	0.0.0.0	0.0.0.0
CCTV	192.168....	255.255....	192.168....	255.255....	100	0.0.0.0	0.0.0.0
Hall	192.168....	192.168....	192.168....	255.255....	254	0.0.0.0	0.0.0.0
IT	192.168....	192.168....	192.168....	255.255....	254	0.0.0.0	0.0.0.0
Guest	192.168....	192.168....	192.168....	255.255....	254	0.0.0.0	0.0.0.0
2thClassAP	192.168....	192.168....	192.168....	255.255....	100	0.0.0.0	192.16

<table>
<tr><td>FTP Server</td><td>IPv4 and FTP</td><td>


</td></tr>
<tr><td>Syslog Server</td><td>IPv4 and Syslog</td><td>


</td></tr>
<tr><td>Email Server</td><td>IPv4 and Email</td><td>


</td></tr>
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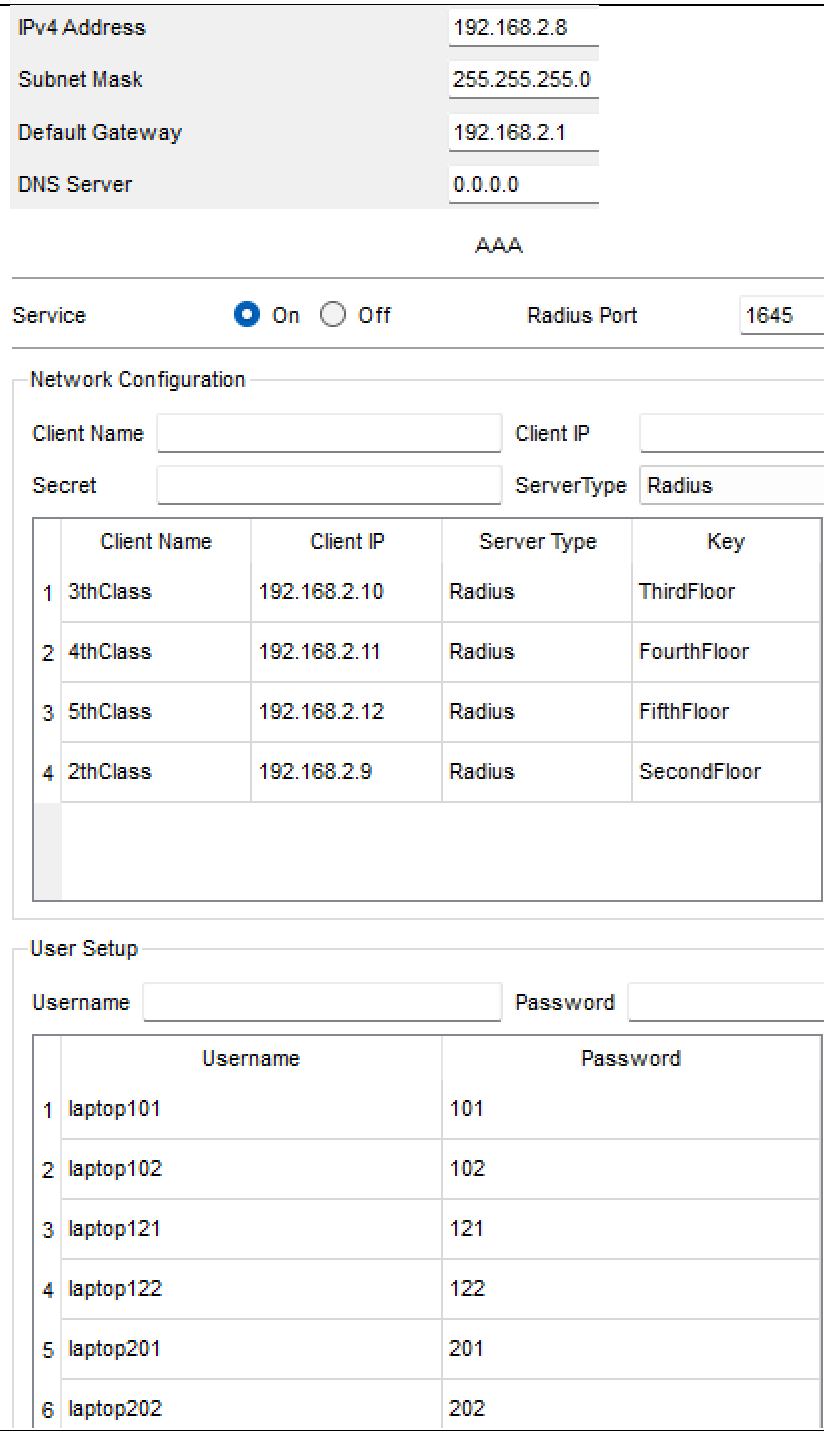
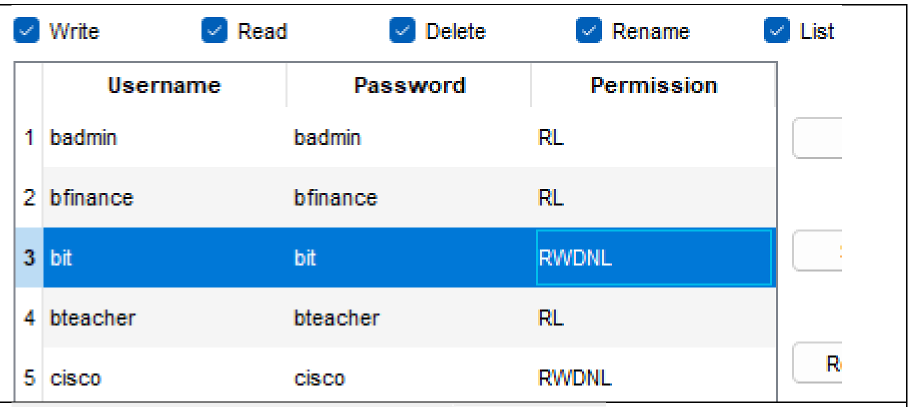
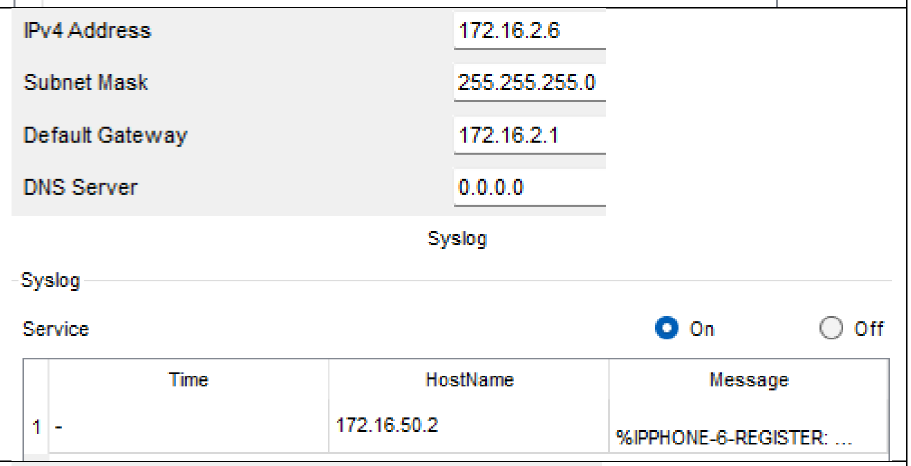
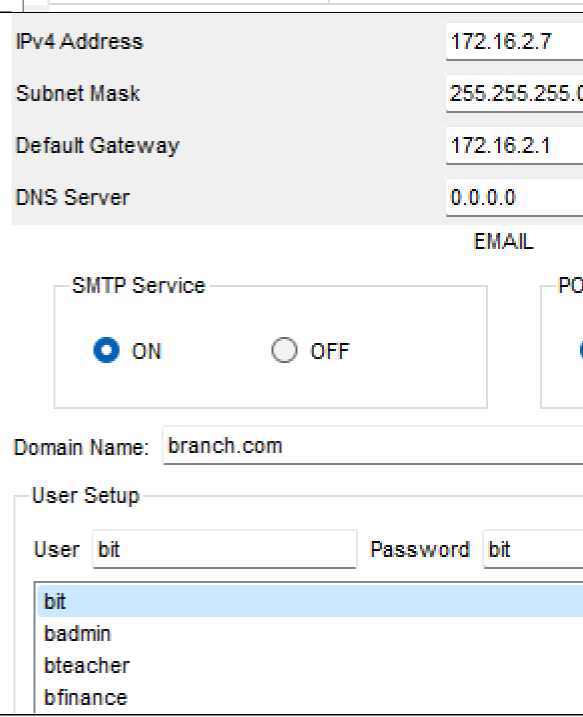
<table>
<tr>
<td>AAA Server</td>
<td>IPv4 and AAA</td>
<td>
</td>
</tr>
</table>

<table>
<tr><th>Device</th><th>Configure</th><th>Command</th></tr>
<tr><td>Branch Web Server</td><td>IPv4</td><td>
<table>
<tr><td>IPv4 Address</td><td>172.16.2.2</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td></tr>
<tr><td>Default Gateway</td><td>172.16.2.1</td></tr>
<tr><td>DNS Server</td><td>0.0.0.0</td></tr>
</table>
</td></tr>
<tr><td>Branch DNS Server</td><td>IPv4 and DNS</td><td>
<table>
<tr><td>IPv4 Address</td><td>172.16.2.3</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td></tr>
<tr><td>Default Gateway</td><td>172.16.2.1</td></tr>
<tr><td>DNS Server</td><td>0.0.0.0</td></tr>
</table>
<table>
<tr><th>No.</th><th>Name</th><th>Type</th><th>Detail</th></tr>
<tr><td>0</td><td>branch.com</td><td>A Record</td><td>172.16.2.7</td></tr>
<tr><td>1</td><td>main.com</td><td>A Record</td><td>192.168.2.7</td></tr>
<tr><td>2</td><td>www.branch.com</td><td>A Record</td><td>172.16.2.2</td></tr>
<tr><td>3</td><td>www.main.com</td><td>A Record</td><td>192.168.2.2</td></tr>
<tr><td>4</td><td>www.outside.com</td><td>A Record</td><td>123.123.123.2</td></tr>
</table>
</td></tr>
<tr><td>Branch DHCP Server</td><td>IPv4 and DHCP</td><td>
<table>
<tr><td>IPv4 Address</td><td>172.16.2.4</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td></tr>
<tr><td>Default Gateway</td><td>172.16.2.1</td></tr>
<tr><td>DNS Server</td><td>0.0.0.0</td></tr>
</table>
<table>
<tr><th>Pool Name</th><th>Default Gateway</th><th>DNS Server</th><th>Start IP Address</th><th>Subnet Mask</th><th>Max User</th><th>TFTP Server</th><th>WL Addre</th></tr>
<tr><td>RFID</td><td>172.16.1...</td><td>172.16.2.3</td><td>172.16.1...</td><td>255.255....</td><td>200</td><td>0.0.0.0</td><td>0.0.0.0</td></tr>
<tr><td>CCTV&Motion&...</td><td>172.16.1...</td><td>172.16.2.3</td><td>172.16.1...</td><td>255.255....</td><td>200</td><td>0.0.0.0</td><td>0.0.0.0</td></tr>
<tr><td>Branch4thAP</td><td>172.16.9...</td><td>172.16.2.3</td><td>172.16.9...</td><td>255.255....</td><td>254</td><td>0.0.0.0</td><td>172.16</td></tr>
<tr><td>Branch3thAP</td><td>172.16.9...</td><td>172.16.2.3</td><td>172.16.9...</td><td>255.255....</td><td>254</td><td>0.0.0.0</td><td>172.16</td></tr>
<tr><td>Branch2thAP</td><td>172.16.9...</td><td>172.16.2.3</td><td>172.16.9...</td><td>255.255....</td><td>254</td><td>0.0.0.0</td><td>172.16</td></tr>
<tr><td>Branch4thClass</td><td>172.16.1...</td><td>172.16.2.3</td><td>172.16.1...</td><td>255.255....</td><td>254</td><td>0.0.0.0</td><td>0.0.0.0</td></tr>
<tr><td>Branch3thClass</td><td>172.16.1...</td><td>172.16.2.3</td><td>172.16.1...</td><td>255.255....</td><td>254</td><td>0.0.0.0</td><td>0.0.0.0</td></tr>
</table>
</td></tr>
<tr><td>Branch FTP Server</td><td>IPv4 and FTP</td><td>
<table>
<tr><td>IPv4 Address</td><td>172.16.2.5</td></tr>
<tr><td>Subnet Mask</td><td>255.255.255.0</td></tr>
<tr><td>Default Gateway</td><td>172.16.2.1</td></tr>
<tr><td>DNS Server</td><td>0.0.0.0</td></tr>
</table>
</td></tr>
</table>

		
Branch Syslog Server	IPv4 and Syslog	
Branch Email Server	IPv4 and Email	

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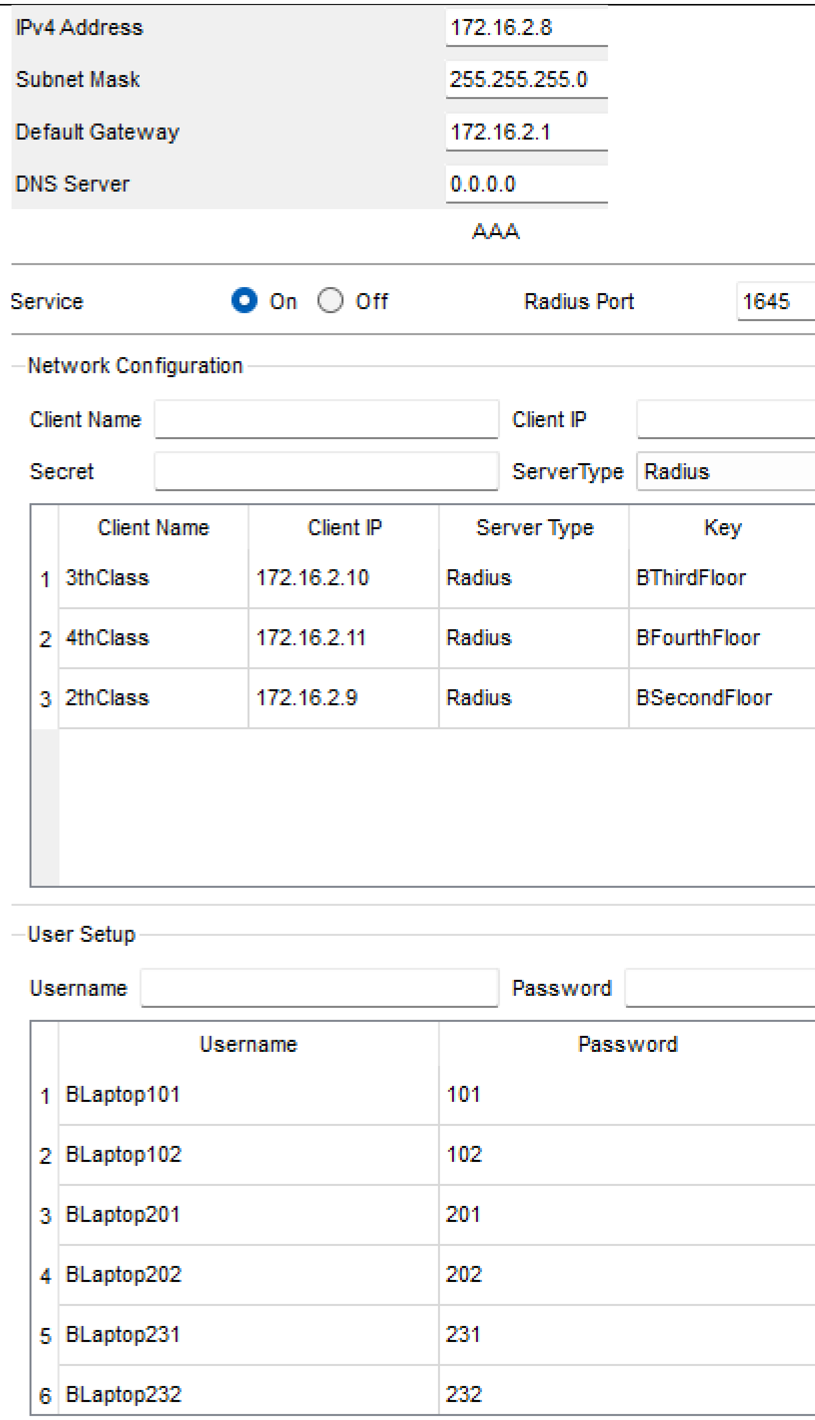
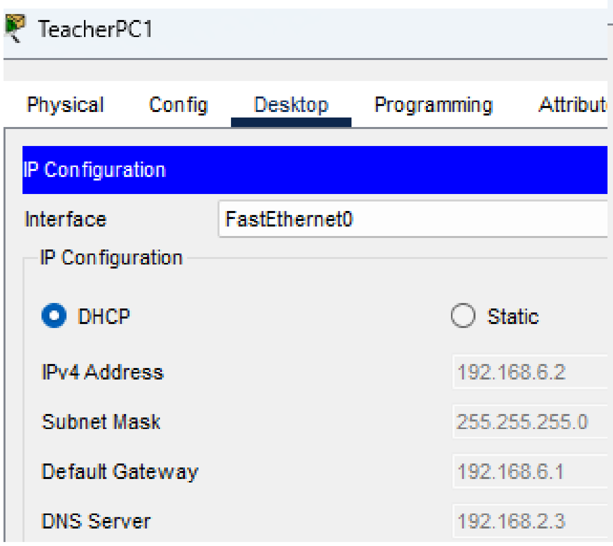
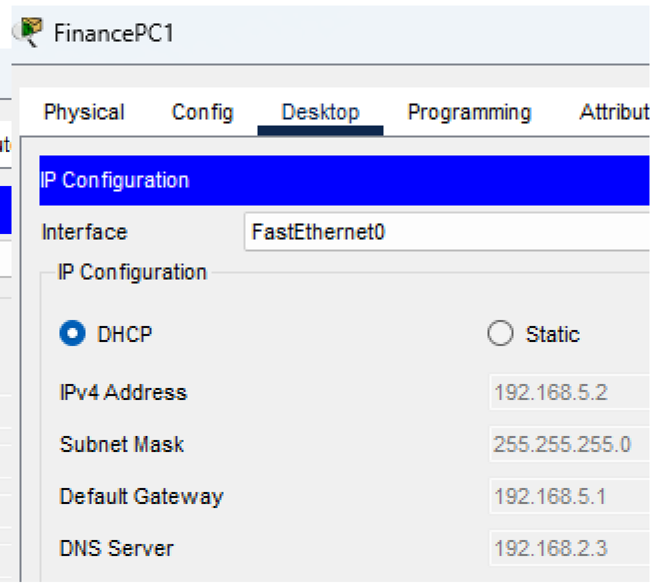
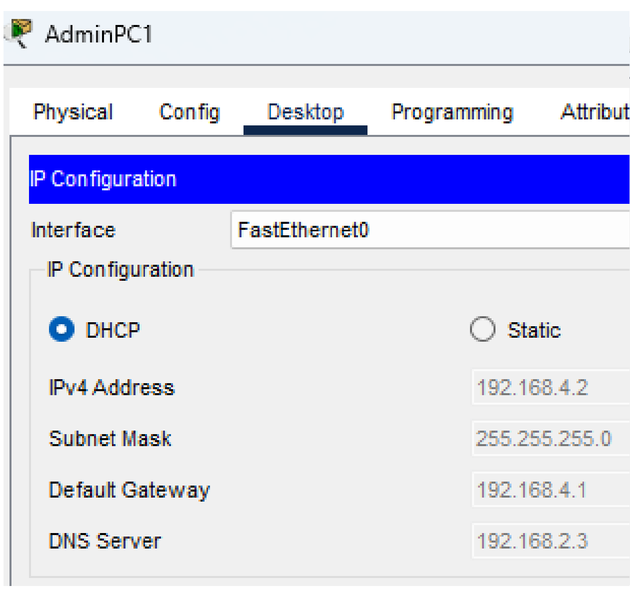
<tr><td>Branch AAA Server</td><td>IPv4 and AAA</td><td>
</td></tr>
</table>

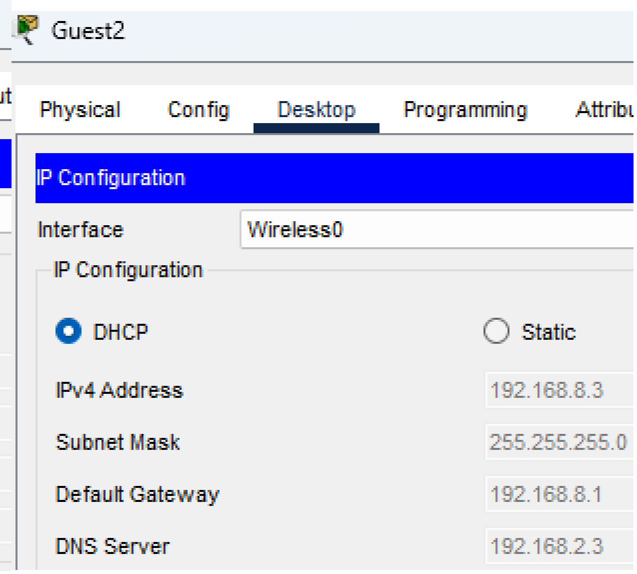
Charter 5: Testing

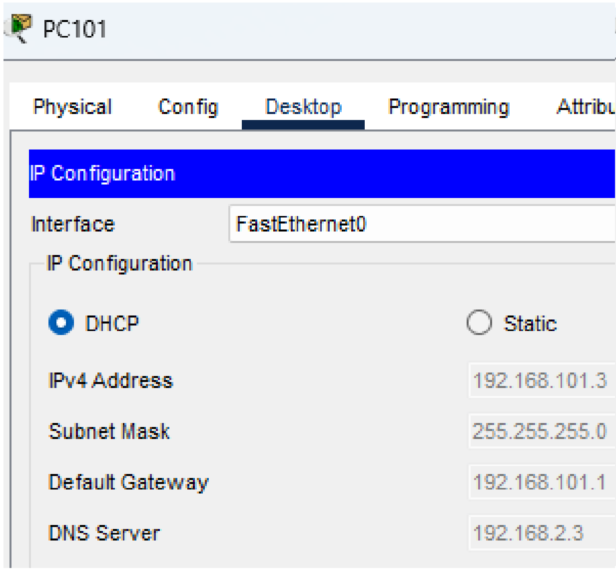
Name	Test whether the DHCP is running normally
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	DHCP Server has Configured DHCP Pool.
Steps	1. Open PC IP Configuration 2. Enable DHCP 3. Monitor result
Expected result	All PC Can obtain IP Address

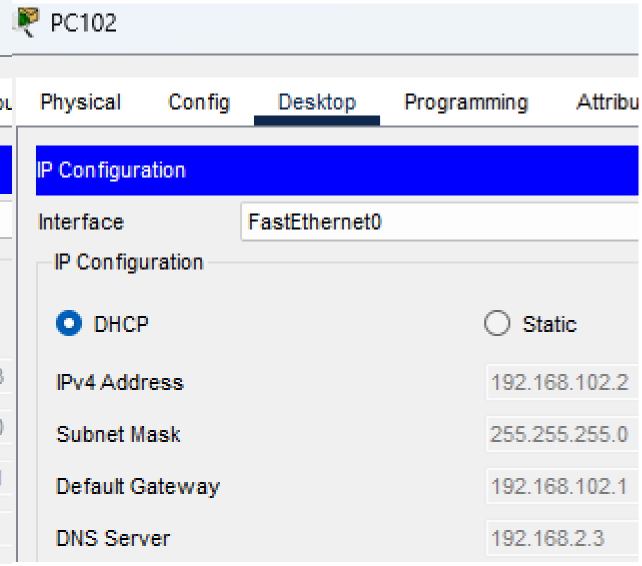


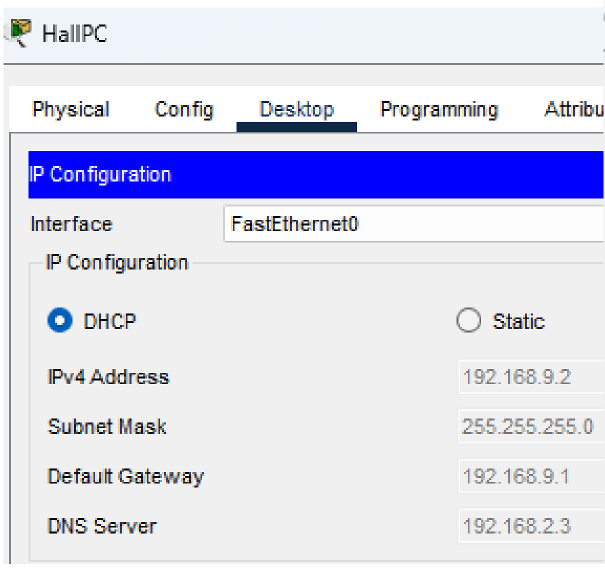


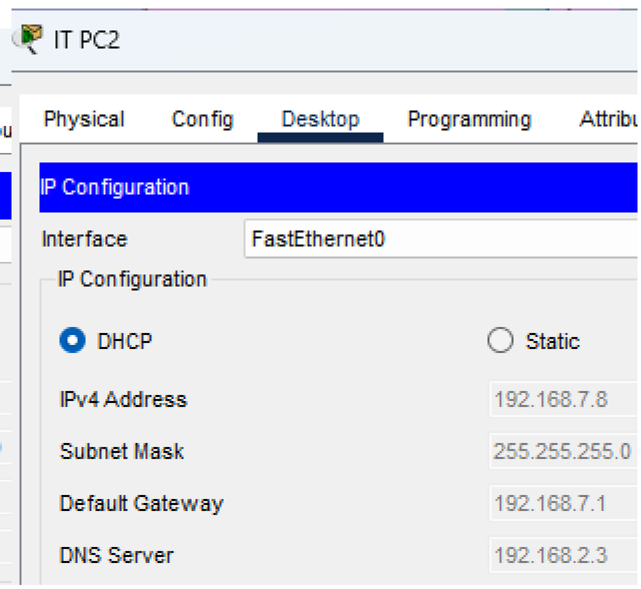


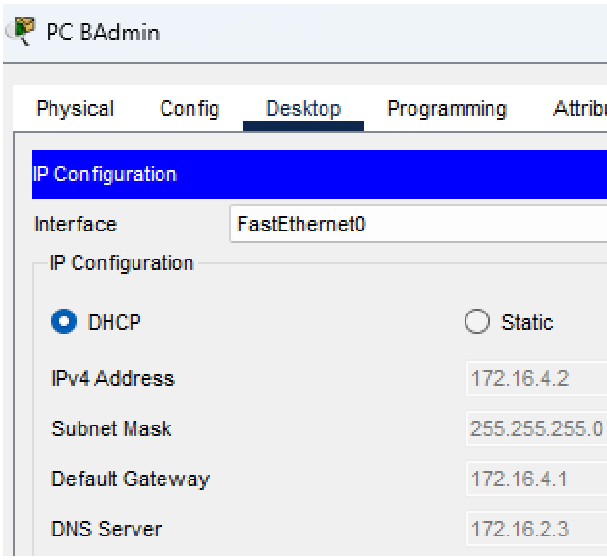


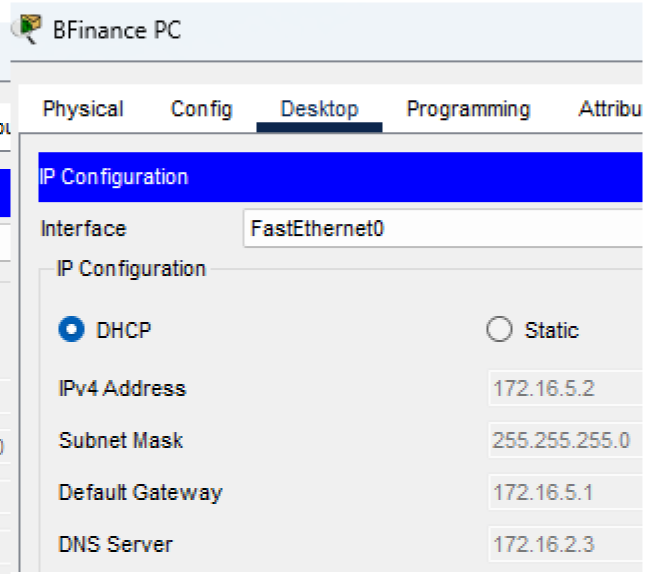

PC101
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.101.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.101.1
DNS Server
192.168.2.3

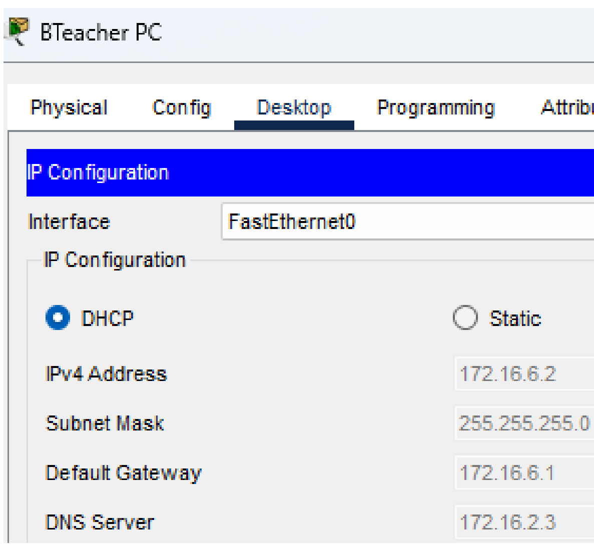

PC102
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.102.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.102.1
DNS Server
192.168.2.3

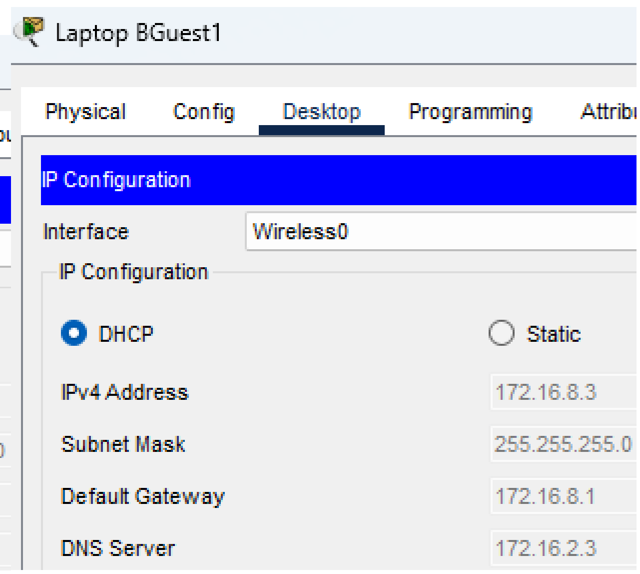

HallPC
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.9.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.9.1
DNS Server
192.168.2.3

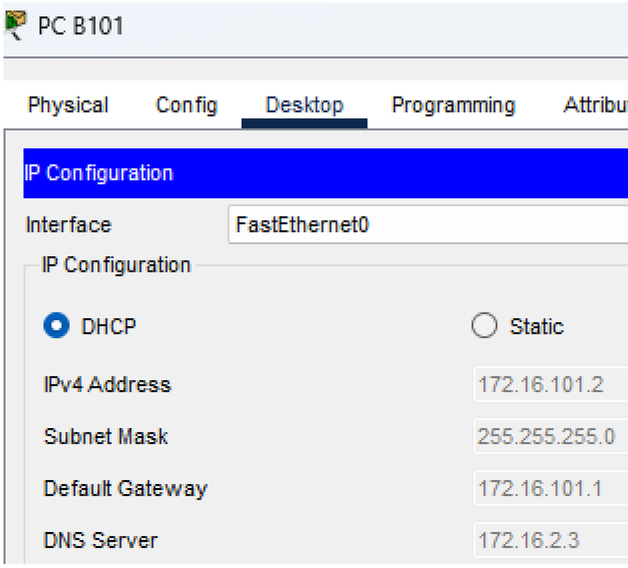

IT PC2
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.7.8
Subnet Mask
255.255.255.0
Default Gateway
192.168.7.1
DNS Server
192.168.2.3

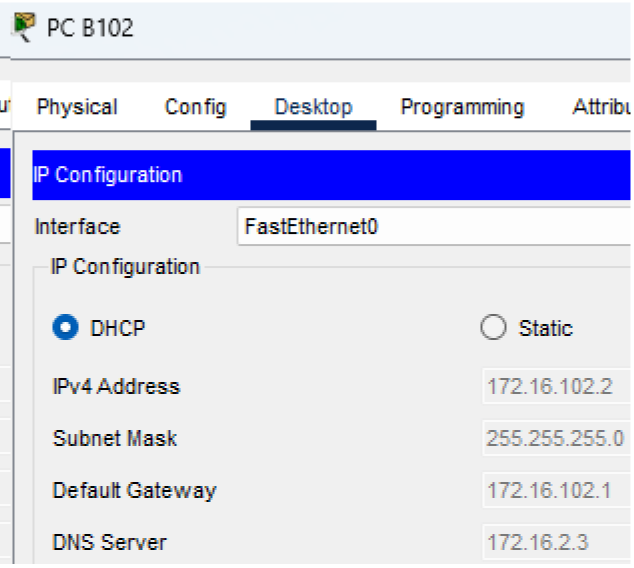

PC BAdmin
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.4.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.4.1
DNS Server
172.16.2.3

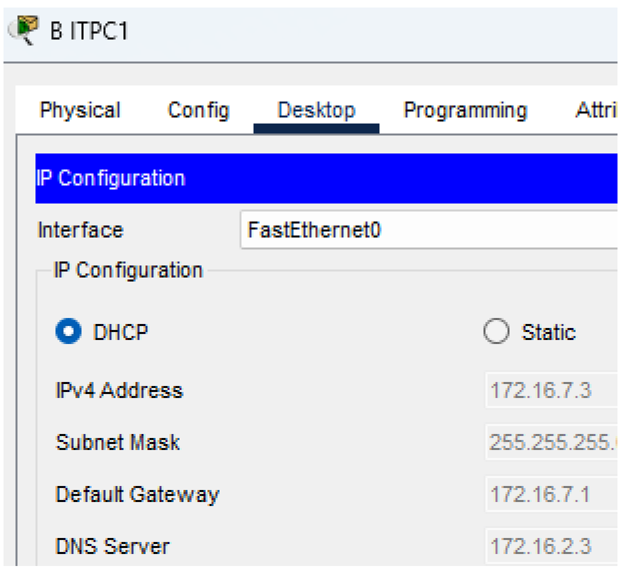

BFinance PC
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.5.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.5.1
DNS Server
172.16.2.3


BTeacher PC
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.6.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.6.1
DNS Server
172.16.2.3

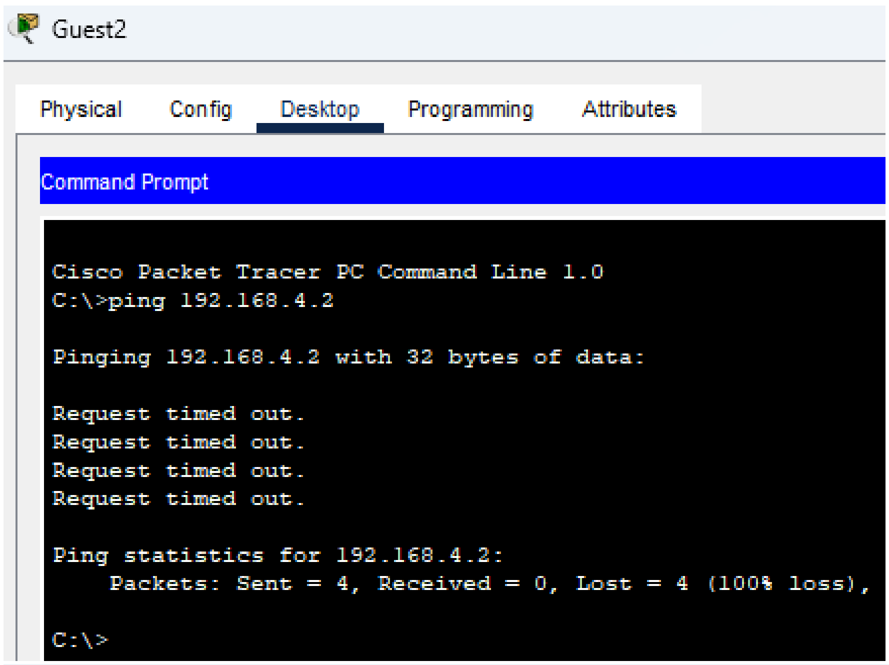

Laptop BGuest1
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
Wireless0
IP Configuration
DHCP
Static
IPv4 Address
172.16.8.3
Subnet Mask
255.255.255.0
Default Gateway
172.16.8.1
DNS Server
172.16.2.3

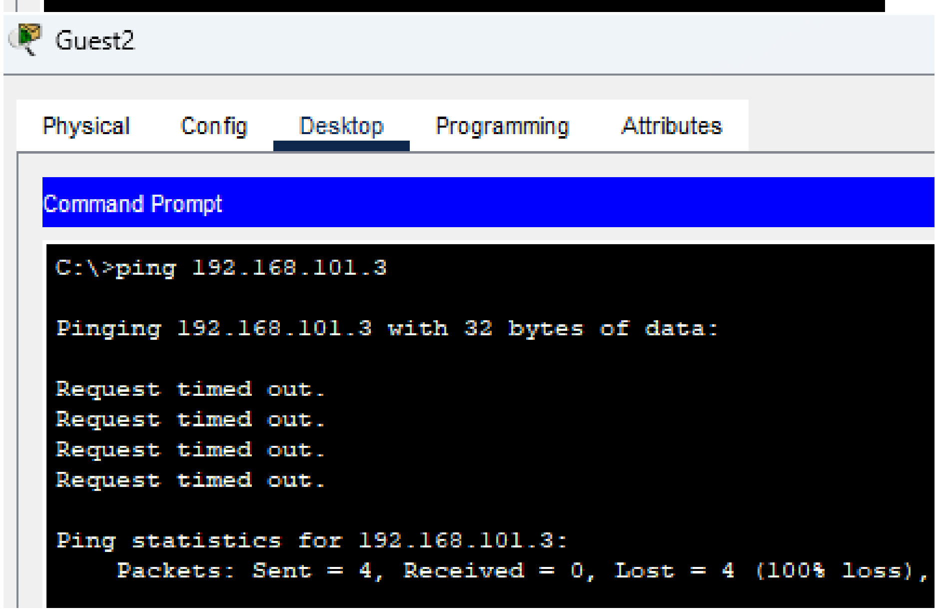

PC B101
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.101.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.101.1
DNS Server
172.16.2.3

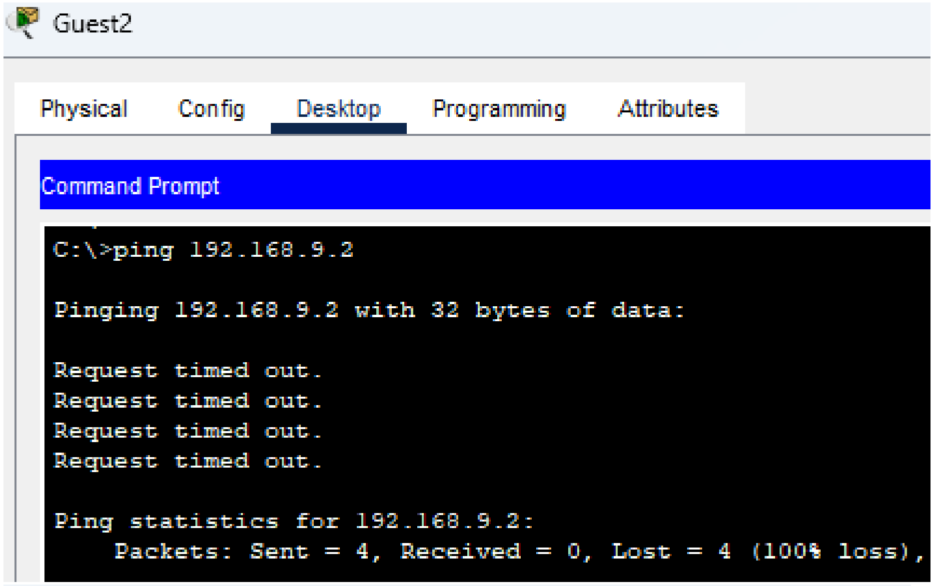

PC B102
Physical
Config
Desktop
Programming
Attribu
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.102.2
Subnet Mask
255.255.255.0
Default Gateway
172.16.102.1
DNS Server
172.16.2.3

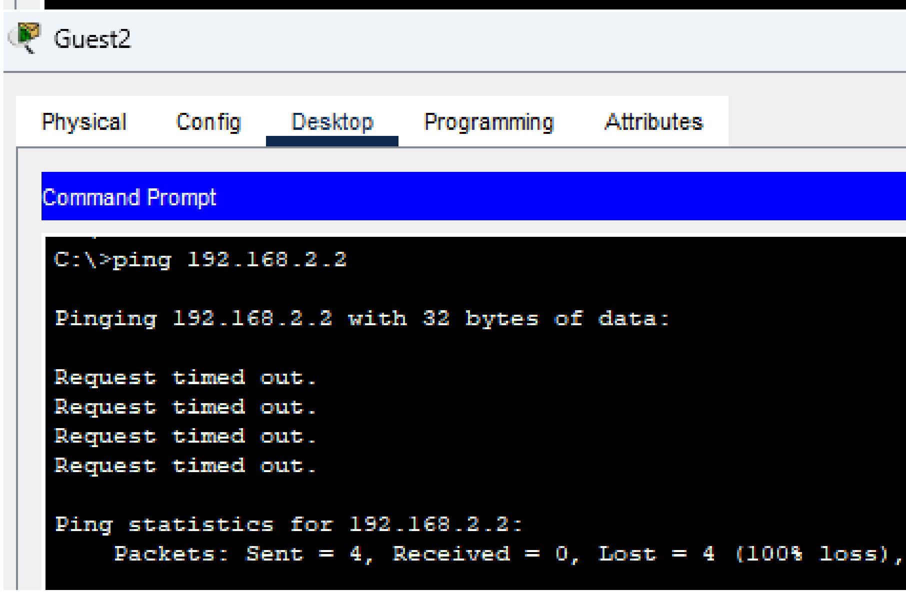

B ITPC1
Physical
Config
Desktop
Programming
Attri
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.7.3
Subnet Mask
255.255.255.
Default Gateway
172.16.7.1
DNS Server
172.16.2.3

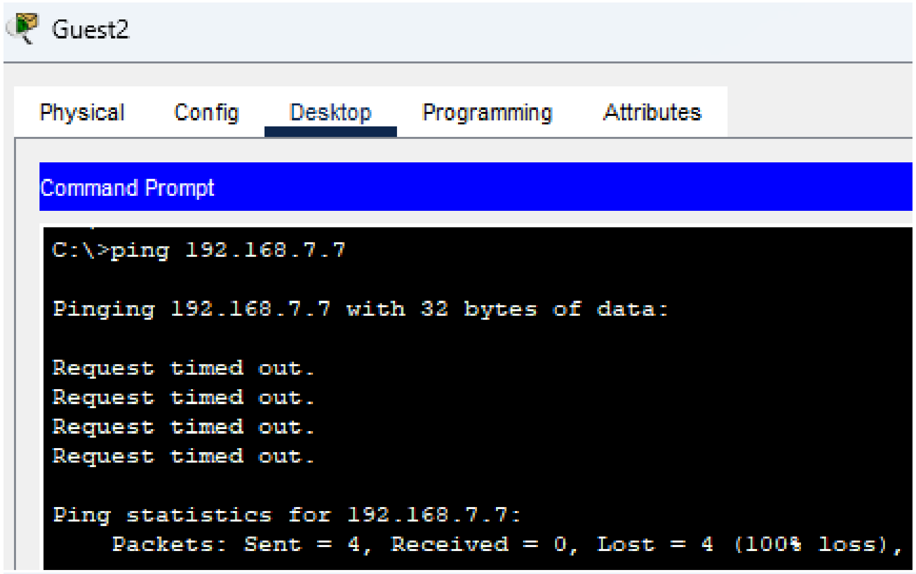
Name	Ping Test for Guest PC
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	All Pc in the network respond to ping requests.
Steps	1. Open Guest Command Prompt 2. Type target IP test of ping from Gues Laptop 3. Monitor the result.
Expected result	Guest cannot ping all pc.

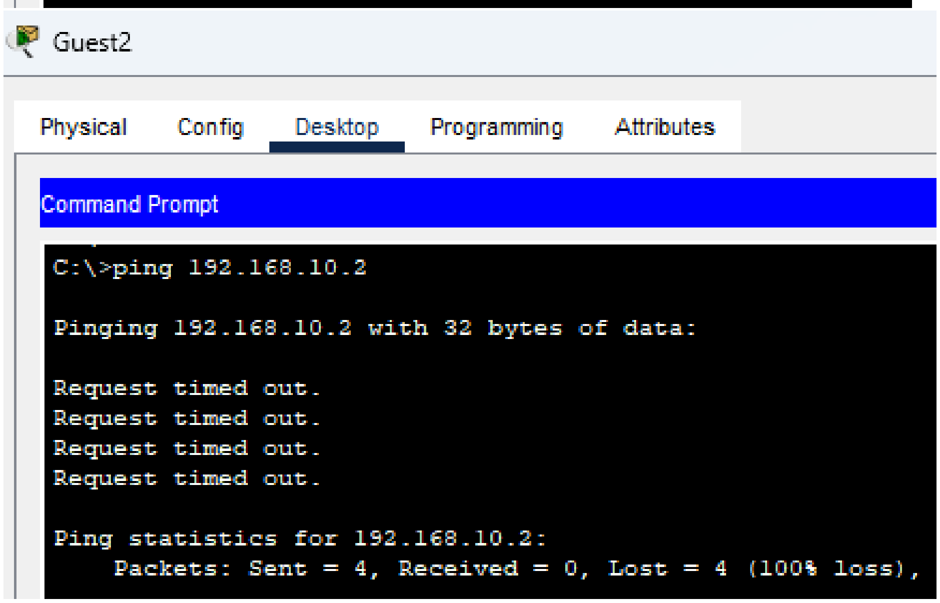




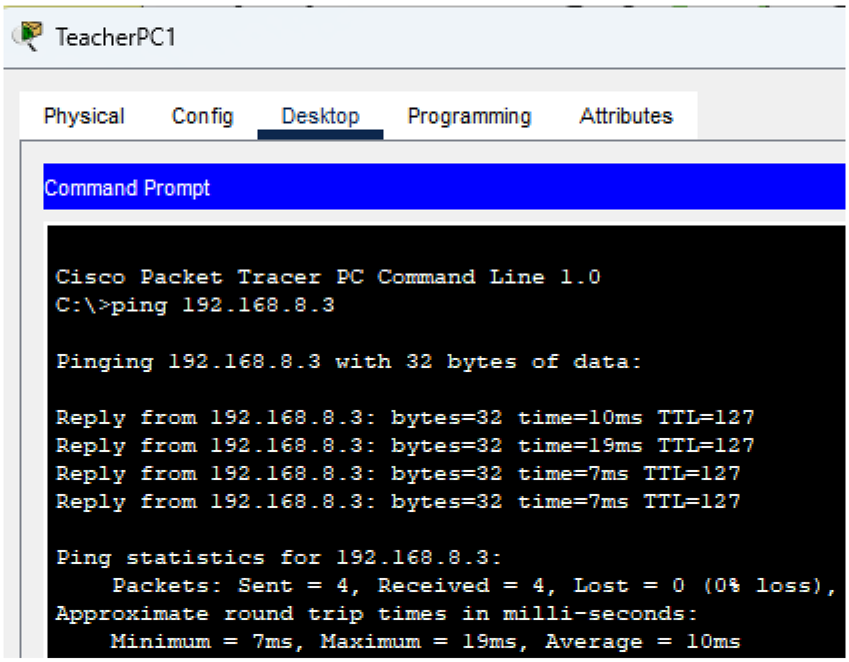

Guest2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.9.2
Pinging 192.168.9.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.9.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

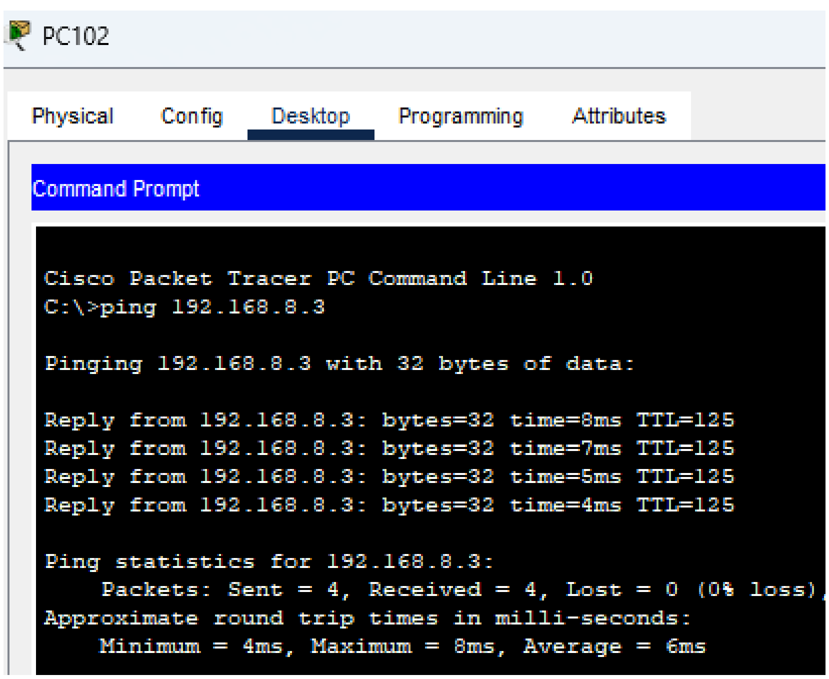

Guest2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

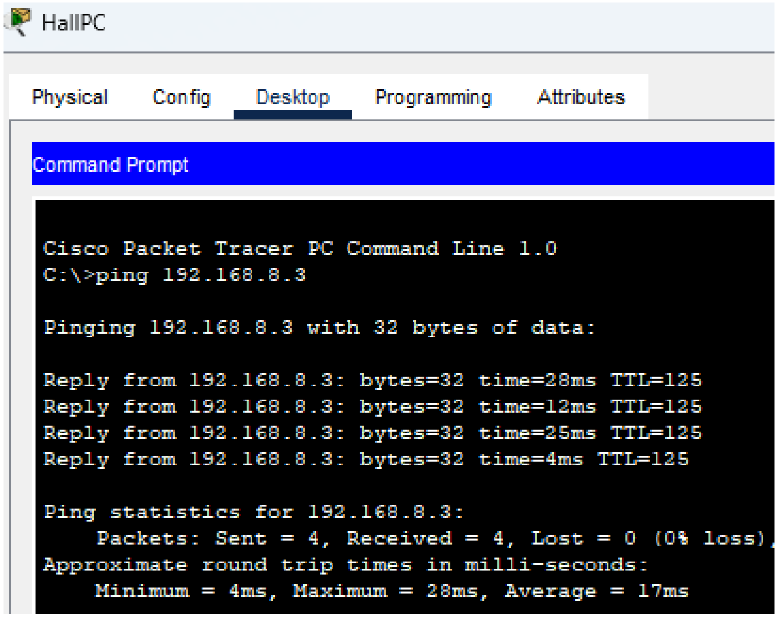

Guest2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.7.7
Pinging 192.168.7.7 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.7.7:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

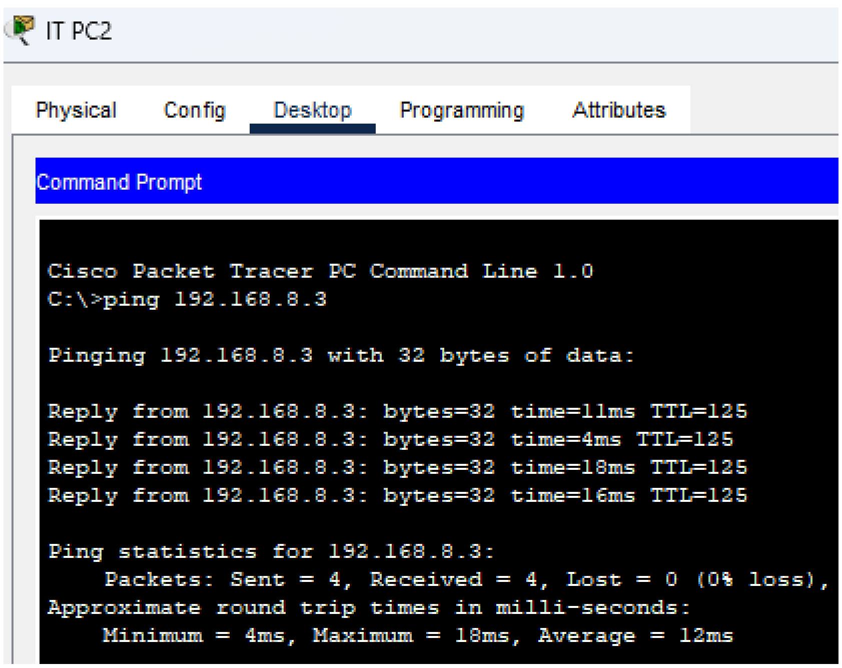

Guest2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

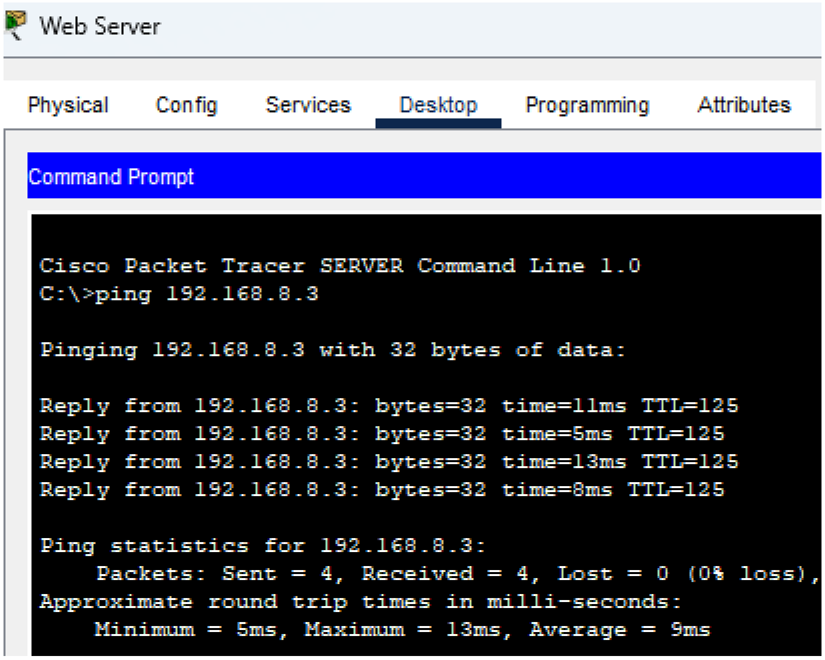
Name	Ping test form all PC to Guest
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	All Pc in the network respond to ping requests.
Steps	1. Open PC Command Prompt 2. Type Guest IP test of ping from PC 3. Monitor the result.
Expected result	All Pc can ping Guest



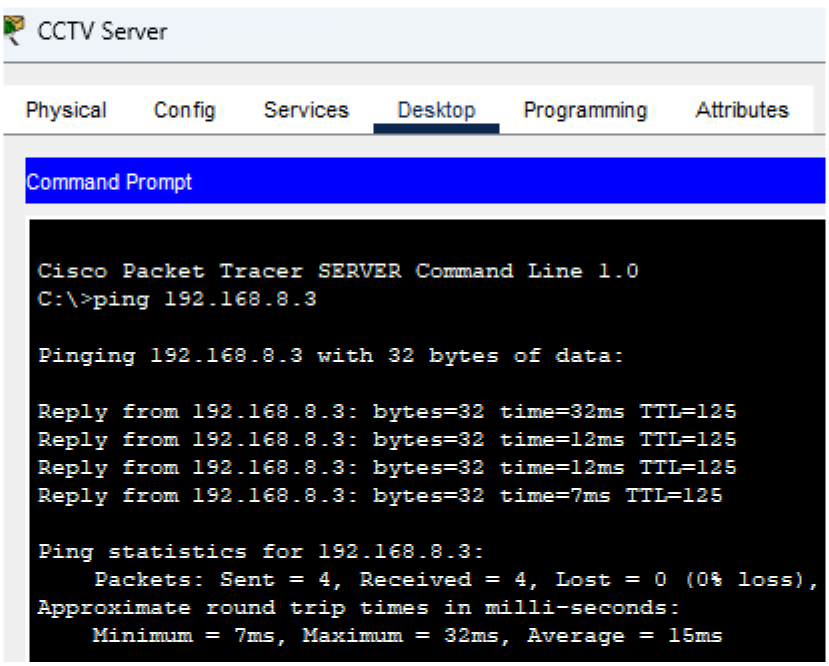



HallPC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.8.3
Pinging 192.168.8.3 with 32 bytes of data:
Reply from 192.168.8.3: bytes=32 time=28ms TTL=125
Reply from 192.168.8.3: bytes=32 time=12ms TTL=125
Reply from 192.168.8.3: bytes=32 time=25ms TTL=125
Reply from 192.168.8.3: bytes=32 time=4ms TTL=125
Ping statistics for 192.168.8.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 28ms, Average = 17ms


IT PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.8.3
Pinging 192.168.8.3 with 32 bytes of data:
Reply from 192.168.8.3: bytes=32 time=11ms TTL=125
Reply from 192.168.8.3: bytes=32 time=4ms TTL=125
Reply from 192.168.8.3: bytes=32 time=18ms TTL=125
Reply from 192.168.8.3: bytes=32 time=16ms TTL=125
Ping statistics for 192.168.8.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 18ms, Average = 12ms

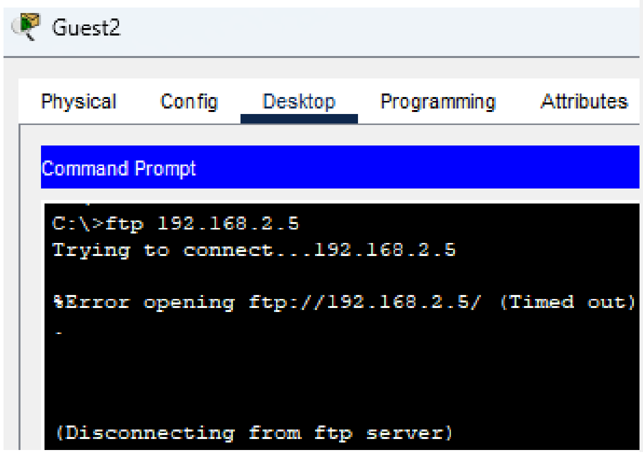


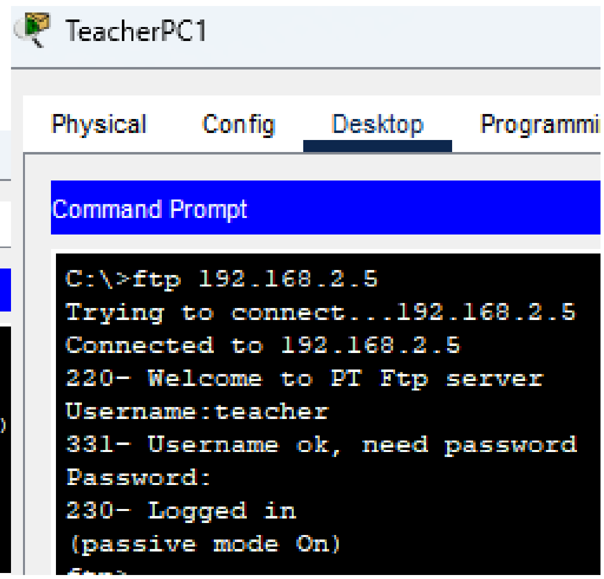

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.8.3

Pinging 192.168.8.3 with 32 bytes of data:

Reply from 192.168.8.3: bytes=32 time=11ms TTL=125
Reply from 192.168.8.3: bytes=32 time=5ms TTL=125
Reply from 192.168.8.3: bytes=32 time=13ms TTL=125
Reply from 192.168.8.3: bytes=32 time=8ms TTL=125

Ping statistics for 192.168.8.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 13ms, Average = 9ms
```

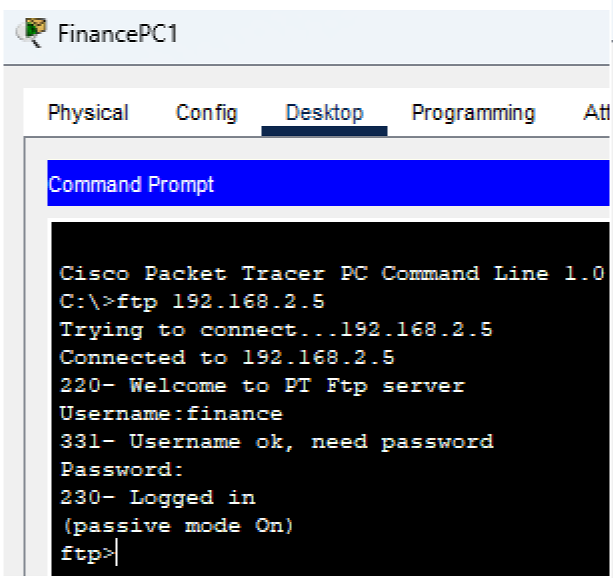


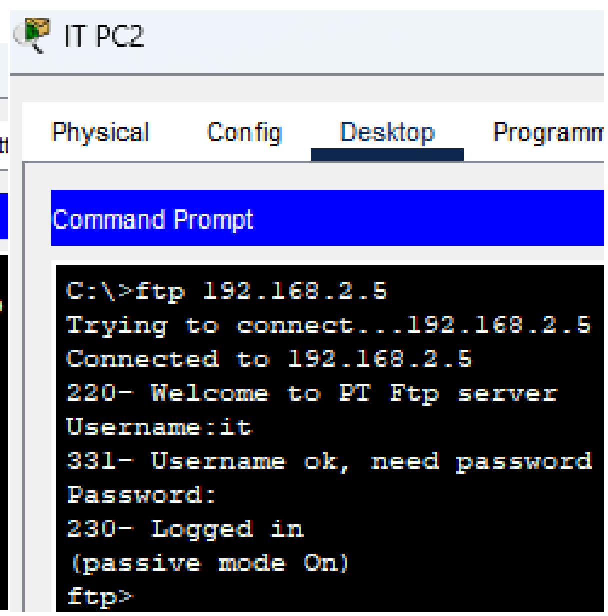

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.8.3

Pinging 192.168.8.3 with 32 bytes of data:

Reply from 192.168.8.3: bytes=32 time=32ms TTL=125
Reply from 192.168.8.3: bytes=32 time=12ms TTL=125
Reply from 192.168.8.3: bytes=32 time=12ms TTL=125
Reply from 192.168.8.3: bytes=32 time=7ms TTL=125

Ping statistics for 192.168.8.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 32ms, Average = 15ms
```

Name	Testing Guest and staff Ftp
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	FTP Server setting is completed, make sure the ACL is set.
Steps	1. Open PC Command Prompt 2. Type ftp command connect to the FTP Server. 3. Monitor the result.
Expected result	Guest: FTP connection attempt fails. Staff: FTP connection successfully.

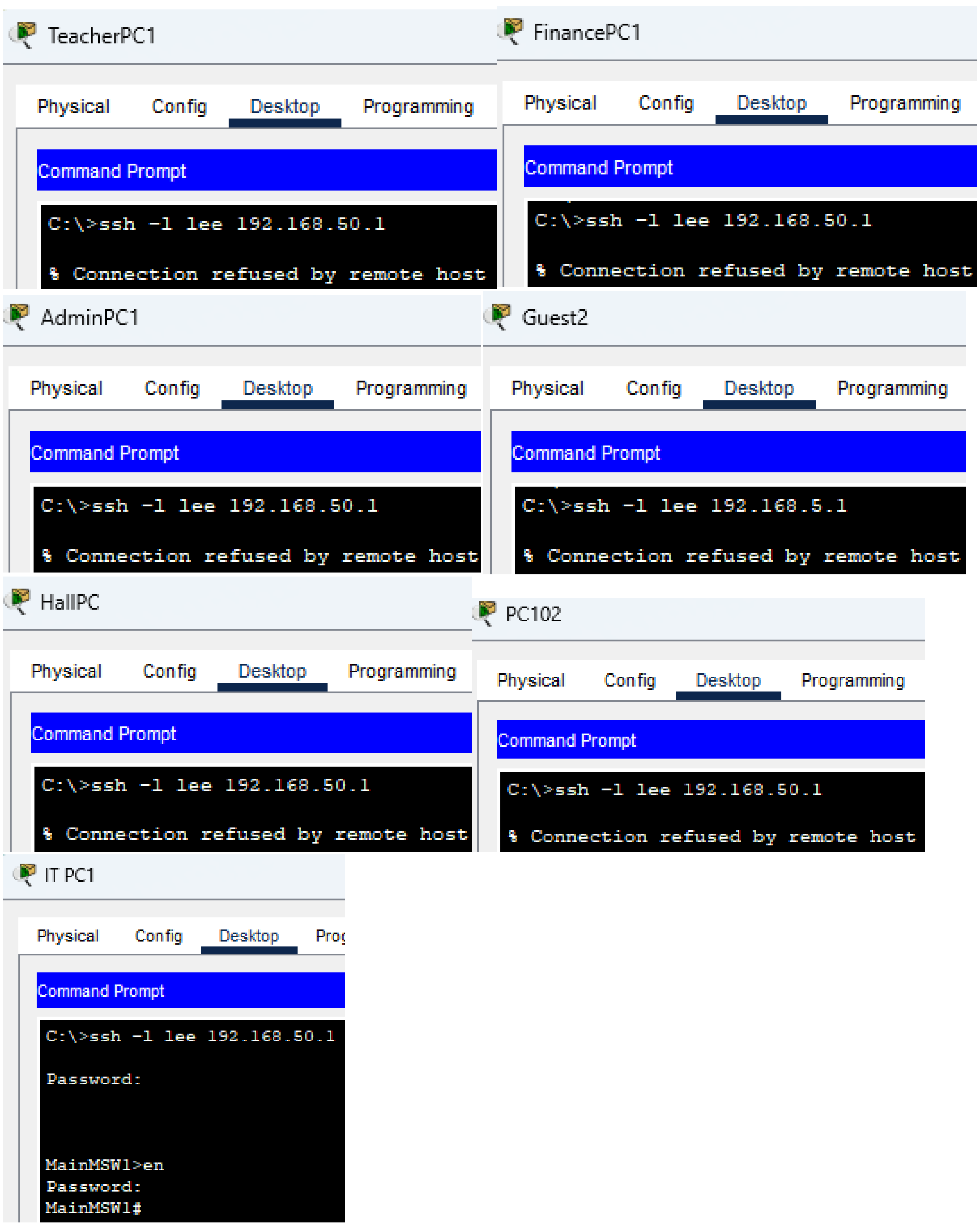








Name	All PC testing SSH
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	Router is configured for ssh and the ACL is Set
Steps	1. Open PC Command Prompt 2. Type SSH command connect to SSH router. 3. Monitor the result.
Expected result	Only IT Department can ssh.

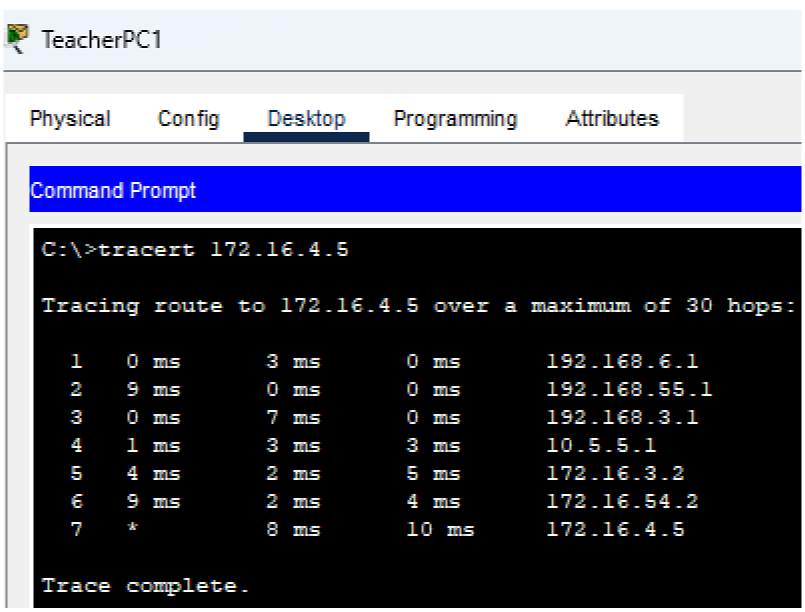


Name	VPN Testing
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	Router VPN configuration is completed
Step	1. Open PC command prompt 2. Traceroute target IP address 3. Monitor the result
Expected result	Data was encrypted

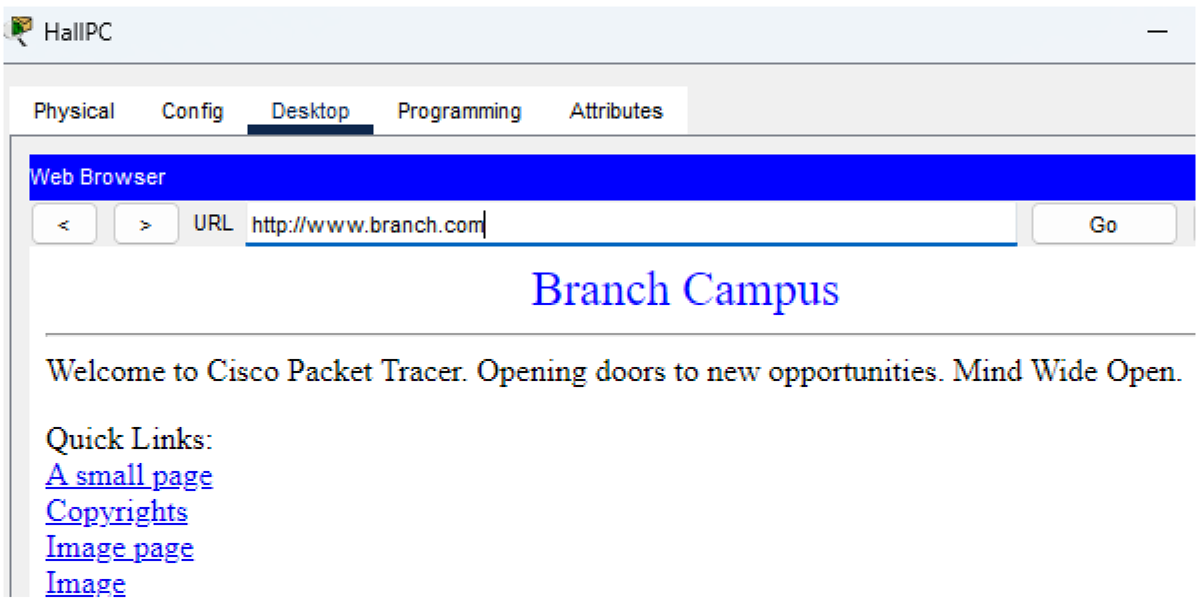
Before VPN Encrypt data

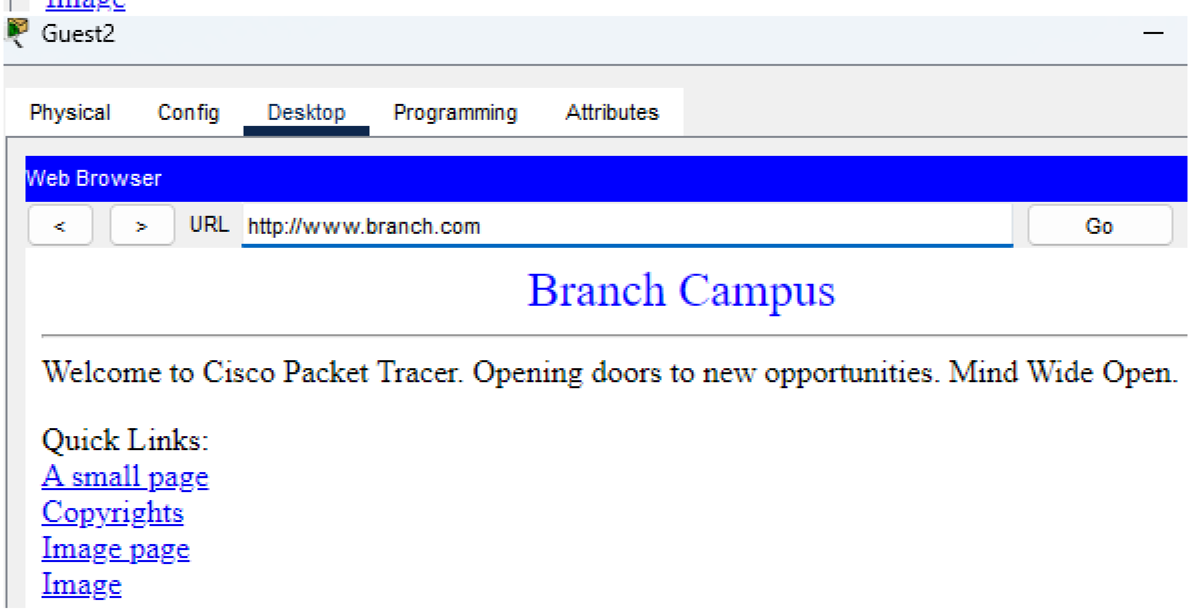
```
C:\>tracert 172.16.4.5

Tracing route to 172.16.4.5 over a maximum of 30 hops:

  1   1 ms      0 ms      1 ms      192.168.6.1
  2   0 ms      0 ms      0 ms      192.168.50.1
  3   0 ms      0 ms      0 ms      192.168.1.1
  4   1 ms      1 ms      0 ms      10.3.3.2
  5   2 ms      1 ms      5 ms      10.5.5.1
  6   1 ms      1 ms      20 ms     172.16.1.2
  7   13 ms     1 ms      1 ms      172.16.50.2
  8   1 ms      1 ms      0 ms      172.16.4.5

Trace complete.

C:\>
```

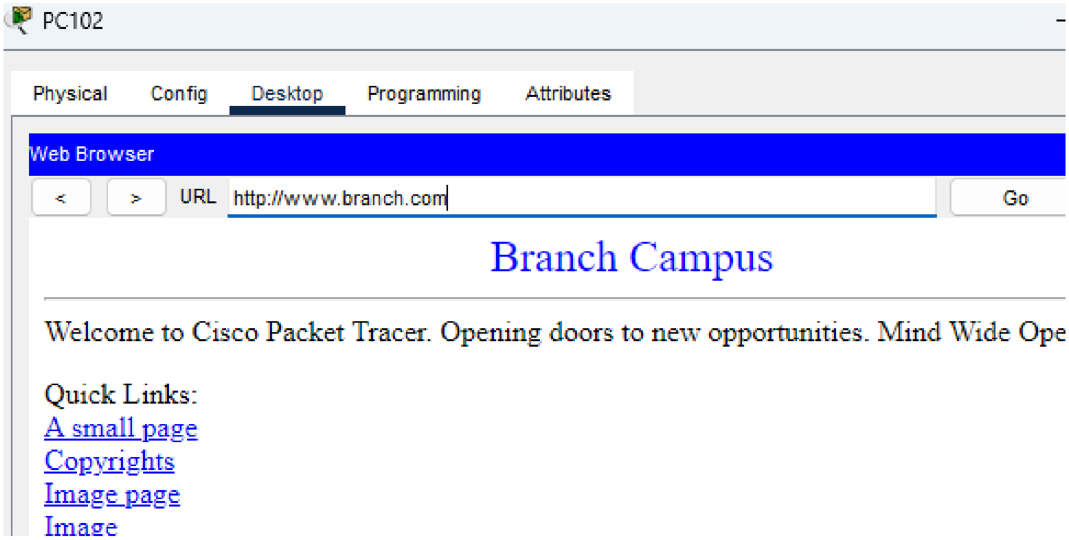
After VPN Encrypt data

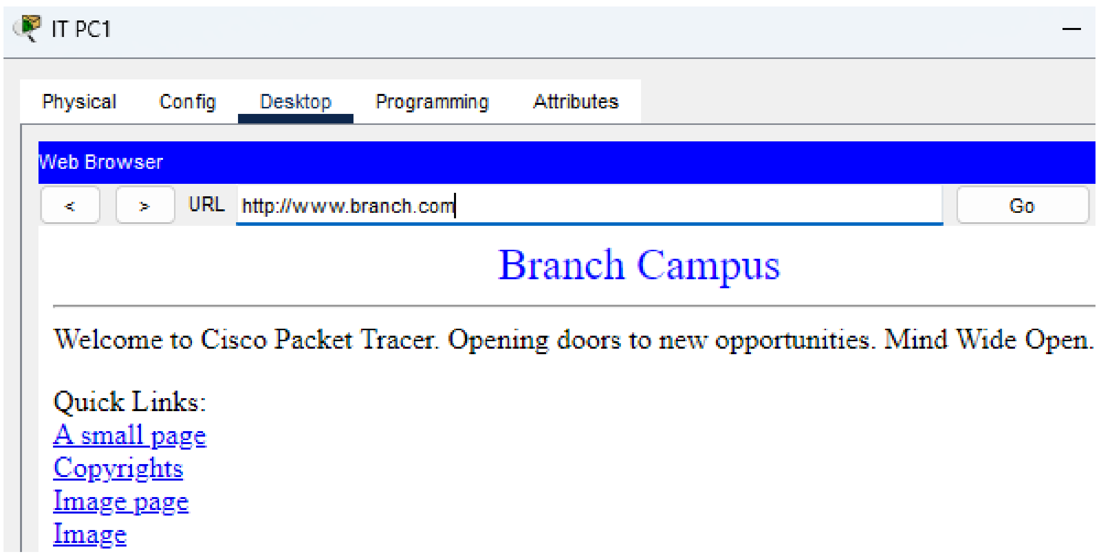


Name	DNS and Web Server Testing
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	DNS Server and Web Server settings are complete and functioning properly.
Step	1. Open PC Web Browser 2. Type DNS name or IP address 3. Monitor the result
Expected result	All PC have internet access.






PC102
Physical
Config
Desktop
Programming
Attributes
Web Browser
<
>
URL
http://www.branch.com
Go
Branch Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Ope
Quick Links:
A small page
Copyrights
Image page
Image


IT PC1
Physical
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Desktop
Programming
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Web Browser
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Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
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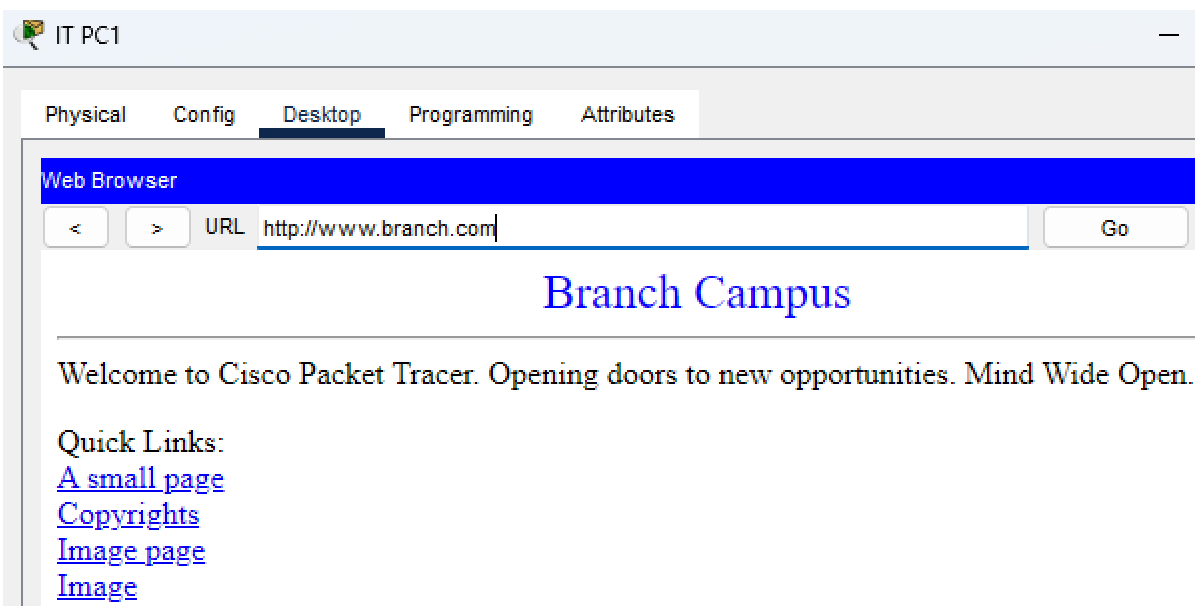
IT PC1
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Branch Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
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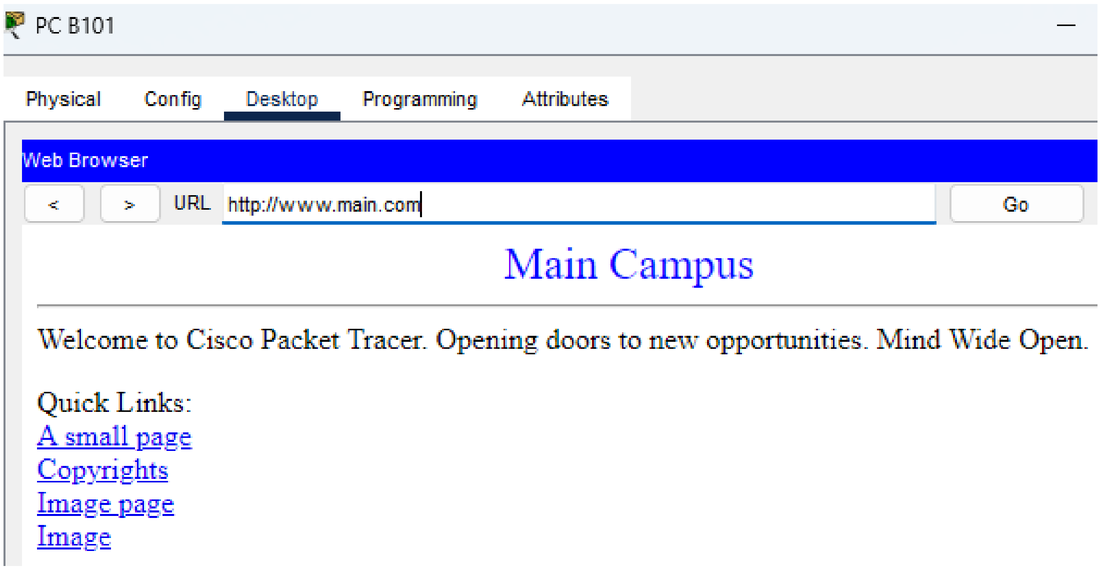
PC B101
Physical
Config
Desktop
Programming
Attributes
Web Browser
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>
URL
http://www.main.com
Go
Main Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
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Image page
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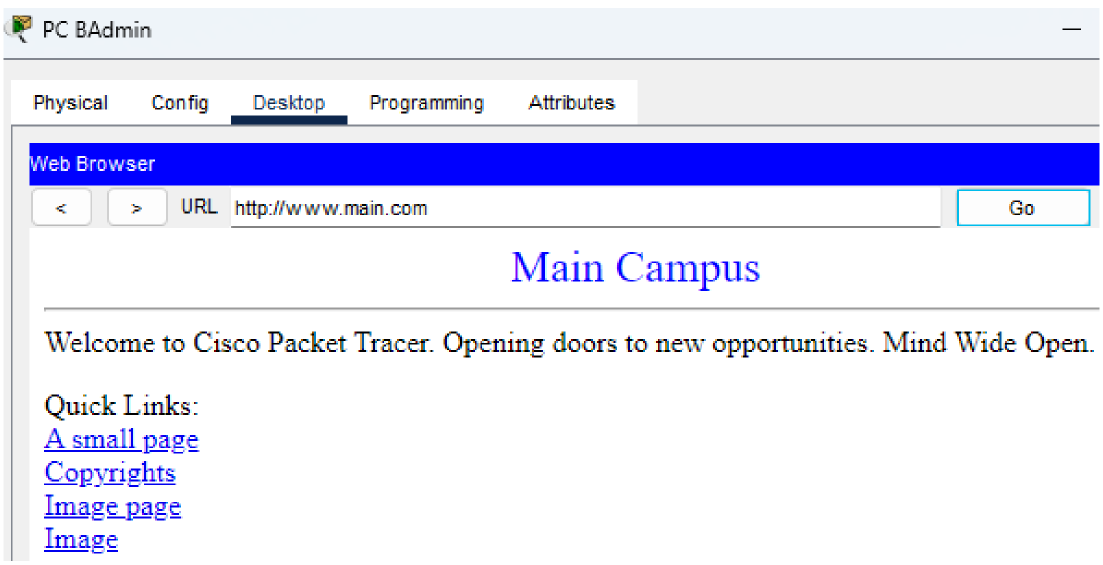
PC BAdmin
Physical
Config
Desktop
Programming
Attributes
Web Browser
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URL
http://www.main.com
Go
Main Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
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Image page
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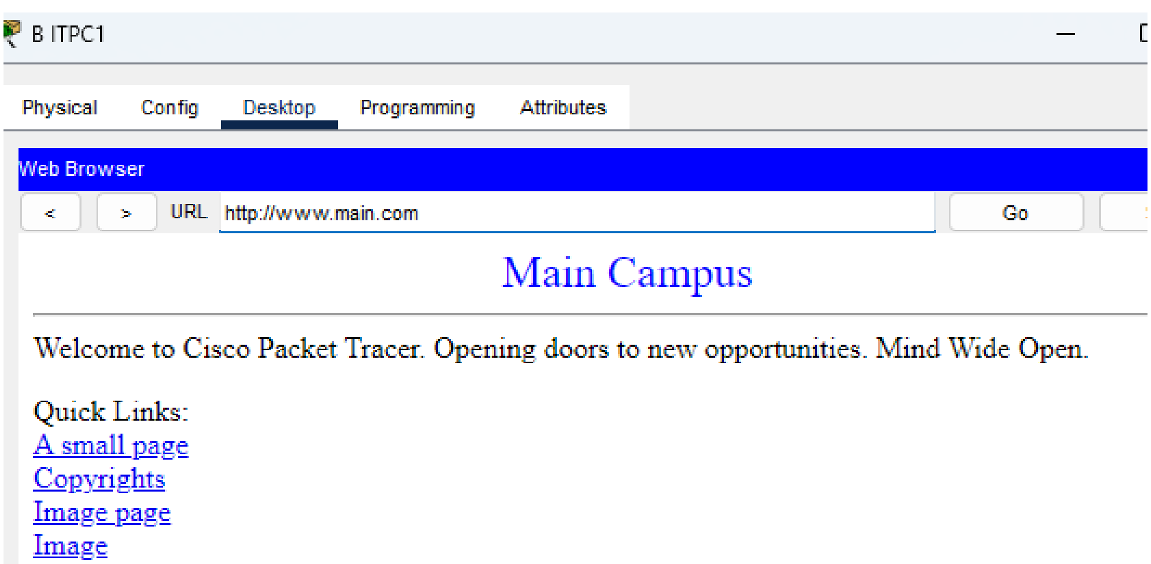
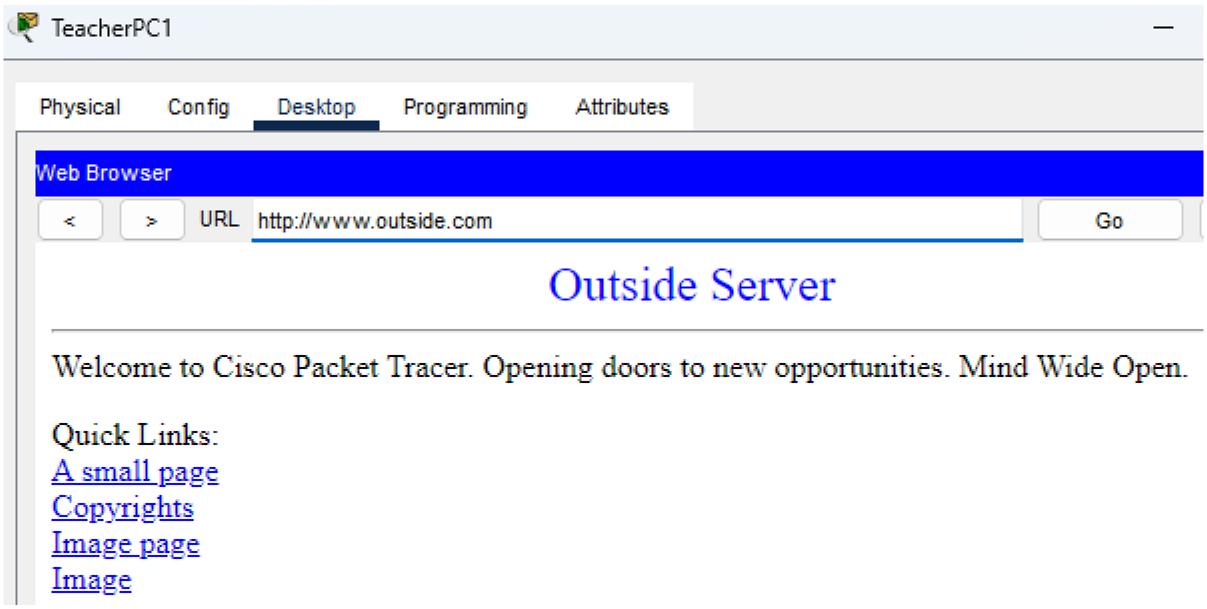
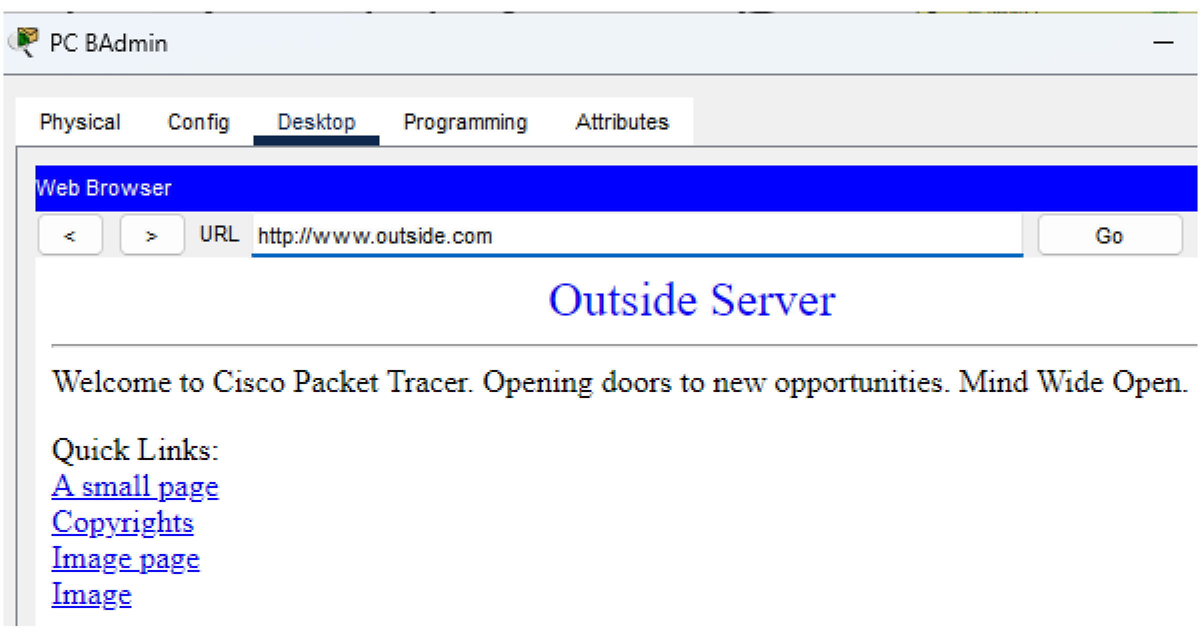
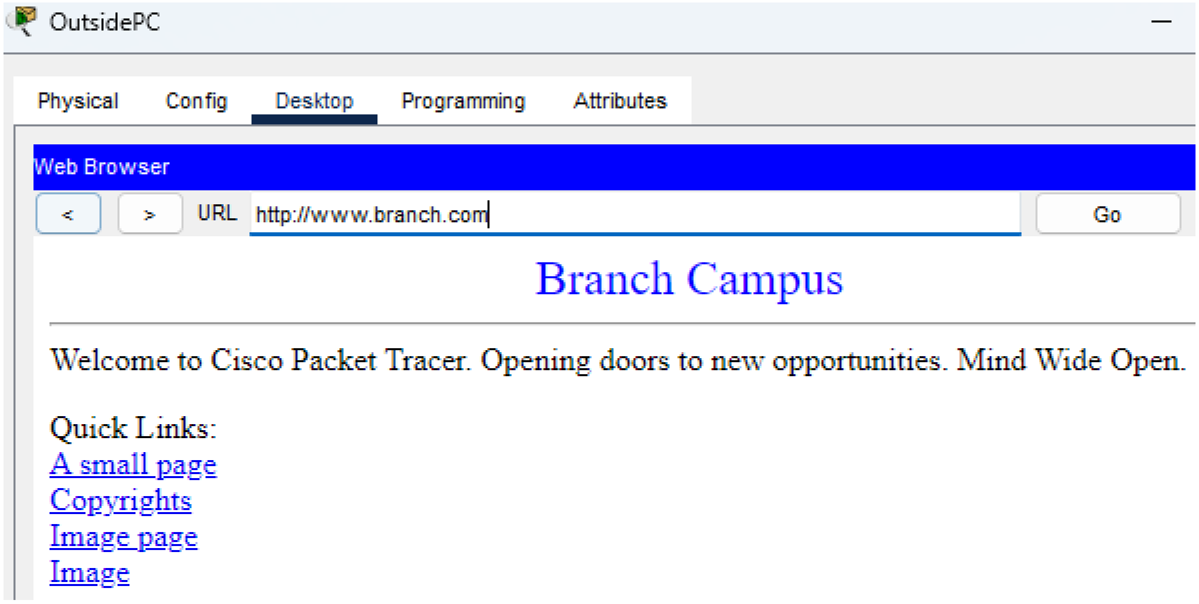

B ITPC1
Physical
Config
Desktop
Programming
Attributes
Web Browser
<
>
URL
http://www.main.com
Go
Main Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

Name	Outside PC Web Server Testing
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	Outside Web Server settings are complete
Step	1. Open PC Web Browser 2. Type DNS name or IP address 3. Monitor the result
Expected result	All PC have internet access, but outside PC only can use virtual IP Internet.






OutsidePC
Physical
Config
Desktop
Programming
Attributes
Web Browser
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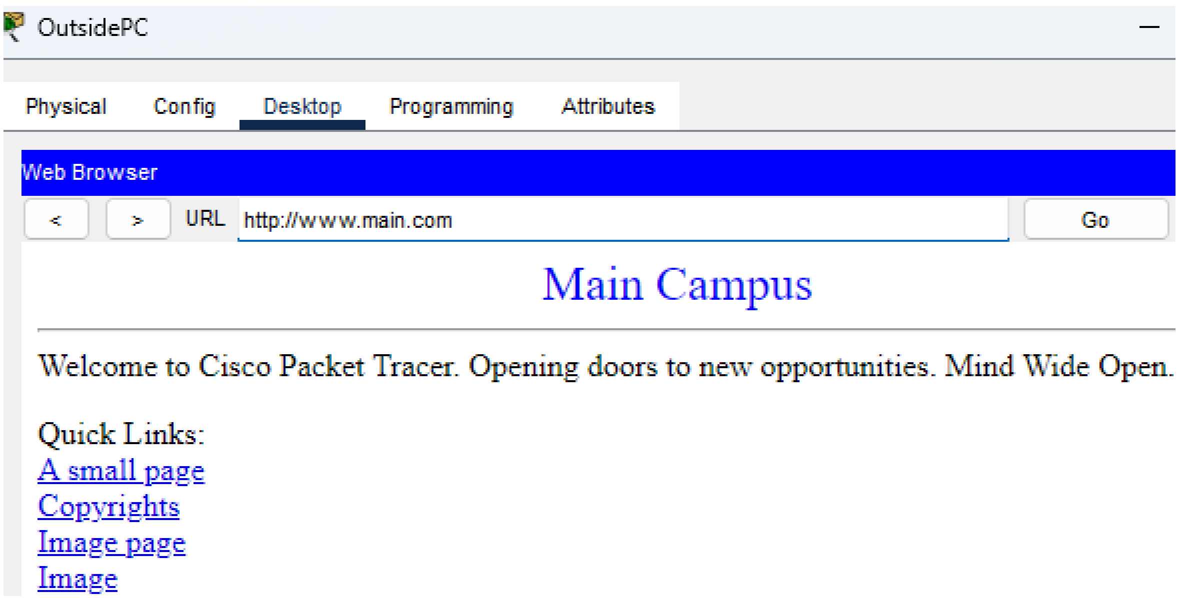
>
URL
http://www.branch.com
Go
Branch Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

OutsidePC
Physical
Config
Desktop
Programming
Attributes
Web Browser
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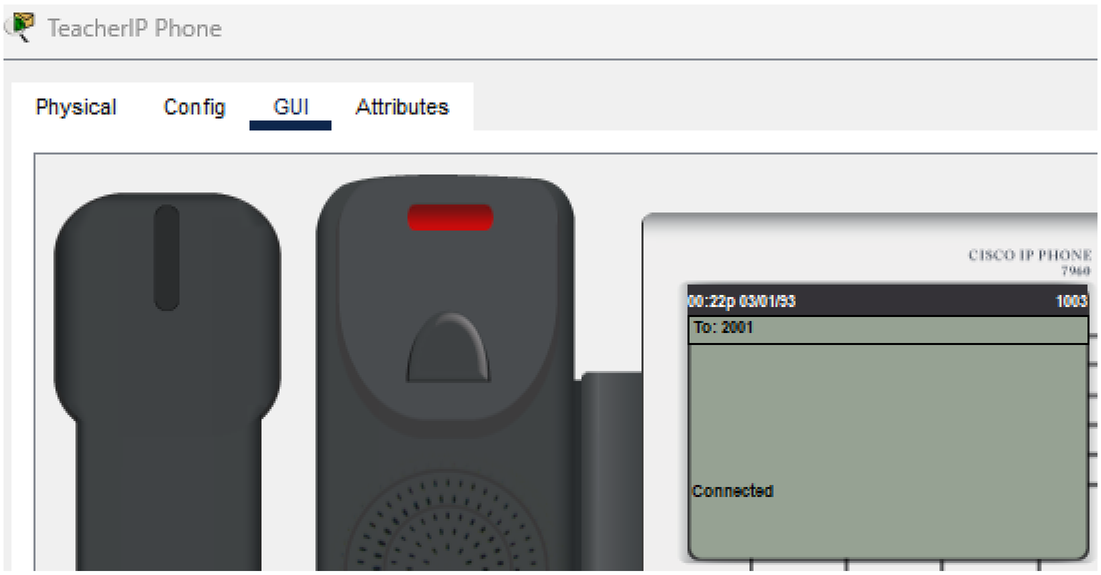
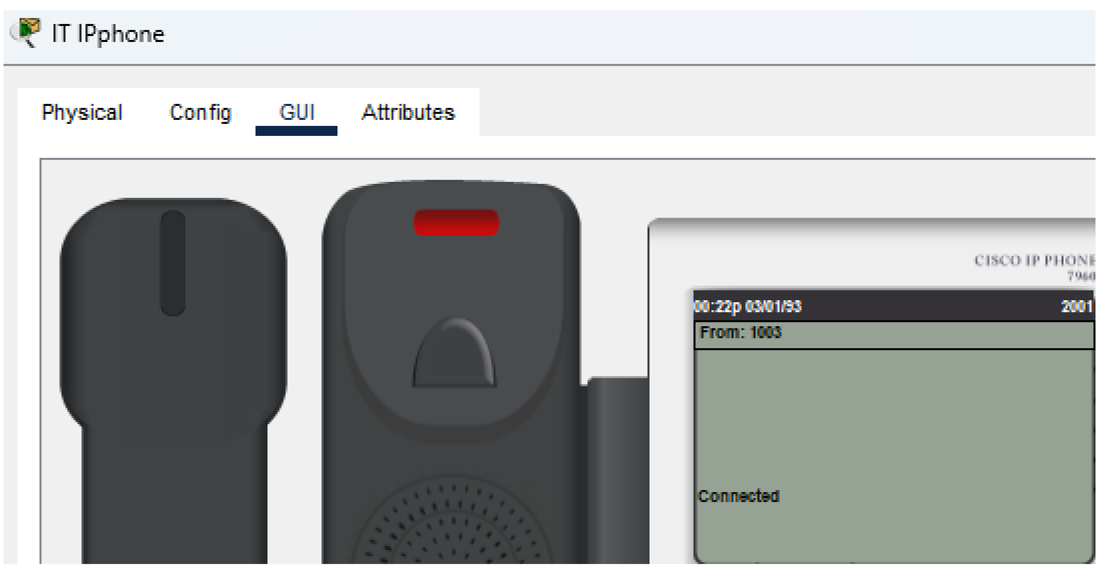
>
URL
http://www.main.com
Go
Main Campus
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

Name	IP Phone testing
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	Dial peers are configured on the router.
Step	1. Open IP Phone GUI 2. Enter the number 3. Monitor the result
Expected result	IP Phone can dial different numbers.

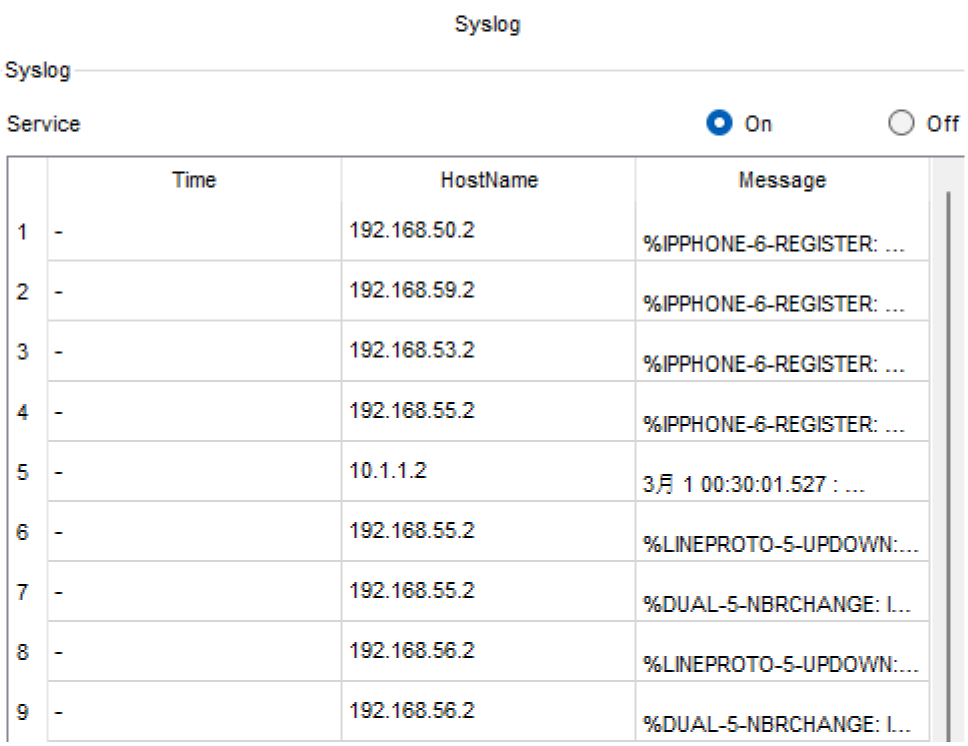





BAdmin IPPhone
Physical
Config
GUI
Attributes
CISCO IP PHONE
00:22p 03/01/93
4002
To: 6001
Connected


Security IP Phone
Physical
Config
GUI
Attributes
CISCO IP PHONE
00:22p 03/01/93
6001
From: 4002
Connected

Name	Syslog testing
Requirement	Cisco Packte Tracer is open All the setting done
Preconditions	Confirm that the Syslog server can receive, and process logs generated by the router.
Step	1. Open the Syslog service of the syslog server. 2. Enable Syslog Service 3. Monitor the result
Expected result	The Syslog Server can collect logs from devices such as routers.


Syslog

Syslog

Service ◉ On ○ Off

	Time	HostName	Message
1	-	192.168.50.2	%IPPHONE-6-REGISTER: ...
2	-	192.168.59.2	%IPPHONE-6-REGISTER: ...
3	-	192.168.53.2	%IPPHONE-6-REGISTER: ...
4	-	192.168.55.2	%IPPHONE-6-REGISTER: ...
5	-	10.1.1.2	3月 1 00:30:01.527 : ...
6	-	192.168.55.2	%LINEPROTO-5-UPDOWN:...
7	-	192.168.55.2	%DUAL-5-NBRCHANGE: I...
8	-	192.168.56.2	%LINEPROTO-5-UPDOWN:...
9	-	192.168.56.2	%DUAL-5-NBRCHANGE: I...

Charter6: Future improvement

In the future, I will want to add HSRP to achieve the effect of backup network, because when the main network device fails, it can be seamlessly switched to the backup device to ensure the stable operation of the network. Reduce downtime and business interruptions due to equipment failure.

Additionally, I would like to adopt fiber connectivity as part of the network infrastructure, because fiber connections, with their high bandwidth and low latency characteristics, can effectively cope with growing network traffic and data demands, This high-performance connectivity will help improve overall network performance and prepare for future access to more devices and services.

Finally, I want to add IPv6 networks, because with the development of the time, the number of devices used by students and teachers has gradually increased, and the IPv4 address space has gradually become insufficient. Therefore, the adoption of IPv4 will provide the network with a larger address space to ensure that each device can obtain a unique identity thereby better supporting the modern network environment.

Charter 7: Summary and Conclusion

The design concept of the project focuses on building a secure and stable network to meet the network need of a total of 5,000 people in the university, covering two buildings on the main campus and branch campus.

In the end, I had a lot of difficulty with this project and was forced to give up some of the techniques I wanted to add to the project. I believe this failure will give me a boost. It also reminded me that not everything is smooth sailing and that perseverance is the most important thing. I also believe that these techniques will help me in the future.

Appendices

Reference

1. "Campus Network Design & Implementation Project on Packet Tracer | Enterprise Network Project #4." Www.youtube.com, www.youtube.com/watch?v=qIbhkmTB8Q8.

2. "How to Configure VPNs Using Cisco Packet Tracer (Overview)." Www.youtube.com, https://www.youtube.com/watch?v=mMhCtLlRfDM&list=PLK-Bs6BGQBEOyEorKVVfwDJusFwV61LbM. Accessed 22 Jan. 2024.

3. "Configuring an ASA Firewall on Cisco Packet Tracer - Part One." Www.youtube.com, https://www.youtube.com/watch?v=SLZS1mSc_VY&list=PLK-Bs6BGQBEP4HUmB1g27aIL4EPyXeLNx. Accessed 22 Jan. 2024.

4. "How to Configure Syslog Server in Cisco Packet Tracer | Technical Hakim #SyslogConfiguration CCNA." Www.youtube.com, https://www.youtube.com/watch?v=aB1KJi7IkXw. Accessed 22 Jan. 2024.

5. "Create SMTP(Two Email Gmail & Yahoo) in Cisco Packet Treacher." Www.youtube.com, https://www.youtube.com/watch?v=72qOBD-6of8&t=353s. Accessed 22 Jan. 2024.

Project Proposal Form

Student name: Lee Han Liang

Student ID No: **Email Address:**

Project title: University Network Design

Supervisor: Mr. Gan

Brief description of project:

Networking is more important on campus, and university campus networking needs depend on a variety of activities, such as studies, lectures and other events. A robust network infrastructure is essential to support seamless data flow and to ensure that students and faculty can access resources and collaborate effectively.

I propose to design a university network in my final year project. I plan to design two campuses, a 7-storey building on the main campus and a 5-storey building on the branch campus. The main campus has 32classrooms, faculty, admin, and Finance offices on the first floor, a school auditorium on the sixth floor, and a server room and CCTV room on the top floor. There are 24 classrooms on the branch campus. Faculty and administrative offices and a CCTV room are located on the first floor.

I needed to design a brand new network system for a university that required a security-oriented network, a monitoring room, firewalls, and a network that could handle 5,000 people.

Objectives :
In this project, I will use VPN to connect two campus networks.. DHCP to automatically assign all network address, WI-FI for everyone, VLAN differentiate between faculty, staff , students, administrators, visitors.. ACL is used to restrict the movement of faculty, students and guest and CCTV system is used to protect everyone on campus.

Hardware:
In this project I will use these hardware: firewalls, servers, IP phones, switches, multilayer switches, routers, access points, RFID cards, cameras, laptops, PCs.

Software:
In this project I will be using Cisco packet tracer, Draw.io, Microsoft Word and Microsoft PowerPoint software to complete my final year project.

Student Signature: ....

Project Manager Signature: ....

Date: ....

PROJECT SPECIFICATION FORM (PSF)

Student name: Lee Han Liang **Student ID No:**

Intake: Feb 2022 **Email Address:**

Project title: University Network Design

Supervisor: Mr. Gan

1. Brief description of project background.

Nowadays, networking needs are very important in offices, universities, and shopping malls, all of which require Internet access. In this project, I will be designing a network for faculty, students, and visitors to use. The reason I am doing this project is because I want to know how the network environment really works in universities and colleges. There are two campuses, separated by about five kilometers. My goal for this project is to interconnect and secure the networks of the main and branch campuses.

2. **Brief description of project objectives.**

After completing my project, my system can meet the targets below:

a) Implement a VPN to enable University employees and students to securely access the Internet and internal resources remotely.
b) Create staff, student, admin, and guest VLANs and separate employee VLANs by department.
c) CCTV on campus, classrooms, and corridors.
d) DHCP automatically assigns IP addresses to each floor.
e) RFID door access.
f) WI-FI available to all on campus.
g) Firewalls protect networks and servers.
h) Nat Convert Private IP to Public IP

3. **Brief description of the resources needed by the proposal.**

Hardware:
The hardware I will be using in this project includes laptops and PCs.

Software:
In this project, I will be using software including Cisco packet tracer, Draw.io, Microsoft Word, and Microsoft PowerPoint software to complete my final project.

Access to information/expertise:
Internet – Google.com/Youtube.com/Git hub and ChatGPT
Interview-network engineer/friend/lecturer

User involvement
Project Supervisor – Mr. Gan
Project member – Lee Han Liang

4. **Academic research being carried out and other information, techniques being learnt. (i.e. What are the names of the books you are going to read / data sets you are going to use and why.)**

Main reference :
https://www.youtube.com/watch?v=e1cD2KIme-E
The first reference completes the network of the sub-campus.

Other reference:
https://www.youtube.com/watch?v=rYZ9KFpdS4s&list=PLZ-4ZnYRIJpI6ij6OWzroxPNfHW8k-tfX
The second completes the configuration of HSRP

5. **Brief description of the development plan for the proposed project. (i.e. which software methodology and why, the major areas or functions to be developed, and the order in which they will be developed)**

a) Search- ChatGPT, internet reference.
b) Write a project proposal.
c) Designing the university network environment.
d) Draw university network topology.
e) Drawing of university floor plans
f) Creating a University Network with Cisco Packet Tracer
g) Troubleshooting and testing
h) Project paperwork
i) Presentation

6. **Brief description of the evaluation and test plan for the proposed project**.

DHCP- Used to auto-assign IP.

WI-FI - Give them wireless Internet access.

VLAN - Networks used to segregate sectors.

ACL - You can block unknown IP addresses or prevent insiders from stealing information.
CCTV System - It reduces crime on campus and keeps people safe.

VPN - It is possible for the two campuses to establish a private channel that will minimize the leakage of confidential documents.

RFID – RFID cards can be used to open the door for more security.

Firewall - It can protect the campus network security and defend against unrecognized IP addresses.

Nat- Nat can convert a private IP to a public IP for increased security.

The Supervisor and Project Manager should sign in the space provided below to indicate that the necessary resources (including supervision time/expertise) are available for the proposal.

Supervisor’s signature: ....

Date: ....

Project Weekly Report

Project Weekly Report 1

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/11/15** **Sheet no.1**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Do online research.**
2. **Writing project proposal**
3. **Discuss the project theme and tittle with supervisor**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 2

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/11/16** **Sheet no. 2**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Writing project specification**
2. **Designing a University Network with Cisco Packet Tracer**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 3

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: Lee Han Liang **Date:** 2023/11/29 **Sheet no.** 3

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. Writing Weekly Report.
2. Completion of the main campus design.
3. Configure VLAN for the main campus network.

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 4

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/12/06** **Sheet no.4**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure Home Router**
2. **Finding and resolving errors**
3. **Do online search**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 5

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/12/14** **Sheet no.5**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Home Router switched to WLC to set up WIFI.**
2. **Configure DHCP IP.**
3. **Finding and resolving errors**
4. **Do Online Research**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 6

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/12/20** **Sheet no.6**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure VoIP Phone**
2. **Configure IOT Devices**
3. **Configure Backup link on Multilayer Switch**
4. **Finding and resolving**
5. **Do Online Research**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 7

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2023/12/28** **Sheet no.7**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Network configuration completed at branch campus.**
2. **Finding and Resolving Port Channel Problems**
3. **Do Online Research**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 8

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2024/01/03** **Sheet no.8**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure VPN**
2. **Finding and Resolving VPN Problems**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 9

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2024/01/09** **Sheet no.9**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure SSH**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 10

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2024/01/09** **Sheet no.10**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure ACL**

Student Signature: ....

Supervisor Signature: ....

Date: ....

Project Weekly Report 11

Notes on use of the weekly report

1. This report is designed for meeting of more than 10 minutes duration, of which there should be at least ten during the course of the project.
2. The student should prepare for the supervisory meeting by deciding which questions he or she needs to ask the supervisor and what progress has been made (if any) since the last meeting and noting these in the relevant sections of the form, effectively forming an agenda for the meeting;
3. The business of the meeting should be noted briefly in the relevant section of the form;
4. The actions on the student (and, perhaps the supervisor), which should be carried out before the next meeting should be noted briefly in the relevant section of the form;
5. The report is one of the deliverables from the project and is an important record of the student's organisation and learning experience. The student should ensure that it is handed in as an appendix of the report, with sheets dated and numbered consecutively;

Student's name: **Lee Han Liang** **Date: 2024/01/09** **Sheet no.11**

Project title: University Network Design

Supervisor's name: Mr. Gan

Report:

1. **Configure NAT**

Student Signature: ....

Supervisor Signature: ....

Date: ....